

MACROECONOMICS I

ECON-2031



MODULE

February, 2023
Arba Minch, Ethiopia

CHAPTER ONE

INTRODUCTION TO MACROECONOMICS

1.1. Introduction

- ❖ Dear students, welcome to the first chapter of the course Macroeconomics I. In this chapter we examine what macroeconomics is and the evolutions of macroeconomics. The chapter also presents the prominent Schools of Thoughts in Macroeconomics.

❑ Chapter Objectives

In this chapter, we will highlight the basics of macroeconomics analysis. Hence, after a successful completion of this chapter, students will be able to:

- *Define macroeconomics, and distinguish it from microeconomics,*
- *Appreciate the major goals of macroeconomists for the macro-economy,*
- *Appreciate the major policy instruments (macroeconomists may choose from to alter the macro-economy), and the framework of macroeconomic analysis,*
- *Understand the evolution of macroeconomics thinking, &*
- *Explain on the principles and policy implications of various macroeconomic thoughts.*

1.1. What Macroeconomics is about?

Economics is the study of the economy and the behaviour of people in the economy. Traditionally, economics is divided into **microeconomics**, which studies the behaviour of individuals and organizations (consumers, firms and the like) at a disaggregated level, and **macroeconomics**, which studies the overall or aggregate behaviour of the economy. Since our interest here is with macroeconomics, we seek to explain phenomena such as inflation, unemployment, and economic growth and we are not concerned with, say, the demand for or supply of a specific commodity.

Macroeconomics is concerned with the behaviour of the economy as a whole- with respect to booms and recessions, the economy's total output of goods and services and the growth of output, the rate of inflation and unemployment, the balance of payments, and exchange rates. Macroeconomics focuses on the economic behaviour and policies that affect consumption and investment, trade balance, the determinants of changes in wages and prices, monetary and fiscal policies, the money stock, government budget, interest rate, and national debt.

In macroeconomics, we do *two things*.

- ✓ *First*, we seek to understand the economic functioning of the world we live in; and,
- ✓ *Second*, we ask if we can do anything to improve the performance of the economy.

That is, we are concerned with both explanation and policy prescriptions. Explanation involves an attempt to understand the behaviour of economic variables, both at a moment in time and as time passes. Modern macroeconomics recognizes that it is important to focus on more than just short period of time, and so has an explicitly dynamic focus. We thus try to explain the behaviour of economic variables over time. This means that we wish to explain the behavior of the economy both in the long run and in the short run.

1.2. Macroeconomic Goals, Instruments and Methods of Analysis

Macroeconomic analysis deals with the behaviour of the economy as a whole with respect to output/income, employment, general price level, and distribution of income and wealth, balance of payments, etc. To bring about desirable changes in these variables, countries (both developed and less developed) adopt various macroeconomic policies. These policies do vary from one country to another depending on the economic condition prevailing in the respective economies.

Macroeconomic Goals

Macroeconomic policy makers design various policies to achieve desirable outcomes in the macroeconomic goals that are generally desired by societies. The primary goals for the macro-economy (among others) are:

- ✓ Economic growth
 - ✓ Low unemployment or Full employment
 - ✓ Price stability or low inflation
 - ✓ Equitable Distribution of Income
-
- ❖ ***Economic growth*** ultimately determines the prevailing standard of living in a country. Economic growth is measured by the percentage change in real (inflation-adjusted) gross domestic product.
 - A growth rate of more than 3% is considered as good.
 - ❖ ***Unemployment***, as measured by the unemployment rate, is the percentage of people in the labor force who do not have a job. When people lack jobs, the economy is wasting a precious resource-labor, and the result is lower goods and services produced. Besides, unemployment represents people's livelihoods. While unemployment is unlikely to be zero, a measured unemployment rate of 6% or less is considered low (good).

❖ ***Inflation*** occurs when there is a sustained increase in the overall level of prices, and is measured more commonly by the consumer price index. If many people face a situation where the prices that they pay for food, shelter, and health care are rising much faster than the wages they receive for their labor, there will be widespread unhappiness as their standard of living declines as their purchasing power declines. Besides, rapidly rising prices reduces business expectation and thereby investment. For such reasons, the objective of any nation and/or macroeconomists is to keep the inflation rate as low as possible. Inflation rate of lower than 5% is a major goal.

❖ ***Equitable Distribution of Income:*** nations try to narrow the gap between the higher income and the lower income groups. This is to ensure that all people are equal in terms of standard of living. Taxation (progressive) is one of the methods of achieving equitable distribution of income.

❖ ***Framework / Method of Macroeconomic Analysis***

As you learn in introductory economics, principal tools used by economists are theories and models. In microeconomics, we use theories of supply and demand; likewise, in macroeconomics, we use the theories and/or models of aggregate demand (AD) and aggregate supply (AS). There are two major perspectives on macroeconomics: the Classical and the Keynesian perspective, each of which has its own version of AD and AS model. Based on the two perspectives, macroeconomists try to show what drives the macro economy.

❖ ***Policy Instruments***

National governments have two major policy instruments for influencing the macro economy or to achieve the macroeconomic goals. The first is ***monetary policy***, which involves managing the money supply and interest rates. The second is ***fiscal policy***, which mainly involves changes in government spending/purchases and taxes.

In summary, we study macroeconomics from three different perspectives as shown in the structure below:

1. What are the macroeconomic goals?
2. What are the frameworks economists can use to analyze the macro-economy?
3. What are the policy tools governments may use to manage the macro- economy?

Goals

- *Economic Growth*
- *Low Unemployment*
- *Low Inflation*
- *Equitable Distribution of Income, etc.*

Framework

- *Aggregate Demand/ Aggregate Supply*
- *Keynesian Model*
- *Classical Model*

Policy Tools

- *Fiscal Policy*
- *Monetary Policy*
- *Income Policy*

- *This chart shows what macroeconomics is about and how macroeconomists do with the macro-economy.*

1.3. The State of Macroeconomics: Evolution and Recent Developments

❖ The Development of Macroeconomics

Partly as a reaction to the Great Depression of the 1930's and with the publication of Keynes's General Theory of Employment, interest and Money in 1936, modern macroeconomics developed as analytical framework for understanding what causes fluctuations in the levels of employment and output.

What economic theory needed during the 1930's was an explanation of the disastrous experience of those years. How could an economy plunge to a predicament in which a quarter of a nation's resources were idle? Keynes provided a theory to explain this phenomenon. It was so successful that it began a Keynesian era in macroeconomic theory that stood in sharp contrast to the classical theory that had prevailed over the preceding century or more.

During the decade, following the appearance of the General Theory of Keynes, economists addressed themselves to refining and building on the pioneer work of Keynes, to analyzing the complex economic processes that determine the actual level of employment. This was a major shift from early held belief that the economy, through forces of competitive market, will remain uninterruptedly at full utilization (full employment).

The accepted economic theory of the pre-Keynesian era argued that full employment was the normal state of affairs and that departure from this were strictly temporary.

❖ Schools of Thoughts in Macroeconomics

Following the appearance of the General Theory of Keynes, macroeconomic theory could be neatly divided into two parts: Classical and Keynesian. Keynes chooses to contrast the ideas he presented in the General Theory with the prevailing ideas by labeling them as **Classical**.

In economic theory, the term “Classical” had been coined by Karl Marks, who used it to cover the theories of David Ricardo, James Mill and their predecessors. Keynes extended the term to include the followers of Ricardo, who perfected the theory of the Ricardian economics, like J.S.Mill, Marshal, Edge Worth, and Pigou. Moreover, New and different formulations of Classical theory were appeared under the heading *neoclassical in 1870s*.

Despite the tremendous success of Keynes work, it did not by any means put an end to the further development of classical economics. Beginning in 1950s another extension of classical macro-economic theory has emerged. This theory is known as *monetarism*, so called because of the critical role it assigns to money as a determinant of what happens in the economy. Finally, in *1970s* came the latest theoretical development with roots in classical theory-the *New Classical Economics*.

As these developments were occurring on the classical side, there were also changes on the Keynesian side. Over the years following the publication of the General theory, economists refined and extended the many insights contained in it and gradually built what is known as Keynesian economics. Keynesian economics has also two groups: *Neo-Keynesian* (members of this school are predominantly British) and *Post-Keynesians* (members are predominantly Americans). The most recent wave of thought with Keynesian root is New Keynesian economics.

So much is about the development of Macroeconomic thought. Based on the current debates, it is possible to classify the schools of thought into two:

- ✓ The New Classical School
- ✓ The New Keynesian School

So far, it has been tried to highlight the evolution of macroeconomic thinking over time. Now, let's consider somewhat deeper analysis of macroeconomic thinking of each school emphasizing major tenets and policy implications of the same.

1. Classical (1776 – 1870)

In this period the distinction between micro and macro was not clear. The ruling principle was the *invisible hand* coined by Alfred Marshall. The Classics have made an ample contribution to the development of economic science. They felt the *market was self-adjusting*. *That is, for them school markets (be it, commodity, factor, and money) works best if left to themselves.*

- With regard to the labor market, they contend that labor demand and labor supply are brought into equilibrium by the real wage. As a result, there is no involuntary unemployment.
- With regard to the financial market, for Classical economists saving and investment are brought into equilibrium by the interest rate and investment responds to the interest rate.
- In the money market, money demand is simply a transaction demand and money has no any effect on the real economy and hence raising the money supply simply pushes up prices (i.e. inflationary). *That is what we call **Classical dichotomy***: the quantity of money affect only nominal variables (i.e. money wages, and nominal GNP), and have no influence whatsoever on the real variables of the economy such as real GNP, level of employment and real wage rate.
- *In general, for classical economists, both Fiscal policy and monetary policy are useless. Because, for them;*
 - *Fiscal policy raises interest rate and crowds out investment.*
 - *Monetary policy raises prices and does nothing to “real” things.*

Implication: *No need for macroeconomic policy.*

Government has no any role in the economy through its fiscal and monetary policy. As such classical economists are advocates of “*laissez-faire*” –*free market system*.

Influential Classical Economists Include:

✓ *David Hume (1711-1776), Adam Smith (1723-90), Thomas Malthus (1766-1834), David Ricardo (1772-1823), John Stuart Mill (1806-1873), Alfred Marshall (1842-1924) and Arthur Pigou (1877-1959).*

2. **Neo-classical 1870–1936**: Basically, the neoclassical school is not different from the classical school. The main distinction is the tool of analysis, such as the marginal analysis. In the area of growth theory, they gave us neoclassical growth models. Eg: Solow growth model.

3. **Keynesian Macroeconomics (1936 – 1970s)**

Influential Keynesian Economists Include:

✓ *John Maynard Keynes (1883-1946), Paul Samuelson (1915-2009), James Tobin (1918-2002), and Franco Modigliani (1918- 2003).*

*The main theme of the Keynesian stream is that the economy is subjected to failure so that it may not achieve full employment level. Thus, government intervention is inevitable. This school views the labor market in that workers and firms bargain for a money wage, not for real wage. Money wages adjust slowly and workers resist any drop in the money wage, which is termed as **Nominal***

wage rigidity. Unlike the Classicals, for Keynesians saving and investment are brought into equilibrium by changes in income. *Investment is not influenced by a mere change in interest rate;* rather it is affected by expectations of the future, which is uncertain. Money demand is affected by transactions, but also by other things, in particular fear, which may lead to a “speculative demand” for high money balances. *With regard to the role of the government, Keynes argued that the government role was needed to preserve capitalism because without management, a modern capitalist economy is so unstable that it may threaten the social compact that it rests on.*

❑ **In short, for the Keynesian school:**

- ❖ The self-correcting feature of the market, which is of course the hallmark of classical theory, does not work. They believe that prices and especially wages respond slowly to changes in supply and demand, resulting in shortages and surpluses,
- ❖ AD is influenced by a host of economic decisions – both public and private.
- ❖ Changes in AD, whether anticipated or unanticipated, has its greatest short-run impact on real output and employment, not on prices.
- ❖ Keynesians are more concerned about combating unemployment than about conquering inflation.

❖ **Policy Implication**

- ✓ **During recessions,** government should satisfy peoples demand for more cash preventing the downward spiral of shrinking income and shrinking spending via ***monetary expansion***. But Keynes worried that even this might sometimes not be enough, particularly if a recession had been allowed to get out of hand and become a true depression. Once the economy is deeply depressed, households and especially firms may be unwilling to increase spending no matter how much cash they have; they may simply add any monetary expansion to their hoarding. Such a situation, in which monetary policy has become ineffective, has come to be known as a “***liquidity trap***”. In such a case, the government has to do what the private sector will not: spend. ***When monetary expansion is ineffective, fiscal expansion must take its place.*** Such a fiscal expansion can break the vicious circle of low spending and low incomes and getting the economy moving again.

4. The Neo-Keynesian synthesis (1950s & 1960s)

The neo-Keynesian synthesis was developed by neoclassical economists who allowed the economy for a short run behavior with Keynesian properties and a long run with classical

properties. Since it contains classical and Keynesian elements, the approach is often referred to as the neoclassical synthesis. They pick best elements of Classical and Keynesian approaches. For them the Economy is “Keynesian” in the short run but “Classical” in the long run. Long-run AS curve is vertical while short-run AS curve is upward sloping.

✓ ***The Neo-Keynesian synthesizers include:*** Paul Samuelson (1915-2009), James Tobin (1918-2002), Franco Modigliani (1918-2003), Robert Solow (1924-) *plus virtually all economists in 1950s and 1960s except Milton Friedman (1912-2006).*

5. 1970s – Present.

There was no dominant school of thought of macroeconomics. There have been two main intellectual traditions in macroeconomics. One school of thought believes that government intervention can significantly improve the operation of the economy; the other believes that markets work best if left to themselves. In the 1960s, the debate on these questions involved *Keynesians*, including Franco Modigliani and James Tobin, on one side, and *monetarists*, led by Milton Friedman, on the other. In the 1970s, the debate on much the same issues brought to the fore a new group- the *new classical macroeconomists*, who by and large replaced the monetarists in keeping up the argument against using active government policies to try to improve economic performance. On the other side are the *new Keynesians*; they may not share some of the detailed belief of Keynesians, except the belief that government policy can help the economy perform better.

6. Monetarism (1970s)

Influential Monetarists Include:

✓ *Milton Friedman (1912-2006), & Karl Brunner (1916-), Allan Meltzer (1928-), & David Laidler (1938-)*

Monetarism, as advocates of free market, started challenging Keynes’s theory in the 1970s. Milton Friedman, the founder of monetarism, attacked Keynes idea of smoothing business cycle on the ground that such active policy is not only unnecessary but actually harmful, worsening the very economic instability that it is supposed to correct, and should be replaced by simple, mechanical monetary rules. This is the doctrine that came to be known as “**monetarism.**” *Friedman began with a factual claim: most recessions, including the huge slump that initiated the Great Depression, did not follow Keynes’s script.* That is, they did not arise because the private sector

was trying to increase its holdings of a fixed amount of money. Rather, they occurred because of a fall in the quantity of money in circulation. With regard to the labor market, while not putting forward his own theory of the labor market, Friedman argues that people do tend to think in real terms and not in nominal amounts. Friedman had a strong faith in the ability of the private sector to produce growth and stability if it is not constrained by government. *Friedman is a libertarian, opposed to government interference on principle.*

❖ ***Policy Rule under Monetarism***

If economic slumps begin when people spontaneously decide to increase their money holdings, then the monetary authority must monitor the economy and pump money in when it finds a slump is imminent. If such slumps are always created by a fall in the quantity of money, then the monetary authority need not monitor the economy; it need only make sure that the quantity of money doesn't slump. In other words, a straightforward rule- ***“Keep the money supply steady”- is good enough,*** so that there is no need for a “discretionary” policy of the form, “Pump money in when your economic advisers think a recession is imminent.” ***Thus, for Monetarists, depressions were the consequence of a temporary shortage of money and this implied that monetary policy was of prime importance in determining the aggregate level of output and employment.***

7. The New Classical School (1980s to the present)

Influential New Classical Economists include:

- *Robert Lucas (1937-), Thomas Sargent (1943-), Edward Prescott (1940-), Robert Barro (1944-).*

The new classical macroeconomics remained influential in the 1980s. This school of macroeconomics shares many policy views with Friedman. It sees the world as one in which individuals act rationally in their self-interest in markets that adjust rapidly to changing conditions. The government is claimed, likely only to make things worse by intervening. The central working assumptions of the new classical school are:

1. ***Economic agents maximize.*** Households and firms make optimal decisions given all available information in reaching decisions and that those decisions are the best possible in the circumstances in which they find themselves.

2. ***Expectations are rational***, which means they are statistically the best predictions of the future that can be made using the available information. Rational expectations imply that people will eventually come to understand whatever government policy used, and thus that it is not possible to fool most of the people all the time or even most of the time.
3. ***Markets clear***. There is no reason why firms or workers would not adjust wages or prices if that would make them better off. Accordingly, prices and wages adjust in order to equate supply and demand; in other words, market clear. For instance, any unemployed person who really wants a job will offer to cut this or her wage until the wage is low enough to attract an offer from some employer. Similarly, anyone with an excess supply of goods on the shelf will cut prices so as to sell. The essence of the new classical approach is the assumption that markets are continuously in equilibrium.

8. *The New Keynesian Macroeconomics (1980s)*

Influential New Keynesians include:

- ✓ *Edmund Phelps (1933-), David Romer, Greg Mankiw (1958-), Stanley Fischer (1943-),*
- ✓ *John B. Taylor (1946-), Olivier-Jean Blanchard (1948-).*

New Keynesian economics is the school of thought in modern macroeconomics that evolved from the ideas of John Maynard Keynes. Keynes wrote *The General Theory of Employment, Interest, and Money* in the thirties, and his influence among academics and policymakers increased through the sixties. In the seventies, however, new classical economists called into question many of the principles of the Keynesian revolution. The label "*new Keynesian*" describes those economists who, in the eighties, responded to this *new classical* critique with adjustments to the original Keynesian tenets.

The new classical group remains highly influential in today's macroeconomics. But a new generation of scholars, the *new Keynesians*, mostly trained in the Keynesian tradition but moving beyond it, emerged in the 1980s. They do not believe that markets clear all the time but seek to understand and explain exactly why markets fail.

The new Keynesians argue that markets sometimes do not clear even when individuals are looking out for their own interests. Both information problems and costs of changing prices lead to some price rigidities, which help cause macroeconomic fluctuations in output and employment.

The primary disagreement between new classical and new Keynesian economists is over how quickly wages and prices adjust. New classical economists build their macroeconomic theories on the assumption that wages and prices are flexible. They believe that prices "clear" markets—balance supply and demand—by adjusting quickly. New Keynesian economists, however, believe that market-clearing models cannot explain short-run economic fluctuations, and so they advocate models with "sticky" wages and prices. *New Keynesian theories rely on this stickiness of wages and prices to explain why involuntary unemployment exists and why monetary policy has a strong influence on economic activity.*

A long tradition in macroeconomics (including both Keynesian and monetarist perspectives) emphasizes that monetary policy affects employment and production in the short run because prices respond sluggishly to changes in the money supply. According to this view, if the money supply falls, people spend less money, and the demand for goods falls. Because prices and wages are inflexible and don't fall immediately, the decreased spending causes a drop in production and layoffs of workers. New classical economists criticized this tradition because it lacked a coherent theoretical explanation for the sluggish behavior of prices. Much new Keynesian research attempts to remedy this omission.

A. Menu Costs and Aggregate-Demand Externalities

One reason that prices do not adjust immediately to clear markets is that adjusting prices is costly. To change its prices, a firm may need to send out a new catalog to customers, distribute new price lists to its sales staff, or in the case of a restaurant, print new menus. These costs of price adjustment, called **menu costs** cause firms to adjust prices intermittently rather than continuously.

Economists disagree about whether menu costs can help explain short-run economic fluctuations. Skeptics point out that menu costs usually are very small. They argue that these small costs are unlikely to explain recessions, which are very costly for society. Proponents reply that small does not mean inconsequential. Even though menu costs are small for the individual firm, they could have large effects on the economy as a whole.

Proponents of the menu-cost hypothesis describe the situation as follows. To understand why prices adjust slowly, one must acknowledge that *changes in prices have externalities*—that is,

effects that go beyond the firm and its customers. For instance, a price reduction by one firm benefits other firms in the economy. When a firm lowers the price it charges, it lowers the average price level slightly and thereby raises real income. (Nominal income is determined by the money supply.) The stimulus from higher income, in turn, raises the demand for the products of all firms. This *macroeconomic impact of one firm's price adjustment on the demand for all other firms' products is called an "aggregate-demand externality."* *Firms which do not lower their prices causes real aggregate demand to be lower in recessions than it would be otherwise; if all firms were to lower their prices together, real balances would rise and real demand for all firms' output would increase. Thus, aggregate demand externality due to firms' inability to reduce prices explains fluctuations in aggregate output and employment.*

In the presence of this aggregate-demand externality, small menu costs can make prices sticky, and this stickiness can have a large cost to society. Suppose that General Motors announces its prices and then, after a fall in the money supply, must decide whether to cut prices. If it did so, car buyers would have a higher real income and would, therefore, buy more products from other companies as well. But the benefits to other companies are not what General Motors cares about. Therefore, General Motors would sometimes fail to pay the menu cost and cut its price, even though the price cut is socially desirable. This is an example in which sticky prices are undesirable for the economy as a whole.

B. The Staggering of Prices

New Keynesian explanations of sticky prices often emphasize that *not everyone in the economy sets prices at the same time*. Instead, the adjustment of prices throughout the economy is staggered. Staggering complicates the setting of prices because firms care about their prices relative to those charged by other firms. Staggering can make the overall level of prices adjust slowly, even when individual prices change frequently.

Consider the following example. Suppose, first, that price setting is synchronized: every firm adjusts its price on the first of every month. If the money supply and aggregate demand rise on May 10, output will be higher from May 10 to June 1 because prices are fixed during this interval. But on June 1 all firms will raise their prices in response to the higher demand, ending the three-week boom.

Now suppose that price setting is staggered: Half the firms set prices on the first of each month and half on the fifteenth. If the money supply rises on May 10, then half the firms can raise their prices on May 15. Yet because half of the firms will not be changing their prices on the fifteenth, a price increase by any firm will raise that firm's relative price, which will cause it to lose customers. Therefore, these firms will probably not raise their prices very much. (In contrast, if all firms are synchronized, all firms can raise prices together, leaving relative prices unaffected.) If the May 15 price setters make little adjustment in their prices, then the other firms will make little adjustment when their turn comes on June 1, because they also want to avoid relative price changes. And so on. The price level rises slowly as the result of small price increases on the first and the fifteenth of each month. Hence, staggering makes the price setting sluggish, because no firm wishes to be the first to post a substantial price increase.

C. Coordination Failure

Some new Keynesian economists suggest that recessions result from a failure of coordination. *Coordination problems can arise in the setting of wages and prices because those who set them must anticipate the actions of other wage and price setters.* Union leaders negotiating wages are concerned about the concessions other unions will win. Firms setting prices are mindful of the prices other firms will charge. *In the real world, coordination is often difficult because the number of firms setting prices is large.*

D. Efficiency Wages

Another important part of new Keynesian economics has been the development of new theories of unemployment. Persistent unemployment is a puzzle for economic theory. *Normally, economists presume that an excess supply of labor would exert a downward pressure on wages.* A reduction in wages would, in turn, reduce unemployment by raising the quantity of labor demanded. *Hence, according to standard economic theory unemployment is a self-correcting problem.*

However, New Keynesian economists often turn to theories of what they call efficiency wages to explain why this market-clearing mechanism may fail. *These theories hold that high wages make workers more productive. The influence of wages on worker efficiency may explain the failure of firms to cut wages despite an excess supply of labor.* Even though a wage reduction would lower a firm's wage bill, it would also—if the theories are correct—cause worker productivity and the firm's profits to decline.

Let's mention three of the various reasons about how wages affect worker productivity.

1. ***High wages reduce labor turnover.*** Workers quit jobs for many reasons—to accept better positions at other firms, to change careers, or to move to other parts of the country. The more a firm pays its workers, the greater their incentive to stay with the firm. By paying a high wage, a firm reduces the frequency of quits, thereby decreasing the time spent hiring and training new workers.
2. ***The average quality of a firm's work force depends on the wage it pays its employees.*** If a firm reduces wages, the best employees may take jobs elsewhere, leaving the firm with less productive employees who have fewer alternative opportunities. By paying a wage above the equilibrium level, the firm may avoid this adverse selection, improve the average quality of its work force, and thereby increase productivity.
3. ***A high wage improves workers' effort.*** This theory posits that firms cannot perfectly monitor the work effort of their employees and that employees must themselves decide how hard to work. Workers can choose to work hard, or they can choose to shirk and risk getting caught and fired. The firm can raise worker effort by paying a high wage. The higher the wage, the greater is the cost to the worker of getting fired. By paying a higher wage, a firm induces more of its employees not to shirk and, thus, increases their productivity.

❖ **Policy Implications**

New Keynesian economics suggests—in contrast to New Classical theories—that recessions do not represent the efficient functioning of markets. The elements of New Keynesian economics, such as menu costs, staggered prices, and efficiency wages, represent substantial departures from the assumptions of classical economics. *In New Keynesian theories, recessions are caused by some economy-wide market failure.* Thus, they provide a rationale for government intervention in the economy, such as countercyclical monetary or fiscal policy.

✚ **Review Exercises**

1. Define Macroeconomics and differentiate it from Microeconomics?
2. Briefly explain the major goals of Macroeconomics? What instruments policy makers employ to achieve these goals?

3. Write a short description on the principles and implications of the schools of thought in Macroeconomics: Classical, Monetarism, Keynesian, Neo-Keynesian Synthesizers, New classical and New Keynesian Economics.
4. Explain the distinction between the major schools of thought (I.e., Classical Vs Keynesian, New Classical Vs New Keynesians) focusing on their tenets and policy implications?
5. Explain on how Menu costs and Efficiency wage hypothesis can cause economic recessions.

CHAPTER TWO

NATIONAL INCOME ACCOUNTING

National income is considered as the most comprehensive measure of how well an economy is performing. To have an idea of the performance of the economy, measuring the national income of a country is extremely important. But measuring national income is an extremely complicated large task. However, economists have devised various ways of estimating national income. In fact, national income estimates are made in every country these days. In this chapter, we will discuss the various concepts related to national income accounting and the methods of measuring national income.

What is National Income (Output)?

- ✓ The national income of a country in a year is the value, expressed in monetary terms of the net contribution of the factors of production through the production units in the country and abroad in the year.
- ✓ It is thus, the monetary expression of the current flow of net final goods and services resulting from the production activities of the normal residents of a country during the year.
- ✓ Concretely, national income is the net domestic product (or net domestic income) of a country plus net income from abroad.
- ✓ If all the national income is in fact consumed then the national income consists of consumer's goods and services only. If only a part of the national income is consumed in the year then the part that remains accumulate as a stock of goods or as capital stocks. National income then would consist of consumer's goods and services plus increase in stocks of goods, i.e., of consumption plus investment. Net investment includes increase in the stocks of fixed capital goods and working of capital goods.
- ✓ If however, people consume more than the national income then it would probably mean that the stock of capital goods has been eaten in to or that the people have received gifts from abroad. Gifts from abroad refer to current transfer from abroad of consumer goods and services.
- ✓ The national income is, thus the goods and services available to the normal residents in a country, as a result of their production efforts in the year, to consume or to invest.

*In simple words, **national income is the aggregate factor income (i.e., earning of labour and capital, etc), which arises from the current production of goods and services by the nation's economy.** The nation's economy refers to the factor of production (labor and capital, etc) supplied by the normal residents of the national territory.*

From the above definitions it is clear that the concept of national income has three types of interpretations:

- ✓ National income represents a receipt total,
- ✓ National income represents an expenditure total, and
- ✓ National income represents a total value of current production.

This three-fold interpretation takes place because of the fact that every expenditure is at the same time a receipt, and if goods or services bought are valued at their sales prices, we have a three-fold identity that the value received equals the value paid, equal the value of goods and services given in exchange.

Points to note from National income Definition

First, national income refers to the income of a country, say, Ethiopia.

Second, its measurement refers to a specified period of time, say, a year.

Third, national income includes all types of goods and services, which have an exchange value, counting each one of them only once.

When we calculate the national income of a nation, we should remember that no commodity or service should be counted twice; otherwise the results will not be accurate. For example, if raw cotton has been evaluated in agricultural production, it should not be included while calculating the total value of cotton textiles. In the same manner, the value of raw cotton shall have to be kept out while calculating the total value of readymade garments, such as, bush-shirts and trousers made out of cotton textiles. Generally, two statistical methods are used **to avoid double counting or multiple counting** in the calculation of national income: Final products method, and Value-added method.

1. Final products method

- *We add up the value of final products only.*
- We first take the total value of the final consumer goods and producer goods produced in the country during the year.
- This will provide us the aggregate value of all the final goods produced in the country during the year.
- Collective and government services (evaluated at cost price) are also to be added to this aggregate value of the final goods in order to arrive at the total product of the nation concerned.

2. Value-added method

- ✓ We add the values created at each stage in the manufacturing of a commodity.
- ✓ Then all such values accruing at all processes in the manufacturing of all commodities are added up together to arrive at the national income of the country.

Example:

Let us suppose that a ready-made bush-shirt passes through four Stages of production. In the first stage, the farmer produces raw cotton from the soil. In the second stage, the raw cotton is converted in to cotton cloth in a factory. In the third stage, the cotton cloth is converted in to a bush-shirt by the ready-made garments firm. In the fourth stage, the ready-made bush-shirt is sold to a retailer from whom the customer finally purchases it. Now, money value of all the transactions, which take place in the various stages of production of a bush-shirt, should not be added up while making an estimate of the national income of the country; otherwise, there will be multiple counting and the result will not be accurate.

❖ Factors Determining National Income

There are a number of influences that determine the size of the national income in a country. It's on account of these influences that one country may have a larger national income than another. Following are the three main influences:

1. **Quantity and Quality of Factors of production.** The quantity and quality of a country's stocks of the factors of production is one of the most important influences on its national income. The quantity and quality of land, the climate, the rainfall etc., determine the quantity and quality of agricultural production, and hence, the size of national income. The quantity of a labour has doubled influences since labour is at the same time both a factor of production as well as the consumer of what is produced. The quality of labour, depending upon inborn intelligence, education and training, also influences the volume of industrial production. Capital may comprise simple, primitive tools or the most modern type of industrial equipment. The quantity and quality of capital is one of the greatest influences on total output.
2. **The state of technical know-how:** Another influence on output and national income is the state of technical know-how in the country. A country with a poor technical knowledge cannot have a large-sized national income, because it will not be in a position to make the best possible use of its resources.
3. **Political Stability:** Political stability is an essential prerequisite for maintaining production at the highest level.

❖ What is National Income Accounting?

- ✓ It is an accounting record of the level of economic activities of an economy.
- ✓ The instruments that help us to measure the national income of the country.

- ✓ The various types of national income accounts helps us to measure the level of production in the economy at some point of time, and explain the immediate causes of the level of performance.

Various aggregates and concepts concerning domestic and/or national income are used in national income accounting. The following are some of the most important totals relating to national income accounts:

- Gross national product (GNP)
- Gross domestic product (GDP)
- Net national product (NNP)
- National income (NI) or net national income at factor cost
- Personal income (PI)
- Disposable personal income (DPI); and Real income (RI)

Instead of giving a single comprehensive estimate of national income, economists use different such aggregates. The reasons for this are:

- ✓ There is disagreement among the economists over what should and what should not be counted as national income. Instead of passing a judgment on this controversy, the statistics department of the government gives some totals leaving it to each user to select the one he/she prefers.
- ✓ Another reason for this is that a certain total may be most useful for one purpose and a different total for another purpose.

2.1 Basic Concepts of GDP and GNP

1. Gross National product (GNP): What is Gross national product?

- It is the nation's total production of goods and services (usually for one year) evaluated in terms of the market prices of goods and services produced.
- It includes all the economic productions in the economy during one year.
- Strictly speaking, then ***the GNP is the money value of the total national production for any given period.***

2. Gross Domestic Product (GDP)

- The other most important measure of overall economic performance is *GDP*.
- GDP is an attempt to summarize all economic activity over a period of time in terms of a single number; it is a measure of the economy's total output and of total income.
- In other words, ***GDP is the value of all final goods and services produced with in the territorial boundary of an economy in a given time period*** (NB: GDP is a flow not a stock concept).

GDP Vs GNP

Have you got the difference between GDP and GNP from their definitions?

- ❖ GNP is the value of final goods and services produced by domestically owned factors of production within a given period.
 - ❖ While GDP is the value of final goods and services produced within the country's territory within a given period, the factors of production used might not be domestically owned.
 - ❖ *A country's GDP measures the strength of its local economy whereas the GNP measures the overall economic strength of a country or shows how the nationals of a country are doing economically.*
 - ❖ The difference between GDP and GNP corresponds to the net income earned by foreigners. When GDP exceeds GNP residents of a given country are earning less abroad than foreigners are earning in that country. That is, GNP can be higher than GDP, less than GDP or equal to GDP depending on the magnitude of the net factor income /NFI= factor income earned by residents less factor income paid for foreigners/
I.e., $GNP = GDP + NFI$
- ✓ *In simple words, GDP is territorial while GNP is national.*

Two points to note:

First, we must take in to account the ***money value of the final goods*** (and services) produced in the economy to avoid double or multiple counting. It should be remembered that final goods and services are those, which are finally consumed by the consumers. Such goods and services do not enter in to the manufacture of other goods. As against this, intermediate goods and services are those goods and services, which do enter in to the production of other goods and services (Bread, for example, is final good, but flour is an intermediate good). Intermediate products are to be excluded from GDP or GNP.

Second, we must take in to account the money value of ***only currently produced goods and services*** while estimating the GDP/GNP of a country. This is due to the fact that the GDP/GNP is a measure of the economy's production during a particular period of time.

2.2 Approaches of Measuring National Income (GDP/GNP)

There are three methods of estimating the national income of a country. In principle, all the methods lead to the same results. The three methods are: ***Expenditure approach, Income approach and Value-Added Approach.***

- 1. Expenditure Approach:*** The GDP can be viewed as the nation's total expenditure on goods and services produced during the year. Each unit of goods and services produced is matched by an expenditure on that unit. The consumers buy most of the good and services produced in the country. But there are some goods and services, which remains unsold. If the unsold

goods and services were regarded as having been bought by the producers who hold them as stocks or inventories, then the monetary value of the total national production would be equal to the total national expenditure.

Under the expenditure approach to GDP, the total national expenditure can be broken down into the following categories:

- A) **Personal Consumption Expenditure (C):** It includes the consumption expenditure made for both **durable goods** (such as, motor-cars, radio-sets, etc., but not houses) and **non-durable goods** (such as, food, drinks, clothing, etc.) produced in the country during the year. This sub-head also includes expenditure on the purchase of a house but it should be treated as investment rather than consumption expenditure.
- B) **Gross Domestic Private Investment (I):** This item includes private investment in 'capital' or 'producer goods', such as, buildings, machinery, plant, equipment, etc. Business firms primarily purchase such goods. Houses are also included in this category of expenditure, because they are so durable that they represent, in fact, capital goods. Three points should be carefully noted here.
- ✓ **First**, this sub-head includes *capital or investment goods* needed not only to replace the existing depreciated capital goods, but also the *capital goods* required to increase the society's production of goods and services.
 - ✓ **Second**, the term '*investment*' here means *real investment* in the Keynesian sense rather than financial investment. It means the purchase of real investment goods, such as, buildings, machinery, plant, etc. produced during the year.
 - ✓ **Third**, '*investment*' here does not include mere financial transfer, such as, the purchase of existing stocks and shares on the stock exchange. The purchase of existing stocks and shares does not represent new investment for purposes of the national income accounts since it does not involve any production.
- C) **Governments' Purchases of Goods and Services (G):** The governments-central, state and local-purchase from the market: **consumer goods**, such as, paper, stationery, cloth, etc, as well as **investment goods**, such as machinery, equipment, plant, etc, for their own enterprises. In addition, the governments also purchase a number of different services-military, police, secretarial, etc. Transfer payments should be excluded.
- D) **Net Export /Net Foreign Demand (N_x):** As is well known, the entire production of a country is not sold within the country. At the same time, the country imports some finished goods from other countries during the year. To make proper allowance for such exports and imports, the value of imports should be deducted from the value of exports. Net exports represent the net expenditure from abroad on our goods and services, which provides income for domestic producers.

It is; thus, clear that if the entire production of a country is purchased at market prices, the amount so spent will represent the ***GDP of the country at market price***. Therefore, to estimate the GDP, we have to add the above four categories of expenditure.

Illustration

$$\begin{array}{c}
 \textbf{Gross Domestic Product (GDP)} \\
 \textbf{Equals} \\
 \textbf{Personal consumption expenditure (C)} \\
 \textbf{Plus} \\
 \textbf{Gross domestic private investment (I)} \\
 \textbf{Plus} \\
 \textbf{Government purchases of goods and services (G)} \\
 \textbf{Plus} \\
 \textbf{Net export (N}_x\text{)}
 \end{array}$$

Symbolically: $\textbf{GDP} = \textbf{C} + \textbf{I} + \textbf{G} + \textbf{N}_x$

Having the above method of calculating GDP, how can we get the value of GNP?

We know the difference between GDP and GNP is the net factor income from abroad (*NFI*) which is the difference between Income received from abroad by the citizen of the country (*IR*), and the income paid to foreigners (*IP*):

i.e., $\textbf{NFI} = \textbf{IR} - \textbf{IP}$

Then, Gross national product is the sum of *GDP* and net factor income from abroad.

Symbolically:

$$\textbf{GNP} = \textbf{GDP} + \textbf{NFI}$$

I.e., $\textbf{GNP} = \textbf{C} + \textbf{I} + \textbf{G} + \textbf{N}_x + \textbf{NFI}$

If there is no net factor income received from abroad:

$$\textbf{GNP} = \textbf{GDP} = \textbf{C} + \textbf{I} + \textbf{G} + \textbf{N}_x$$

2. *Income Approach:* The expenditure incurred on purchasing goods and services produced in a country during the year also becomes the income of the various factors, which collaborated in the production of those goods and services. We can group these factor-incomes in the following categories:

- A. Wages and salaries of the employees (or compensation to employees),
- B. Incomes of non-company business
- C. Rental incomes of persons
- D. Corporate profits, and
- E. Incomes from Interest

- The first category, as said above, includes the *wages and salaries received by the employees during the year plus certain supplements*. These supplements are the contributions, which the employers make to **social security** and other **pension funds of the workers**.
- The second category includes the incomes earned by individual proprietors, parents and self-employed persons.
- The third category comprises rental income earned by individuals on agricultural and non-agricultural property.
- The fourth category includes corporate profits earned by business corporations before the payment of corporate profit taxes or the payment of dividends to the shareholders. Thus, the corporate profits, used in calculating the GNP, are equal to the sum of corporate profit taxes plus dividends paid to the shareholders plus undistributed corporate profits.
- The fifth category contains net interest earned by individuals or businesses.

An aggregate of the above five categories of incomes will not be equal to the GDP as estimated by the Expenditure Method. The reason is that a part of the total expenditure incurred by the community does not become available to the other factors of production in the form of incomes. There are two such leakages.

- ✓ *First*, indirect taxes levied by the government on goods and services; and
- ✓ *Second*, depreciation of machinery, plants and buildings.

The expenditure incurred by the households (factors) on goods and services includes the indirect taxes levied by the government. The income from these indirect taxes goes to the government and is not available to the households (factors). Likewise, while calculating the *GDP*, we include the depreciation (loss in value suffered by machinery, equipment, buildings, etc.). *Like the indirect taxes, the payment on account of depreciation does not become available to the households (factors) in the form of income*. In other words, depreciation is not part of the factor incomes. Therefore, while estimating the GDP by the Income Method, we have to add indirect taxes and depreciation charges to the factor incomes.

Symbolically: $GDP = W/S + I + R + \Pi + r + IT + D$

Therefore, $GNP = W/S + I + R + \Pi + r + IT + D + NFI$

Or $GNP = GDP + NFI$

3. **Value-Added Approach:** this approach measures the value added (contribution) by each producing entity in the production process. Value added is defined as the difference between total value of the output of a firm and the value of inputs bought from other firms. For the economy as a whole, the sum of all value added must equal the value of all final goods and services. **Thus, GDP is the total value added of all firms in the economy.** Adding the net

factor income earned from abroad on the GDP (at market price)-(computed by adding the value additions of each enterprise in the domestic economy) yields GNP (at market price).

 Generally, the following Precautions should be taken in Measuring GDP/GNP

There are some possible difficulties in measuring GDP which needs precaution:

1. Double counting should be avoided,
2. Unproductive activities should be excluded e.g. transfer payments-local transfer, pension, scholarships, and unemployment allowance.
3. Second hand and intermediate goods should be excluded, because they are not currently produced goods & services.
4. Purchase of new/old shares should be excluded-because it represents a mere transfer of property.

 Is GDP a perfect measure of National Income?

GDP measures the market values of all goods and services produced in the given period defined. Although most goods and services are valued at their market prices when computing GDP, some productions are not sold in the marketplace and therefore do not have market prices. The following are some of the instances.

- A. Housing Services:** A person who rents a house is buying housing services and providing income for the landlord; the rent is part of GDP, both as expenditure by the renter and as income for the landlord. Many people, however, live in their own homes. Although they do not pay rent to a landlord, they are enjoying housing services. *Thus, the value of such service is left out of GDP.*
- B. Home productions:** some of the output of the economy is produced and consumed at home and never enters the marketplace. For example, meals cooked at home are similar to meals cooked at a restaurant, *yet the value added in meals at home is left out of GDP.*
- C. The Underground Economy:** - GDP fails to account for the value of goods and services sold in the underground economy. *(The underground economy is the part of the economy that people hide from the government either because they wish to evade taxation or because the activity is illegal).* The illegal drug trade is an example.
- D. The Informal Economy:** Finally, GDP fails to consider the value of production in the informal sector (both legal which includes 'Tela', 'Teji', shoe repair, small scale trade, and illegal – eg: drug dealing and prostitution).

*Because the value of many goods and services is left unrecorded, **GDP is an imperfect measure of economic activity**. These imperfections are most problematic when comparing standards of living across countries. The size of the underground economy, for instance, varies from country to country. Yet as long as the magnitude of these imperfections remains fairly constant over time, GDP is useful for comparing economic activity from year to year.*

2.3. Other Social Accounts

Various aggregates and concepts concerning domestic income and national income are used in national income accounting. We have discussed the meaning and measurements of two of them: GDP and GNP. The following section gives a brief introduction to some other concepts relating to national income.

i) Net Domestic Product (NDP) and Net National Product (NNP)

- **Net Domestic Product** is the net market value of all the final goods and services produced in the domestic territory of a country during a year.
- **Net national product** is defined as the net production of goods and services produced by citizens of a country during a year

As we know, in the course of a year's production, part of the physical plant that was on hand at the beginning of the year is worn out, or it may depreciate in value as it becomes outdated. Buildings gradually deteriorate, machinery wears out or becomes obsolete, and so on. No sooner or later, if depreciated capital is not replaced, the income of the nation will decline, and the economy as a whole will become poorer. If we want to measure the level of income that can be sustained therefore, we must deduct the depreciation of capital from gross output (GNP or GDP) during the period concerned. What is left after this deduction is the net product. Thus,

$$\mathbf{NNP = GNP - Depreciation}$$

$$\mathbf{NDP = GDP - Depreciation}$$

Net products are better concepts than gross products, because it makes proper allowance for the depreciation suffered by capital goods during the period under consideration. These concepts give a proper idea of the net increase in total production of a country. They are, therefore, used in analyzing the long-period problems of maintaining and increasing the supply of capital goods in the country.

In spite of this fact that for some purposes of economic growth, NNP is much more important than GNP, it is more difficult to measure statistically, because we have no accurate record of the amount of depreciation for various capital goods, such as, buildings, equipments, plants, etc. The result is that the deduction that has to be made from GNP to arrive at NNP is a matter of judgment, rather than of exact measurement. Because of this difficulty, the concept of NNP is not fairly used.

ii) Net National Income at Factor Cost (NNI)

National Income at factor cost means the sum of all incomes earned by resource suppliers for their contribution of land, labor, capital and entrepreneurship during the year's net production. In other words, national income at factor cost shows how much it costs society in terms of economic resources to produce that net output. Sometimes, we simply call it national income.

Both GNP and NNP measure the values of goods produced in industry at the prices that those goods actually bring in the market. Here the difficulty is that when we buy these goods at market prices, then the *prices include the respective taxes* on these goods levied by the government. Hence, not all of the payment made for goods goes to the people who produced them. Some part of it goes to the government. It does not constitute a part of the true cost of producing the goods concerned. ***This violates the basic identity that the value of goods produced is equal to the sum of the money incomes received by the producers.*** In other words, the GNP will exceed the sum money incomes received by people of the country (resource suppliers). The excess is explained by the fact that we have included in GNP the entire output of government services, but we have also included part of the payments made for these services in the prices of the products (in terms of sales tax). In other words, a part of the cost of government has been counted twice. In order to get rid of this double counting, we may introduce the concept of net national income at factor cost. It is equal to net national product minus indirect tax plus subsidies.

Mathematically: $NI_{FC} = NNP_{MP} - \text{Indirect taxes} + \text{Subsidies}$

Where, NI_{FC} is national income at factor cost & NNP_{MP} is net national product at market price.

The concept of national income at factor cost occupies an important place in economics. It indicates the nature of distribution of wealth among various factor inputs. This concept is more

satisfactory than the concept of GNP and NNP, because it eliminates the element of double counting inherent in those two concepts. *It accords with the basic principle of economic theory, that the payments received by the factor suppliers equal the value of the goods produced.* Further, the components of national income (at factor cost) are very useful in dealing with certain economic problems. Changing relative shares of different elements in the population from time to time have an important bearing on the fluctuations of economic activity, and on the personal inequality of incomes, which has much to do with economic welfare.

iii) Personal Income (PI)

Personal income is the sum of all incomes actually received by all individuals or households during a given year. National income (that is, total income earned during a year) is different from personal income (That is, the income actually received by households). We know that all, which is produced by industry and government, must necessarily belong to someone, and hence can be regarded as national income received or accrued by the people (of the country), but not all of the value of this product is paid to them as money income. Part of the net earnings of business firms are taken away from them by government in the form of corporate income taxes excess profits taxes, and the like, before being paid to the owners of the business firms. These are direct taxes that do not enter into the prices of commodities. Further, many firms retain part of their earnings in the business, to make addition to their plants or as a reserve for future emergencies, instead of distributing the entire profits to their shareholders. Therefore, these earnings are not allowed to receive in money all of the salaries or wages that they earn. The government charges a withholding tax, which is used to finance social security payments to the aged, the unemployed and certain other persons. Similarly, the government makes some other transfer payments to people of the country.

All these things create a difference between the total incomes received by individuals and the total social income produced. We subtract from national income all undistributed corporate profits, corporate income taxes and social security withholding taxes and then add transfer payments from government and business firms directly to persons; the resulting figure will be the personal income.

Thus,

$$\begin{aligned} \text{Personal Income (PI)} = & \text{Net National Income} - \text{undivided corporate Profits} - \\ & \text{Corporate Income Taxes} - \text{Social Security Contributions} + \\ & \text{Transfer payments.} \end{aligned}$$

The above concept of personal income is useful for certain special purposes. For example, it shows the ability of people to pay taxes. The personal income data are available on monthly basis whereas the data for national product and national income are not published so frequently and, therefore, the accurate idea cannot be drawn. This is the reason that personal income data are used for analyzing current changes in the economy.

iv) Disposable Income (DI)

The main drawback of personal income is that they do not tell us how much is actually at the disposable of people for their personal expenditure. For this we use the concept of disposable income. After a good part of personal income is paid to government in the form of personal taxes (such as income tax, property tax etc.), what remains of personal income is called Disposable Income. Thus,

$$\text{Disposable Income (DI)} = \text{Personal Income} - \text{Personal Income Taxes}$$

Disposable income data are useful for studying the purchasing power of the consumers. Since disposable income can either be consumed or it can be saved; with its help we may study of consumption and saving that individuals make in the economy.

2.4. Nominal GDP Versus Real GDP

As described before, GDP values the economy's total output of goods and services measured at market prices. But is GDP a good measure of economic well-being? To answer this question, we have to consider what we mean by market prices? When we say market prices, we can measure GDP by the currently prevailing market prices or at some base year prices. *Thus,*

Nominal GDP: *Refers to the market value of goods and services measured at current prices.*

- ✓ *It increases either because prices rise or because quantities rise.*
- ✓ *It is not a good measure of economic well-being.* Because this measure does not accurately reflect how well the economy can satisfy the demands of households, firms, and the government. For instance, if all prices doubled without any change in quantities, GDP would double. Yet it would be misleading to say that the economy's ability to satisfy demands has doubled, because the quantity of every good produced remains the same.

A better measure of economic well-being would include the economy's output of goods and services and would not be influenced by changes in prices. For this purpose, economists use ***real GDP***, which is the value of goods and services measured using a constant set of prices/base-year prices. That is, real GDP shows what would have happened to expenditure on output if quantities had changed but prices had not.

- ✓ Because the prices are held constant, real GDP varies from year to year only if the quantities produced vary. Because economies ability to provide economic satisfaction for its members ultimately depends on the quantities of goods and services produced, ***real GDP provides a better measure of economic performance than nominal GDP.***

2.5. The GDP Deflator and the Consumer Price Index

▪ How can we measure price level?

We can use three indices to measure prices over time: The GDP Deflator, the Consumer Price Index (*CPI*) and the producer price index, among which the GDP deflator and *CPI* are widely applicable.

✚ The GDP Deflator

From nominal GDP and real GDP we can compute a third statistic: the ***GDP deflator***. The GDP deflator is defined as the ratio of nominal GDP to real GDP:

$$\text{GDP Deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}}$$

- ✓ *The GDP deflator reflects what's happening to the overall level of prices in the economy.*

To better understand this, consider an economy with only one good, bread. If P is the price of bread and Q is the quantity sold, then *nominal GDP* is the total number of dollars spent on bread in that year, $P \times Q$. Real GDP is the number of loaves of bread produced in that year times the price of bread in some base year, $P_{base} \times Q$. The GDP deflator is the price of bread in that year relative to the price of bread in the base year P/P_{base} .

The definition of the GDP deflator allows us to separate nominal GDP into two parts: one part measures quantities (*real GDP*) and the other measures prices (*the GDP deflator*). That is,

$$\text{Nominal GDP} = \text{Real GDP} \times \text{GDP Deflator}.$$

Recall that *Nominal GDP* measures the current dollar value of the output of the economy whereas real GDP measures output valued at constant prices. The GDP deflator measures the price of output relative to its price in the base year. We can also write this equation as, ***Real GDP*** = $\frac{\text{Nominal GDP}}{\text{GDP Deflator}}$

- ✓ *The GDP deflator is used to deflate (i.e, take inflation out of) nominal GDP to yield real GDP.*

The Consumer Price Index (CPI)

The most commonly used measure of the general price level is the consumer price index (CPI). Just as GDP turns the quantities of many goods and services into a single number measuring the value of production, the CPI turns the prices of many goods and services into a single index measuring the overall level of prices. The *CPI* is the cost of basket of goods and services relative to the cost of the same basket in some base year.

For example, suppose that the typical consumer buys 5 apples and 2 oranges every month. Then the basket of goods consists of 5 apples and 2 oranges, and the CPI is

$$CPI = \frac{(5 \times \text{Current Price of Apples}) + (2 \times \text{Current Price of Oranges})}{(5 \times 2002 \text{ Price of Apples}) + (2 \times 2002 \text{ Price of Oranges})}$$

In this CPI, 2002 is the base year. The index tells us how much it costs now to buy 5 apples and 2 oranges relative to how much it cost to buy the same basket of fruit in 2002.

Note: The consumer price index is the most closely watched index of prices, but it is not the only such index. Another is the producer price index, which measures the price of a typical basket of goods bought by firms/producers rather than consumers.

The CPI Versus the GDP Deflator

The GDP deflator and the CPI give somewhat different information about what's happening to the overall level of prices in the economy. There are two key differences between the two measures.

1. The GDP deflator measures the prices of all goods and services produced, whereas the CPI measures the prices of only the goods and services bought by consumers. Thus, an increase in the price of goods bought by firms or the government will show up in the GDP deflator but not in the CPI.
2. The GDP deflator includes only those goods produced domestically. Imported goods are not part of GDP and do not show up in the GDP deflator. Hence, an increase in the price of a Toyota made in Japan and sold in this country affects the CPI, because the Toyota is bought by consumers, but it does not affect the GDP deflator.

2.6. GDP and Welfare

➤ Does a rise in GDP guarantees improvement in living standard of societies?

GDP can be taken as a rough measure of standard of living. This is because “Standard of living” is a broader term than GDP. While GDP focuses on production that is bought and sold in markets,

standard of living includes all elements that affect people's well-being, whether they are bought and sold in the market or not. To illuminate the gap between GDP and standard of living, it is useful to spell out issues that GDP does not cover that are clearly relevant to standard of living. *The following are some of the limitations of GDP as a Measure of society's well-being*

➤ ***GDP is a gross measure and has nothing to say about the distribution of income in society.***

It does not indicate how the total national income is actually distributed among the population of a nation. For instance, if GDP is highly unequally distributed among nationals where a small portion of the population earns the significant portion of national output, mass of the population may remain impoverished despite rapid growth in GDP. So, higher GDP doesn't guarantee improvement in living standard.

➤ While GDP includes what is spent on environmental protection, healthcare, and education, it does not include actual levels of environmental cleanliness, health, and learning.

- ☛ GDP includes the cost of buying pollution-control equipment, but it does not address whether the air and water are actually cleaner or dirtier.

- ☛ GDP fails to consider costs of environmental damage to society: where an increase in GDP may be made possible at a cost of environmental degradation (reduction in forest, depletion of mineral resources & wild lives).

- ☛ GDP includes spending on medical care, but does not address whether life expectancy or infant mortality have risen or fallen.

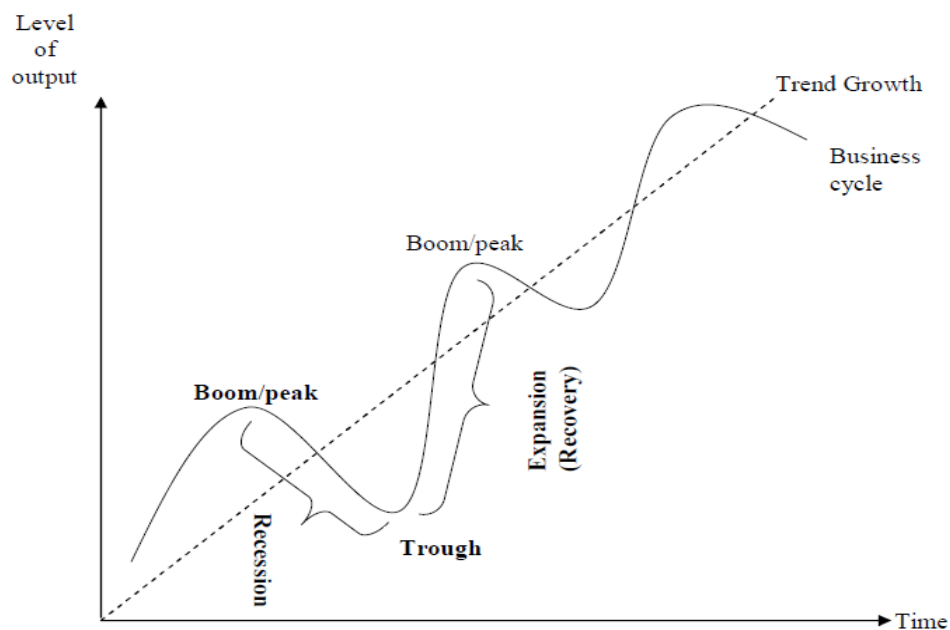
- ☛ Similarly, it counts spending on education, but does not address directly how much of the population are literate.

➤ ***GDP does not include non-market productions.*** For example, hiring someone to clean your house is part of GDP, but doing these tasks yourself is not part of GDP. As considerable number of women are working in the non-market economy like food preparation and child care at home, a considerable amount of production left unrecorded which makes the GDP appear lower.

➤ ***GDP fails to account for Crime rates:*** If people are led by a rising fear of crime, it is hard to believe that an increase in GDP has made them better off.

2.7. The Business Cycle

It refers to the recurrent ups and downs in the level of economic activity. Countries usually experience ups and downs in the level of total output and employment over time. With the fluctuation in the overall economic activity, inflation, and unemployment have also clear cyclical patterns. Therefore, a business cycle is a fluctuation in overall economic activity, which is characterized by the simultaneous expansion or contraction of output in most sectors.



The trend growth of GDP is the path GDP would take if factors of production were fully employed. Over time, real GDP changes for two reasons:

First, more resources become available which allows the economy to produce more goods and services, resulting in a rising trend level of output.

Second, factors are not fully employed all the time. Thus, output can be increased by increasing capacity utilization.

Output is not always at its trend level, that is, the level corresponding to full employment of the factors of production. Rather output fluctuates around the trend level. During expansion (or recovery) the employment of factors of production increased, and that is a source of increased production. Conversely, during a recession unemployment increases and less output is produced than can in fact be produced with the existing resources and technology. **Deviations of output from trend are referred to as the output gap.**

The output gap measures the gap between actual output and the output the economy could produce at full employment given the existing resources. Full employment output is also called potential output.

$$\text{Output gap} = \text{potential output} - \text{actual output}$$

Often a long expansion reduces unemployment too much, causes inflationary pressures, and therefore triggers policies to fight inflation- and such policies usually create recessions.

We can identify four phases in the business cycle:

- a) **Boom/peak:** - it is a phase in which the economy is producing the highest level of output in the business cycle. It is the point, which marks the end of economic expansion and the beginning of recession. At this point, the economy is operating close to full capacity so that the national product and national income correspond to a very high degree of utilization of resources. Because of this, unemployment level is low, business is good; and it is a period of prosperity.
- b) **Recession/Contraction:** - during a recession phase, the level of economic performance generally declines. Total output declines, and business generally decline. As a result, unemployment problem rises. When the recession becomes particularly severe, we say the economy reaches depression or trough. This period can cause hardship on business and citizens.
- c) **Trough/Depression:** - this phase is the lowest point in a business cycle. It marks the end of a recession and the beginning of economic recovery/expansion. During this period, there is an excessive amount of unemployment and idle productive capacity.
- d) **Recovery/Expansion:** - during this phase, the economy starts to grow or recover. In this phase, more and more resources are employed in the production process; output increases, unemployment level diminishes and national income rises. When this expansion of the economy reaches its maximum, the economy once again comes to another boom or peak.

2.8. Inflation: What is inflation? And what causes inflation?

A continuous increase in the overall level of prices is called inflation, and it is one of the primary concerns of macroeconomists and policymakers. It could be measured by an index such as the consumer price index (CPI) or by the implicit price deflator for GDP'. Inflation is frequently described as a state where "too much money is chasing too few goods". When there is inflation, the currency loses purchasing power.

In the definition of inflation, two key words must be borne in mind. **First**, is overall or general, which implies that the rise in prices that constitutes inflation must cover the entire basket of goods

in the economy as distinct from an isolated rise in the price of a single commodity or group of commodities. The implication here is that changes in the individual prices or any combination of the prices cannot be considered as the occurrence of inflation. **Second**, the rise in the aggregate level of prices must be continuous for inflation to be said to have occurred. The aggregate price level must show a tendency of a continuous rise over different time periods. This must be separated from a situation of a one-off rise in the price level.

Broadly, inflation can be grouped into four types, according to its magnitude.

1. **Creeping Inflation:** This occurs when the rise in price is very slow. A sustained annual rise in prices of less than 3% per annum falls under this category. Such an increase in prices is regarded safe and essential for economic growth.
2. **Walking Inflation:** Walking inflation occurs when prices rise moderately and annual inflation rate is a single digit. This occurs when the rate of rise in prices is in the intermediate range of 3 to less than 10 per cent. Inflation of this rate is a warning signal for the government to control it before it turns into running inflation.
3. **Running Inflation:** When prices rise rapidly at the rate of 10 to 20 per cent per annum, it is called running inflation. This type of inflation has tremendous adverse effects on the poor and middle class. Its control requires strong monetary and fiscal measures.
4. **Hyperinflation:** is often defined as inflation that exceeds 50 percent per month, which is just over 1 percent per day. Compounded over many months, this rate of inflation leads to very large increases in the price level. This is a situation where the inflation rate can no longer be measurable and uncontrollable. Such a situation brings a total collapse of the monetary system because of the continuous fall in the purchasing power of money.

Broadly, the causes of inflation can be categorized as demand-pull and cost-push.

I. Demand-Pull Inflation

Demand-pull inflation is caused by an increase in the conditions of demand. Typically, demand-pull inflation becomes a threat when an economy has experienced a strong boom with GDP rising faster than the trend growth of potential GDP. As Keynesian economists explained, when the value of aggregate demand exceeds the value of aggregate supply at the full employment level, the inflationary gap arises. The larger the gap between aggregate demand and aggregate supply, the more rapid is the inflation.

Possible causes of demand pull inflation

1. **A depreciation of the exchange rate** which makes exports more competitive in international markets leading to an increase in demand for domestic goods and a rise in demand for domestic factor resources.
2. **Higher demand from a government (fiscal) stimulus:** for instance, a reduction in direct or indirect taxation or higher government spending and borrowing. If direct taxes are reduced, consumers will have more disposable income causing demand to rise. Higher government spending increases demand.
3. **Monetary stimulus to the economy:** A fall in interest rates may stimulate too much demand – for example in raising demand for loans or in causing rise in house price inflation. Milton Friedman wrote that *“Inflation is always and everywhere a monetary phenomenon.”*
4. **Faster economic growth in other countries** – providing a boost to domestic goods overseas.

II. Cost-Push Inflation

This type of inflation arises from anything that causes the conditions of supply to decrease. In particular, cost-push inflation occurs when businesses respond to **rising costs**, by increasing their prices to protect profit margins. There are many reasons why costs might rise:

1. **Component costs:** An increase in the prices of raw materials and components. This might be because of a rise in global commodity prices such as oil, gas copper and agricultural products used in food processing.
2. **Rising labour costs** - caused by wage increases that exceed improvements in productivity. Wage and salary costs often rise when unemployment is low (creating labour shortages) and when people expect inflation so they bid for higher pay in order to protect their real incomes.
3. **Higher indirect taxes imposed by the government** – for example, a rise in the duty on alcohol, cigarettes and petrol/diesel or a rise in the rate of Value Added Tax. Depending on the price elasticity of demand and supply, suppliers may pass on the burden of the tax onto consumers.
4. **A fall in the exchange rate** – this can cause cost push inflation because it normally leads to an increase in the prices of imported materials and/or products.

2.9. Unemployment

One aspect of economic performance is how well an economy uses its resources. Because an economy's workers are its chief resource, keeping workers employed is a paramount concern of macroeconomists and policymakers. Unemployment is the macroeconomic problem that affects people in various ways. For most people, the loss of a job means a reduced living standard and psychological distress. For a nation, unemployment represents wasted resources. As defined by the International Labor organization (ILO, 1998), unemployment represents the portion of the

nation's labor force (working age group; i.e., ages 15 to 65 in Ethiopia) that is able to work and willing for work or seeking for work but without work.

The unemployment rate measures the percentage of those people from the labor force wanting to work who do not have jobs. The labor force is the sum of the employed and unemployed, and the unemployment rate is defined as the percentage of the labor force that is unemployed. That is,

$$\text{Labor Force} = \text{Number of Employed} + \text{Number of Unemployed},$$

$$\text{Unemployment Rate} = \frac{\text{Number of Unemployed}}{\text{Labour Force}} \times 100.$$

A related statistic is the labor-force participation rate, the percentage of the adult population that is in the labor force: **Labor-Force Participation Rate** $= \frac{\text{Labour Force}}{\text{Adult Population}} \times 100$.

NB: In your macroeconomics-II course, you will study the causes of unemployment, and the policy implications.

Growth-Unemployment Dynamics: Okun's Law

Because employed workers help to produce goods and services and unemployed workers do not, increases in the unemployment rate should be associated with decreases in real GDP. The negative relationship between unemployment and GDP is called **Okun's law**, after Arthur Okun, the economist who first studied it based on US data.

Okun's law states that for every percentage point the unemployment rate rises, real GDP growth typically falls by 2 percent.

Mathematically; $\% \Delta RGDP = 3\% - 2 \times \% \Delta U$

Where, $\% \Delta RGDP$ is percentage change in Real GDP.

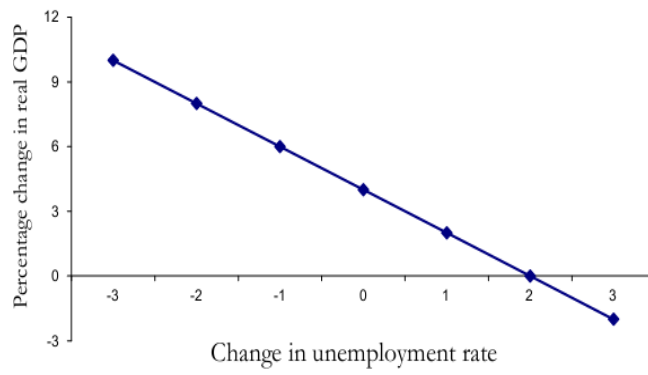
$\% \Delta U$ is percentage change in the unemployment Rate.

That is, if the unemployment rate remains the same, real GDP grows by about 3 percent. But for every percentage point the unemployment rate rises, real GDP growth falls by 2 percent. For instance, if the unemployment rate rises from 6 to 8 percent, then real GDP growth would be:

$$\% \Delta RGDP = 3\% - 2(8\% - 6\%) = -1\%.$$

In this case, Okun's law says that GDP would fall by 1 percent, indicating that the economy is in a recession.

Growth and Unemployment Dynamics



More formally, Okun's Law can be stated as the deviation of output from its natural rate is inversely related to the deviation of unemployment from its natural rate; that is, when output is higher than the natural rate of output, unemployment is lower than the natural rate of unemployment. We can write this as

$$\alpha(Y - \bar{Y}) = -\beta(u - u^n)$$

Where, α measures the amount output deviates from its natural rate
 β measures how responsive inflation is to cyclical unemployment.

Inflation –Unemployment Dynamics: The Phillips Curve

Low inflation and low unemployment are two goals of economic policymakers, but often these goals conflict. For instance, when policymakers use monetary or fiscal policy to expand aggregate demand, then the result would be higher output but a higher price level. The higher output is associated with low unemployment. *The Phillips curve named after New Zealand economist A. W. Phillips (who observed it first) describes this type of empirical relationship between inflation and unemployment: the higher the rate of unemployment, the lower the rate of inflation.*

The **Phillips curve** in its modern form states that the inflation rate depends on three forces:

- Expected inflation, π^e ;
- The deviation of unemployment from the natural rate, called *cyclical unemployment*, $(u - u^n)$ &
- Supply shocks, v .

That is,

$$\pi = \pi^e - \beta(u - u^n) + v$$

where β is a parameter measuring the response of inflation to deviation of unemployment from the natural rate.

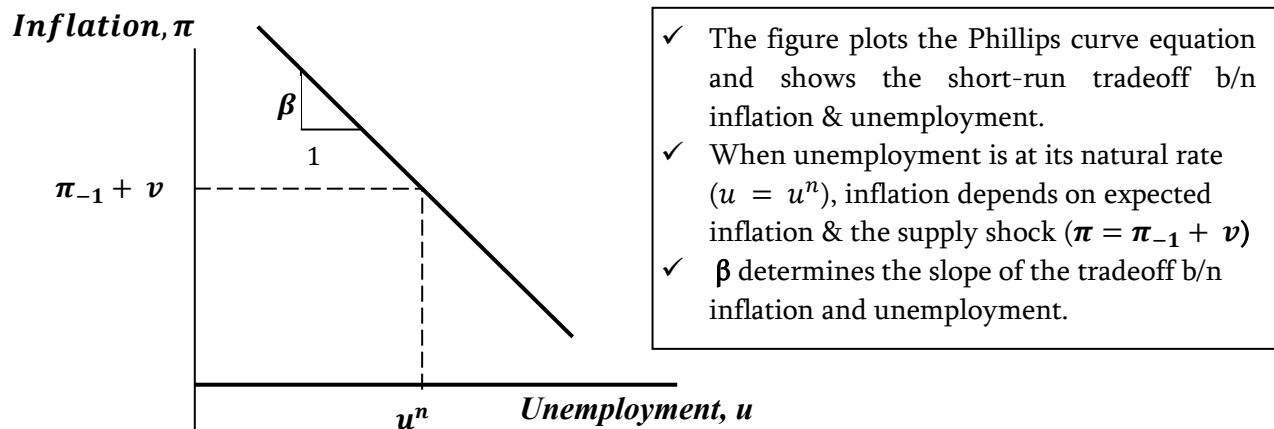
If we reasonably assume that people form their expectations of inflation based on recently observed inflation. For example, suppose that people expect prices to rise this year at the same rate as they did last year. Then expected inflation, π^e equals last year's inflation, π_{-1} :

In this case, we can rewrite the Phillips curve as:

$$\pi = \pi_{-1} - \beta(u - u^n) + v$$

- Note that the minus sign before the cyclical unemployment term implies that high unemployment tends to reduce inflation.
- On the other hand, this equation shows that inflation depends on past inflation, cyclical unemployment (causing **Demand-pull inflation** i.e., *low unemployment pulls the inflation rate up*), and a supply shock (causing **Cost-push inflation**, i.e., *An adverse supply shock, such as a rise in world oil prices, implies a positive value of v and causes inflation to rise*).

Phillips Curve



- ✓ The curve suggests that less unemployment can always be attained by incurring more inflation and that the inflation rate can always be reduced by incurring the costs of more unemployment.

Exercise:

✚ Suppose a hypothetical data of country X (in Billion Birr) is given as follows:

Personal consumption expenditure -----	1140
Compensation of employees-----	1010
Proprietors' income -----	1300
Government purchase of goods and service-----	4000
Public transfer-----	110
Interest income-----	1880
Exports-----	3050
Imports-----	2000
Rental income-----	450
Indirect Business Tax (IBX) -----	2000
Capital consumption allowance/depreciation-----	750
Income earned by foreigners from the domestic economy-----	150
Gross private domestic investment-----	2800
Income earned by nationals from abroad-----	200
Corporate profit-----	1600

➤ Based on this data, calculate:

- A. GDP (using expenditure approach)
- B. GDP (using income approach) &
- C. GNP

CHAPTER THREE

3. AGGREGATE DEMAND IN THE CLOSED ECONOMY

3.1. Foundations of the Theory of Aggregate Demand

Of all the economic fluctuations in world history, the one that stands out as particularly large, painful, and intellectually significant is the Great Depression of the 1930s. During this time, the United States and many other countries experienced massive unemployment and greatly reduced incomes. In the worst year, 1933, one-fourth of the U.S. labor force was unemployed, and real GDP was 30 percent below its 1929 level. This devastating episode caused many economists to question the validity of classical economic theory.

Classical theory seemed incapable of explaining the Depression. According to that theory, national income depends on factor supplies and the available technology, neither of which changed substantially from 1929 to 1933. After the onset of the Depression, many economists believed that a new model was needed to explain such a large and sudden economic downturn and to suggest government policies that might reduce the economic hardship so many people faced.

In 1936 the British economist John Maynard Keynes revolutionized economics with his book *The General Theory of Employment, Interest, and Money*. Keynes proposed a new way to analyze the economy, which he presented as an alternative to classical theory. His vision of how the economy works quickly became a center of controversy. Yet, as economists debated *The General Theory*, a new understanding of economic fluctuations gradually developed.

Keynes proposed that low aggregate demand is responsible for the low income and high unemployment that characterize economic downturns. He criticized classical theory for assuming that aggregate supply alone—capital, labor, and technology—determines national income. Economists today reconcile these two views with the model of aggregate demand and aggregate supply. In the long run, prices are flexible, and aggregate supply determines income. But in the short run, prices are sticky, so changes in aggregate demand influence income.

The model of aggregate demand developed in this chapter, called the **IS–LM model**, the leading interpretation of Keynes's theory. The goal of the model is to show what determines national income for any **given price level**. There are two ways to view this. We model can view the IS–LM model as showing what causes income to change in the short run when the price level is fixed. Or we can view the model as showing what causes the aggregate demand curve to shift. The two

parts of the IS–LM model are, not surprisingly, the **IS curve** and the **LM curve**. IS stands for “investment” and “saving,” and the IS curve represents what’s going on in the market for goods and services. LM stands for “liquidity” and “money,” and the LM curve represents what is happening to the supply and demand for money.

3.2. The Goods Market and the IS Curve

The IS curve plots the relationship between the interest rate and the level of income that arises in the market for goods and services. To develop this relationship, we start with a basic model called the **Keynesian cross**. This model is the simplest interpretation of Keynes’s theory of national income and is a building block for the more complex and realistic IS–LM model.

(A) The Keynesian Cross

In his General Theory, Keynes proposed that an economy’s total income was, in the short run, determined largely by the desire to spend by households, firms, and the government. The more people want to spend, the more goods and services firms can sell. The more firms can sell, the more output they will choose to produce and the more workers they will choose to hire. Thus, the problem during recessions and depressions, according to Keynes, was inadequate spending. The Keynesian cross is an attempt to model this insight.

Planned Expenditure We begin our derivation of the Keynesian cross by differentiating between actual and planned expenditure. **Actual expenditure** is the amount households, firms, and the government spend on goods and services and it equals the economy’s gross domestic product (GDP). **Planned expenditure** is the amount households, firms, and the government would like to spend on goods and services.

Why would actual expenditure ever differ from planned expenditure?

The answer is that firms might engage in unplanned inventory investment because their sales do not meet their expectations. When firms sell less of their product than they planned, their stock of inventories automatically rises; conversely, when firms sell more than planned, their stock of inventories falls. Because these unplanned changes in inventory are counted as investment spending by firms, actual expenditure can be either above or below planned expenditure.

Assuming the economy is closed, so that net exports are zero, we write planned expenditure AD as the sum of consumption(C), investment (I) and government purchase (G)

$$E = C + I + G \dots\dots\dots (1)$$

Consumption is a function of disposable income(Y-T)

$$C = C(Y - T) \dots\dots\dots (2)$$

This equation states that consumption depends on disposable income (Y - T), which is total income Y minus taxes T.

To keep things simple, for now we take planned investment as exogenously fixed:

$$I = \bar{I} \dots\dots\dots (3)$$

,we further assume that fiscal policy—the levels of government purchases and taxes is fixed:

$$\begin{aligned} G &= \bar{G}, \\ T &= \bar{T}. \end{aligned} \dots\dots\dots (4)$$

Combining these four equations, we obtain

$$E = C(Y - \bar{T}) + \bar{I} + \bar{G}.$$

This equation shows that planned expenditure is a function of income Y, the exogenous level of planned investment I, and the exogenous fiscal policy variables G and T

The figure below graphs planned expenditure as a function of the level of income. This line slopes upward because higher income leads to higher consumption and thus higher planned expenditure. The slope of this line is the marginal propensity to consume, the MPC: it shows how much planned expenditure increases when income rises by \$1. This planned-expenditure function is the first piece of the model called the Keynesian cross.

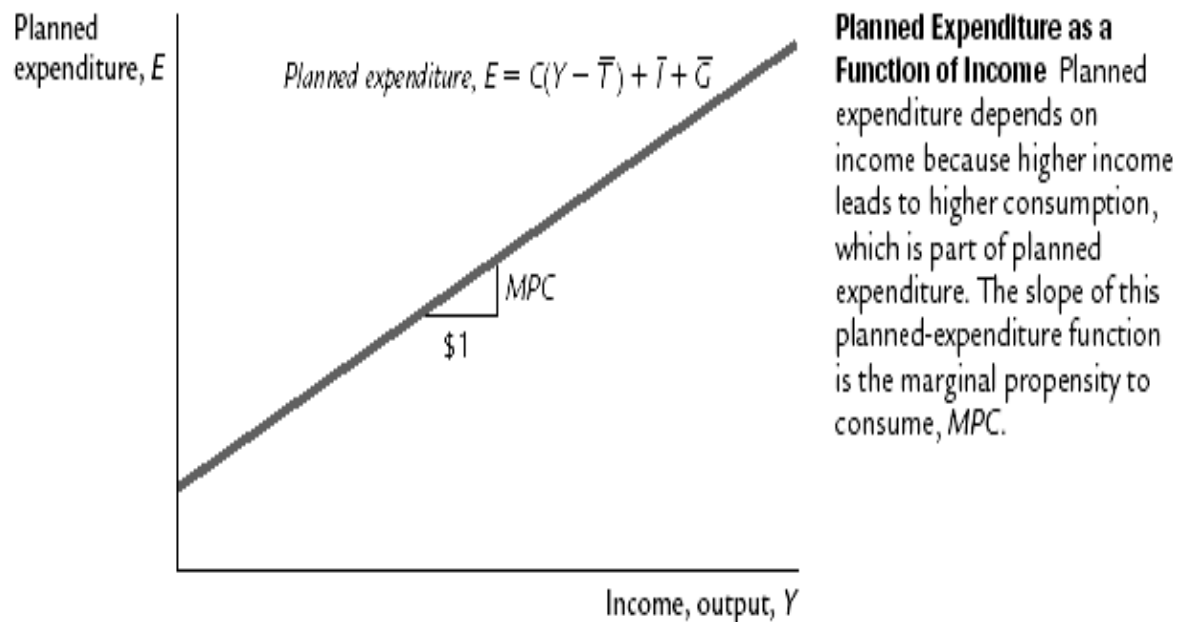


Fig. 3.1 Planned Expenditure as a function of income

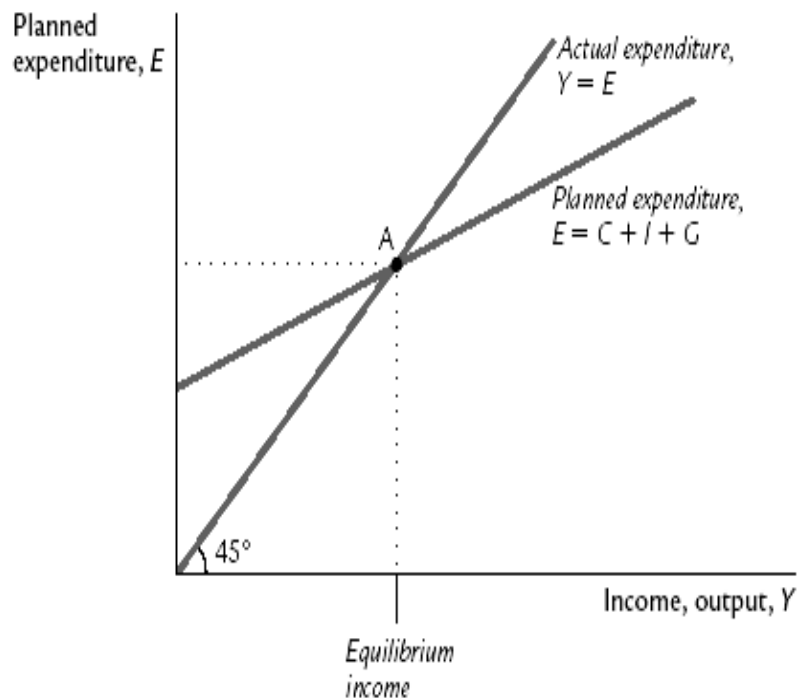
(B) The Economy in Equilibrium

The economy is said to be in equilibrium when Actual expenditure is equal to planned expenditure. This assumption is based on the idea that when people's plans have been realized, they have no reason to change what they are doing. We can write the equilibrium condition as:

$$\text{Actual Expenditure} = \text{Planned Expenditure}$$

$$Y = E$$

In the figure below, the 45-degree line serves as a reference line that translates any horizontal distance in to an equal vertical distance. Thus, anywhere on the line, planned expenditure is equal to out put. The equilibrium of this economy is at point A, where the planned-expenditure function crosses the 45-degree line.



The Keynesian Cross The equilibrium in the Keynesian cross is at point A, where income (actual expenditure) equals planned expenditure.

Fig. 3.2 the Keynesian cross

(C) How does the economy get to the equilibrium?

The equilibrium output would be achieved through inventory adjustment. Unplanned changes in inventories induce firms to change production levels, which in turn changes income and expenditure.

For example, suppose the economy were ever to find itself with GDP at a level greater than the equilibrium level, such as the level Y_1 in Figure below. In this case, planned expenditure E_1 is less than production Y_1 , so firms are selling less than they are producing. Firms add the unsold goods to their stock of inventories. This unplanned rise in inventories induces firms to lay off workers and reduce production, and these actions in turn reduce GDP. This process of unintended inventory accumulation and falling income continues until income Y falls to the equilibrium level. Similarly, suppose GDP were at a level lower than the equilibrium level, such as the level Y_2 in Figure 10-4. In this case, planned expenditure E_2 is greater than production Y_2 . Firms meet the high level of sales by drawing down their inventories. But when firms see their stock of inventories decline, they hire more workers and increase production. GDP rises and the economy approaches the equilibrium.

In summary, the Keynesian cross shows how income Y is determined for given levels of planned investment I and fiscal policy G and T . We can use this model to show how income changes when one of these exogenous variable's changes.

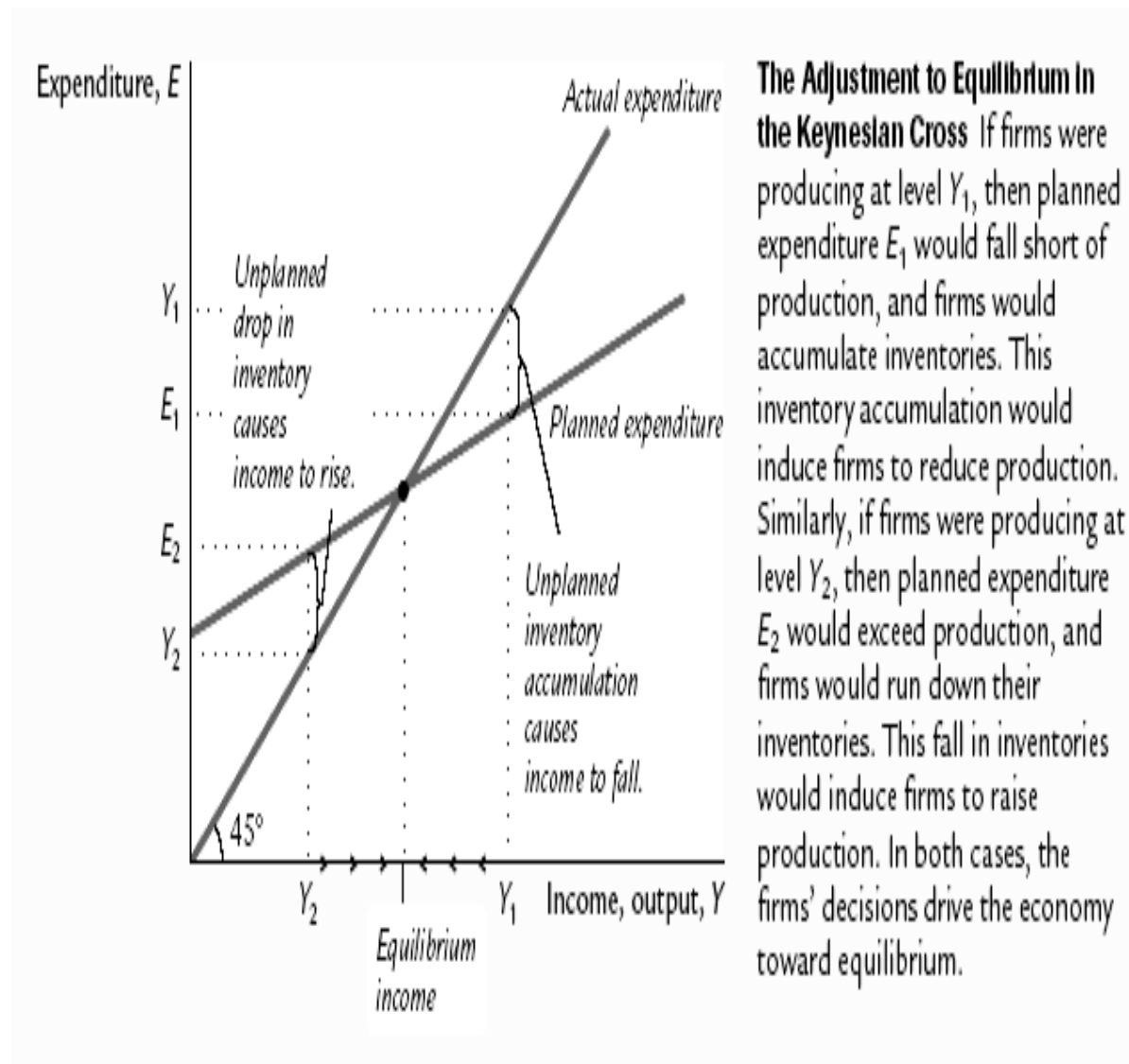


Fig. 3.3 Adjustment in the Keynesian cross

(D) Fiscal Policy and the Multiplier: Government Purchases

Consider how changes in government purchases affect the economy. Since government purchases are one component of expenditure, higher government purchases result in higher planned expenditure for any given level of income. If government purchases rise by ΔG , then the planned expenditure curve shifts upward by ΔG , as shown on the figure below.

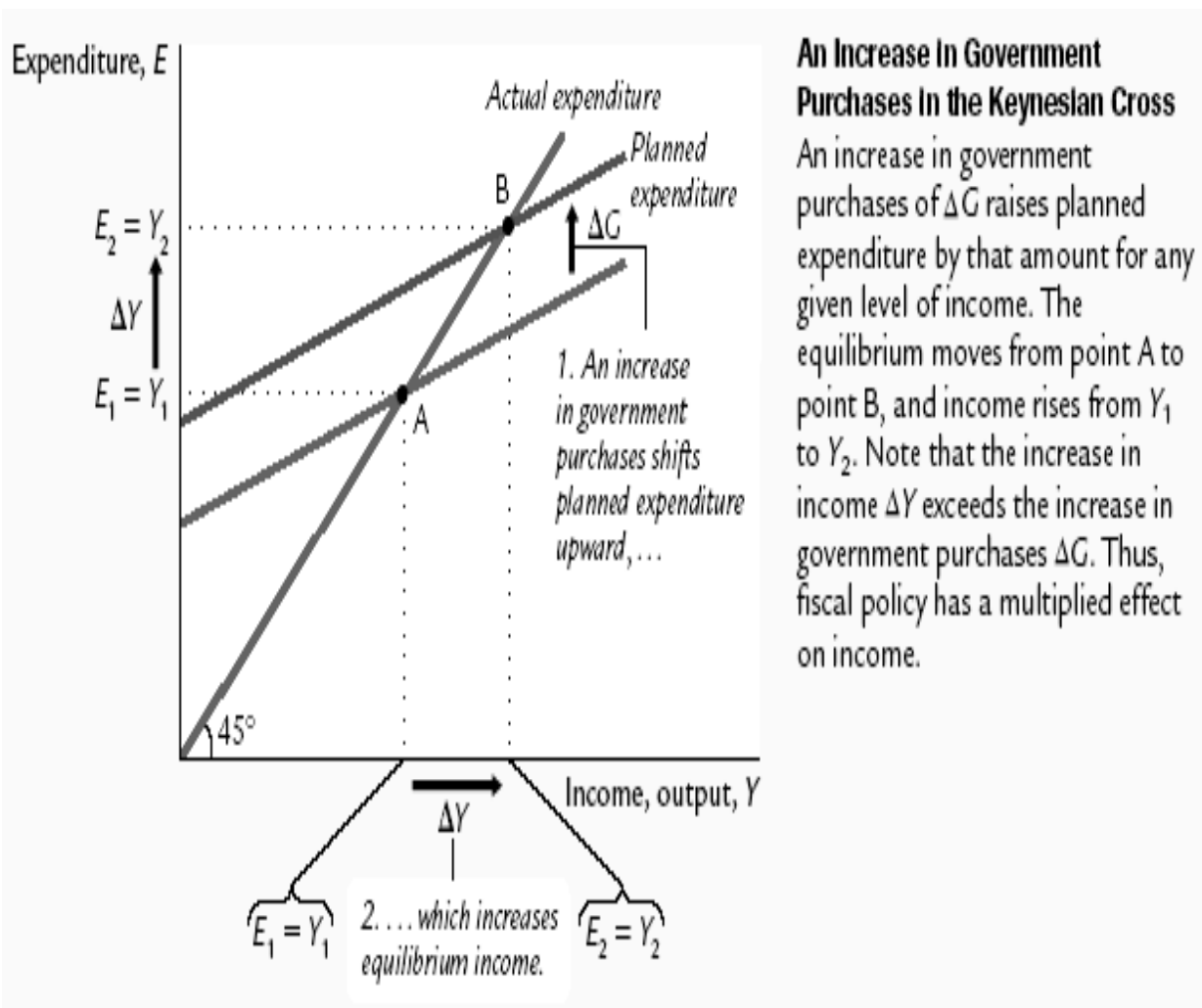


Fig. 3.4 Effect of government purchase on output in Keynesian cross

The graph shows that, an increase in government purchase leads to an even greater increase in income. That is, $\Delta Y > \Delta G$. The ratio $\Delta Y / \Delta G$ is called the **government purchases multiplier**; it tells us how much income rises in response to a \$1 increase in government purchases. An implication of the Keynesian cross is that the government-purchases multiplier is larger than one.

Why does fiscal policy have a multiplied effect on income?

The reason is that, according to the consumption function $C = C(Y - T)$, higher income causes higher consumption. When an increase in government purchases raises income, it also raises consumption, which further raises income, which further raises consumption, and so on. Therefore, in this model, an increase in government purchases causes a greater increase in income.

How big is the multiplier?

The process begins when expenditure rises by ΔG , which implies that income rises by ΔG as well. This increase in income in turn raises consumption by $MPC * \Delta G$, where MPC is the marginal propensity to consume. This increase in consumption raises expenditure and income once again. This second increase in income of $MPC * \Delta G$ again raises consumption, this time by $MPC * (MPC * \Delta G)$, which again raises expenditure and income, and so on. This feedback from consumption to income to consumption continues indefinitely. The total effect on income is

$$\text{Initial Change in Government Purchases} = \Delta G$$

$$\text{First Change in Consumption} = MPC * \Delta G$$

$$\text{Second Change in Consumption} = MPC^2 * \Delta G$$

$$\text{Third Change in Consumption} = MPC^3 * \Delta G$$

$$\dots \dots$$

$$\dots \dots$$

$$\dots \dots$$

$$\Delta Y = (1 + MPC + MPC^2 + MPC^3 + \dots) \Delta G.$$

The government-purchases multiplier is

$$\Delta Y / \Delta G = 1 + MPC + MPC^2 + MPC^3 + \dots$$

This expression for the multiplier is an example of an infinite geometric series. A result from algebra allows us to write the multiplier as²

$$\Delta Y / \Delta G = 1 / (1 - MPC).$$

For example, if the marginal propensity to consume is 0.6, the multiplier is

$$\Delta Y / \Delta G = 1 + 0.6 + 0.6^2 + 0.6^3 + \dots$$

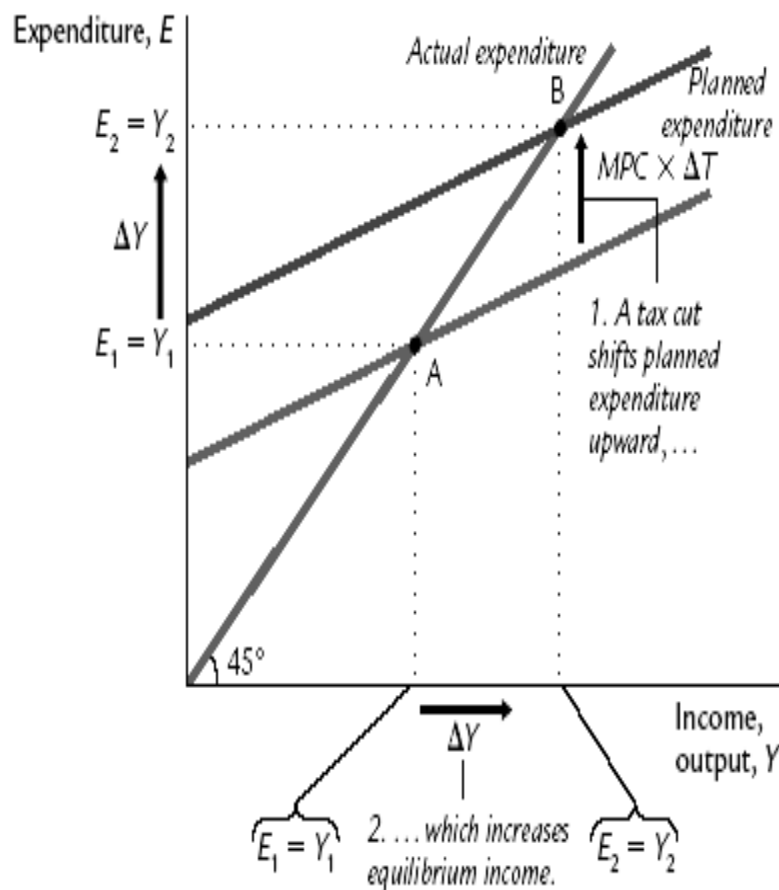
$$= 1 / (1 - 0.6)$$

$$= 2.5.$$

In this case, a \$1.00 increase in government purchases raises equilibrium income by \$2.50.

(E) Fiscal Policy and the Multiplier: Taxes

Consider now how changes in taxes affect equilibrium income. A decrease in taxes of ΔT immediately raises disposable income ($Y - T$) by ΔT and, therefore, increases consumption by $(MPC \times \Delta T)$. For any given (constant) level of income Y , planned expenditure is now higher. As the figure below shows, the planned-expenditure schedule shifts upward by $MPC \times \Delta T$. The equilibrium of the economy moves from point A to point B.



A Decrease in Taxes in the Keynesian Cross A decrease in taxes of ΔT raises planned expenditure by $MPC \times \Delta T$ for any given level of income. The equilibrium moves from point A to point B, and income rises from Y_1 to Y_2 . Again, fiscal policy has a multiplied effect on income.

Fig 3.5 Effect of tax on output in Keynesian cross

Just as an increase in government purchases has a multiplied effect on income, so does a decrease in taxes. As before, the initial change in expenditure, now $MPC \times \Delta T$, is multiplied by $1/(1 - MPC)$. The overall effect on income of the change in taxes is

$$\Delta Y / \Delta T = -MPC / (1 - MPC).$$

This expression is the **tax multiplier**, the amount income changes in response to a \$1 change in taxes. For example, if the marginal propensity to consume is 0.6, then the tax multiplier is

$$\Delta Y/\Delta T = -0.6/(1 - 0.6) = -1.5.$$

In this example, a \$1.00 cut (decrease) in taxes raises equilibrium income by \$1.50.⁴

The multiplier implies that taxes and income are inversely related and a unit decrease in taxes increase income by more than proportionately.

3.3. The Interest Rate, Investment, and the IS Curve

The Keynesian cross is only a stepping stone on our path to the IS–LM model. The Keynesian cross is useful because it shows how the spending plans of households, firms, and the government determine the economy's income. Yet it makes the simplifying assumption that the level of planned investment I is fixed. Here we set an important macroeconomic relationship between planned investment and the interest rate r .

The transition from the Keynesian cross to the IS curve is achieved by noting that if the real interest rate changes, this changes planned investment. The Keynesian cross analysis tells us that changes in planned investment change GDP.

Thus, for example, if interest rate increases, planned investment falls, and so does output. Thus, higher levels of interest rate are associated with lower level of output.

To add this relationship between the interest rate and investment to our model, we write the level of planned investment as

$$I = I(r).$$

Here we derive the IS curve from the Keynesian cross and the investment function, which we can summarize in the following graphical relationship.

Deriving the IS Curve Panel (a) shows the investment function: an increase in the interest rate from r_1 to r_2 reduces planned investment from $I(r_1)$ to $I(r_2)$. Panel (b) shows the Keynesian cross: a decrease in planned investment from $I(r_1)$ to $I(r_2)$ shifts the planned-expenditure function downward and thereby reduces income from Y_1 to Y_2 . Panel (c) shows the IS curve summarizing this relationship between the interest rate and income: the higher the interest rate, the lower the level of income.

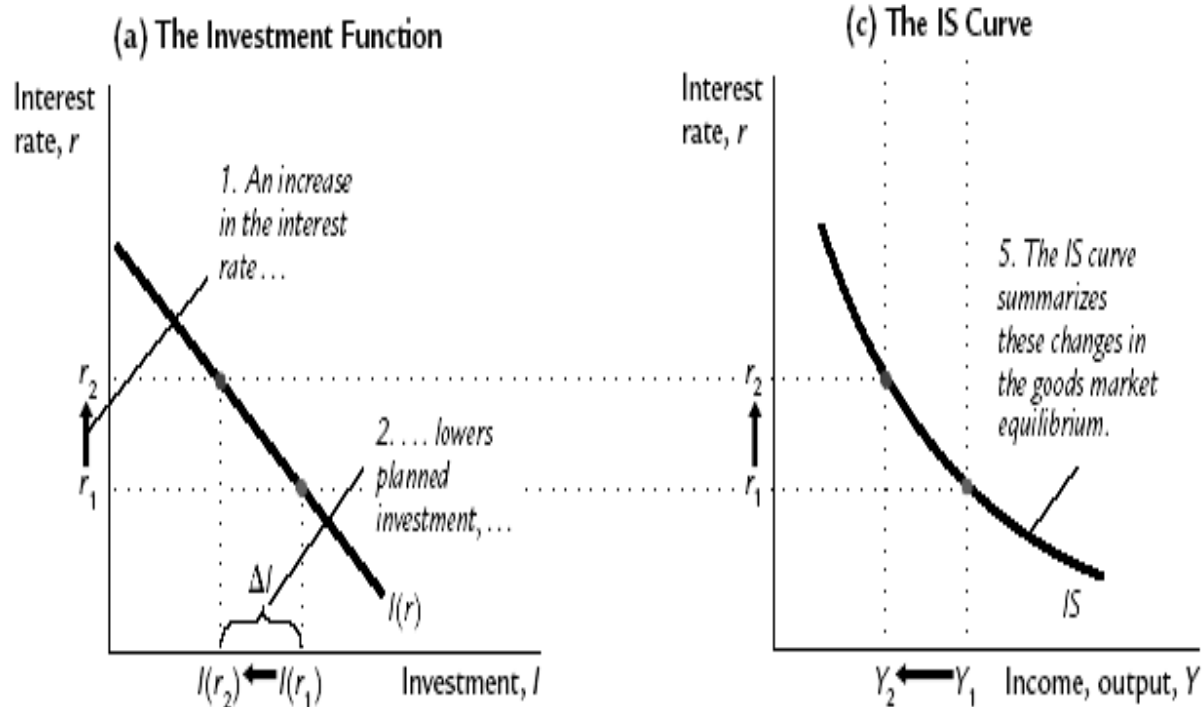


Fig. 3.6 The derivation of IS curve

This investment function is graphed in panel (a) of figure above. Because the interest rate is the cost of borrowing to finance investment projects, an increase in the interest rate reduces planned investment. As a result, the investment function slopes downward.

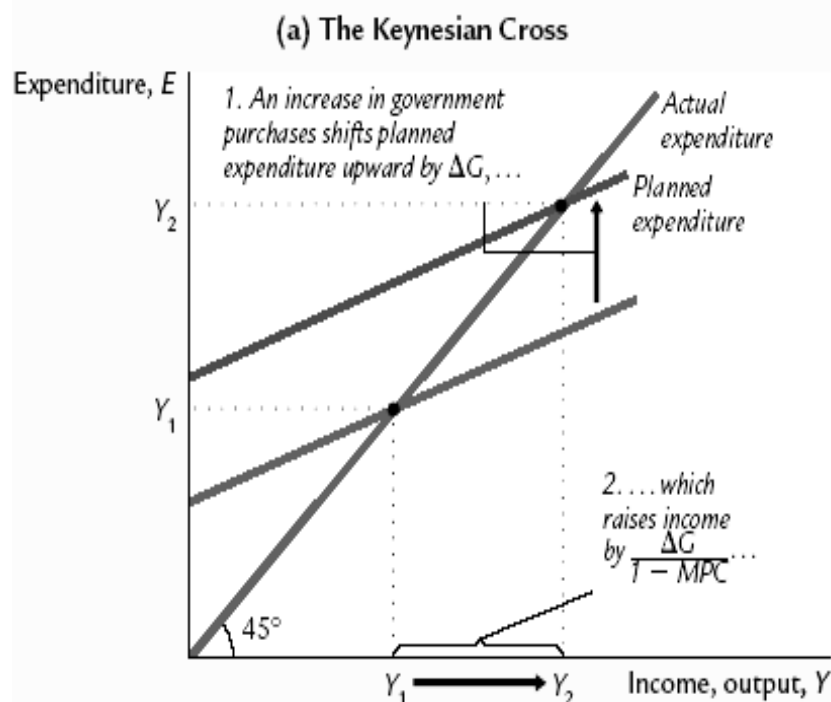
To determine how income changes when the interest rate changes, we can combine the investment function with the Keynesian-cross diagram. Because investment is inversely related to the interest rate, an increase in the interest rate from r_1 to r_2 reduces the quantity of investment from $I(r_1)$ to $I(r_2)$. The reduction in planned investment, in turn, shifts the planned-expenditure function downward, as in panel (b) of figure above. The shift in the planned-expenditure function causes the level of income to fall from Y_1 to Y_2 . Hence, an increase in the interest rate lowers income. The IS curve, shown in panel (c) of Figure, summarizes this relationship between the interest rate and the level of income. In essence, the IS curve combines the interaction between r and I expressed by the investment function and the interaction between I and Y demonstrated by the Keynesian cross. Because an increase in the interest rate causes planned investment to fall, which in turn causes income to fall, the IS curve slopes downward

How Fiscal Policy Shifts the IS Curve

The IS curve is drawn for a given fiscal policy; that is, when we construct the IS curve, we hold G and T fixed. When fiscal policy changes, the IS curve shifts.

Figure below uses the Keynesian cross to show how an increase in government purchases by ΔG shifts the IS curve. This figure is drawn for a given (constant) interest rate and thus for a given level of planned investment. The Keynesian cross shows that this change in fiscal policy raises planned expenditure and thereby increases equilibrium income from Y_1 to Y_2 . Therefore, an increase in government purchases shifts the IS curve outward.

We can use the Keynesian cross to see how other changes in fiscal policy shift the IS curve. Because a decrease in taxes also expands expenditure and income, it too shifts the IS curve outward. A decrease in government purchases or an increase in taxes reduces income; therefore, such a change in fiscal policy shifts the IS curve inward.



An Increase in Government Purchases Shifts the IS Curve Outward Panel (a) shows that an increase in government purchases raises planned expenditure. For any given interest rate, the upward shift in planned expenditure of ΔG leads to an increase in income Y of $\Delta G / (1 - MPC)$. Therefore, in panel (b), the IS curve shifts to the right by this amount.

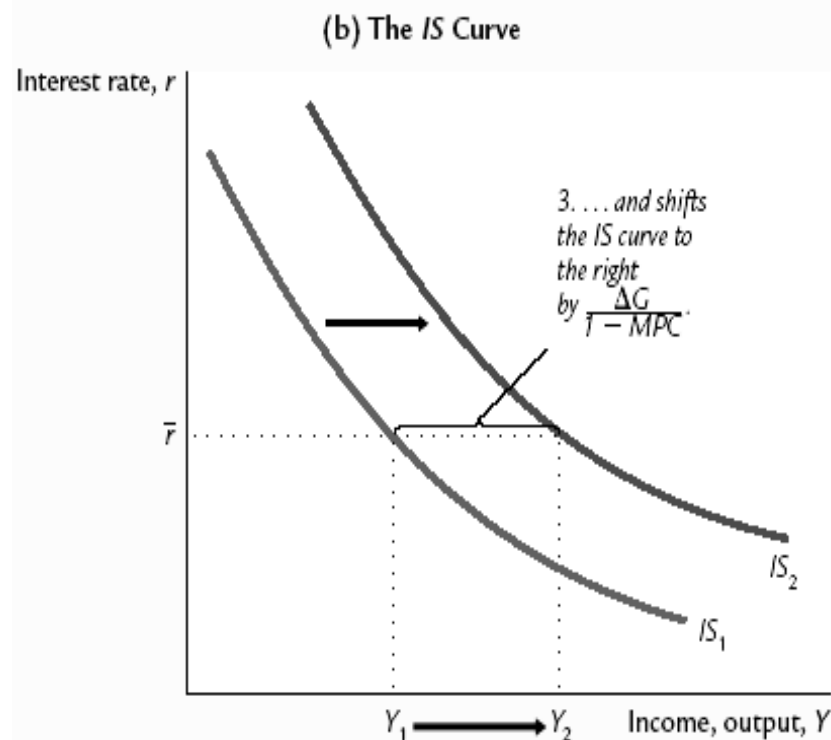


Fig 3.7 Shift of the IS curve

In summary, the IS curve shows the combinations of the interest rate and the level of income that are consistent with equilibrium in the market for goods and services. The IS curve is drawn for a

given fiscal policy. Changes in fiscal policy that raise the demand for goods and services shift the IS curve to the right. Changes in fiscal policy that reduce the demand for goods and services shift the IS curve to the left.

3.4. The Money Market and the LM Curve

The LM curve plots the relationship between the interest rate and the level of income that arises in the market for money balances. To understand this relationship, we begin by looking at a theory of the interest rate, called the **theory of liquidity preference**.

(A) The Theory of Liquidity Preference

This is a theory of interest rate determination in short run. It argues that interest rate, in the short run, is determined in the money market, where the demand and supply of real money balances intersect. Just as the Keynesian cross is a building block for the IS curve, the theory of liquidity preference is a building block for the LM curve.

To develop this theory, we begin with the **supply of real money** balances. If M stands for the supply of money and P stands for the price level, then M/P is the supply of real money balances. The theory of liquidity preference assumes there is a fixed supply of real money balances.

That is,

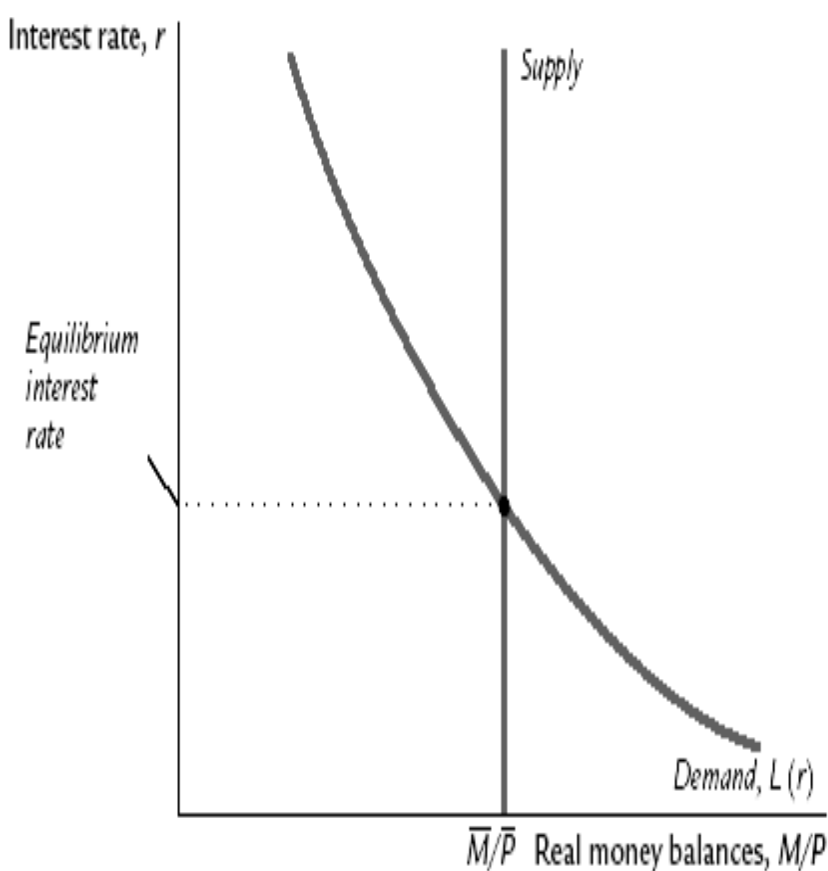
$$(M/P)^s = \bar{M}/\bar{P}.$$

The money supply M is an exogenous policy variable chosen by a central bank, such as the National Bank. The price level P is also an exogenous variable in this model. (We take the price level as given because the IS–LM model—our ultimate goal in this chapter—explains the short run when the price level is fixed.) These assumptions imply that the supply of real money balances is fixed and, in particular, does not depend on the interest rate. Thus, when we plot the supply of real money balances against the interest rate in figure below, we obtain a vertical supply curve.

Next, consider the **demand for real money** balances. The theory of liquidity preference posits that the interest rate is one determinant of how much money people choose to hold. The reason is that the interest rate is the opportunity cost of holding money: it is what you forgo by holding some of your assets as money, which does not bear interest, instead of as interest-bearing bank deposits or bonds. When the interest rate rises, people want to hold less of their wealth in the form of money. We can write the demand for real money balances as

$$(M/P)^d = L(r),$$

where the function $L()$ shows that the quantity of money demanded depends on the interest rate. Thus, the demand curve in figure below slopes downward because higher interest rates reduce the quantity of real money balances demanded. Money demand is best understood here if you consider it as the amount of money people hold in their pocket.



The Theory of Liquidity Preference

The supply and demand for real money balances determine the interest rate. The supply curve for real money balances is vertical because the supply does not depend on the interest rate. The demand curve is downward sloping because a higher interest rate raises the cost of holding money and thus lowers the quantity demanded. At the equilibrium interest rate, the quantity of real money balances demanded equals the quantity supplied.

Fig. 3.7 The money market

According to the theory of liquidity preference, the supply and demand for real money balances determine what interest rate prevails in the economy. That is, the interest rate adjusts to equilibrate the money market.

Q? How does the interest rate get to this equilibrium of money supply and money demand?

The adjustment occurs because whenever the money market is not in equilibrium, people try to adjust their portfolios of assets and, in the process, alter the interest rate. For instance, if the interest rate is above the equilibrium level, the quantity of real money balances supplied exceeds the quantity demanded. Individuals holding the excess supply of money try to convert some of their non-interest-bearing money into interest-bearing bank deposits or bonds. Banks and bond issuers, who prefer to pay lower interest rates, respond to this excess supply of money by lowering the interest rates they offer. Conversely, if the interest rate is below the equilibrium level, so that the quantity of money demanded exceeds the quantity supplied, individuals try to obtain money by selling bonds or making bank withdrawals. To attract now-scarcer funds, banks and bond issuers respond by increasing the interest rates they offer. Eventually, the interest rate reaches the equilibrium level, at which people are content with their portfolios of monetary and non-monetary assets.

We can also use the theory of liquidity preference to show how the interest rate responds to changes in the supply of money. Suppose, for instance, that the National Bank suddenly decreases the money supply. A fall in M reduces M/P , because P is fixed in the model. The supply of real money balances shifts to the left, as in figure below. The equilibrium interest rate rises from r_1 to r_2 , and the higher interest rate makes people satisfied to hold the smaller quantity of real money balances. The opposite would occur if the Fed had suddenly increased the money supply. Thus, according to the theory of liquidity preference, a decrease in the money supply raises the interest rate, and an increase in the money supply lowers the interest rate.

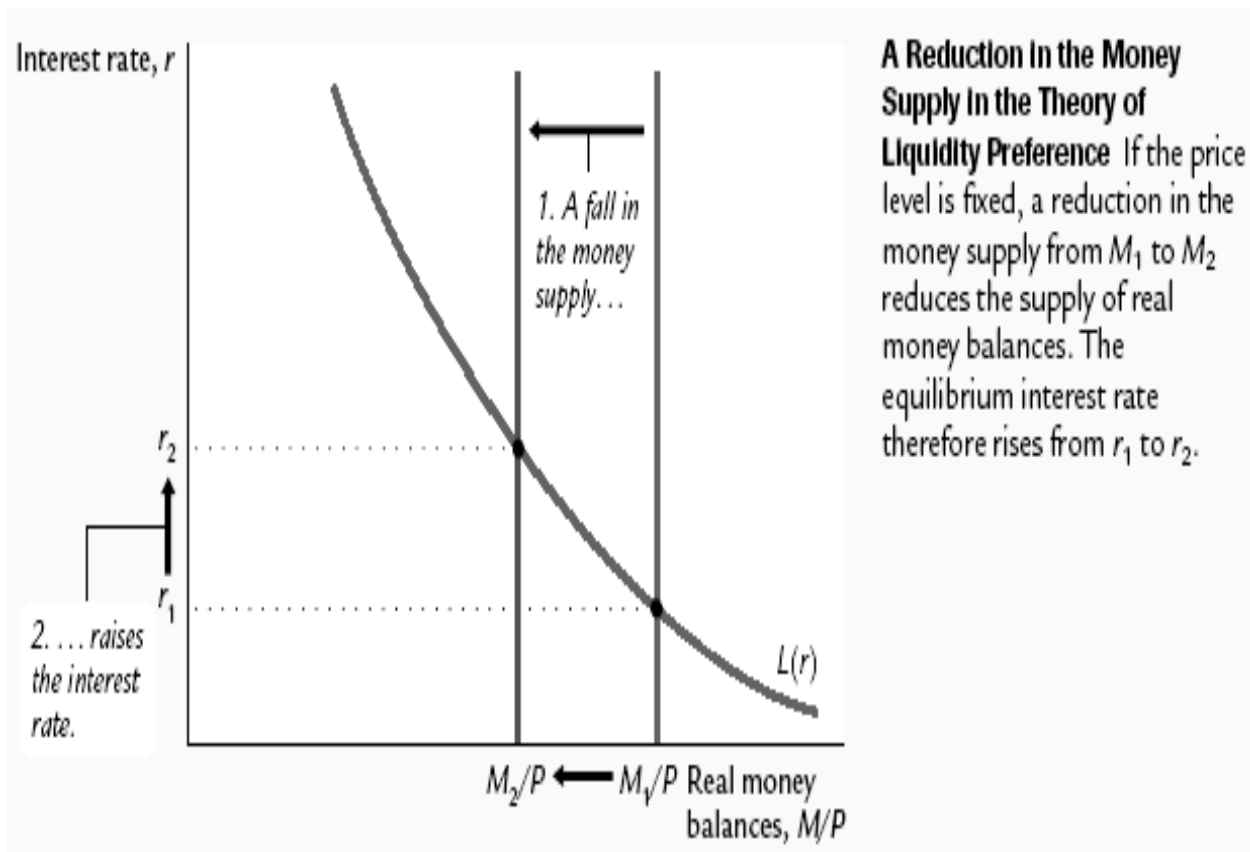


Fig. 3.8 interest rate as an adjustment tool in the money market

(B) Income, Money Demand, and the LM Curve

So far, we assumed that only interest rate determines the quantity of real money balances demanded. But more, realistically, the level of income Y also affects money demand.

Q? How does a change in the economy's level of income Y affect the market for real money balances?

When income is high, expenditure is high, so people engage in more transactions that require the use of money. Thus, greater income implies greater money demand. We can express these ideas by writing the money demand function as

$$(M/P)^d = L(r, Y).$$

The quantity of real money balances demanded is negatively related to the interest rate and positively related to income.

Using the theory of liquidity preference, we can see what happens to the interest rate when the level of income changes. For example, consider what happens when income increases from Y_1 to Y_2 .

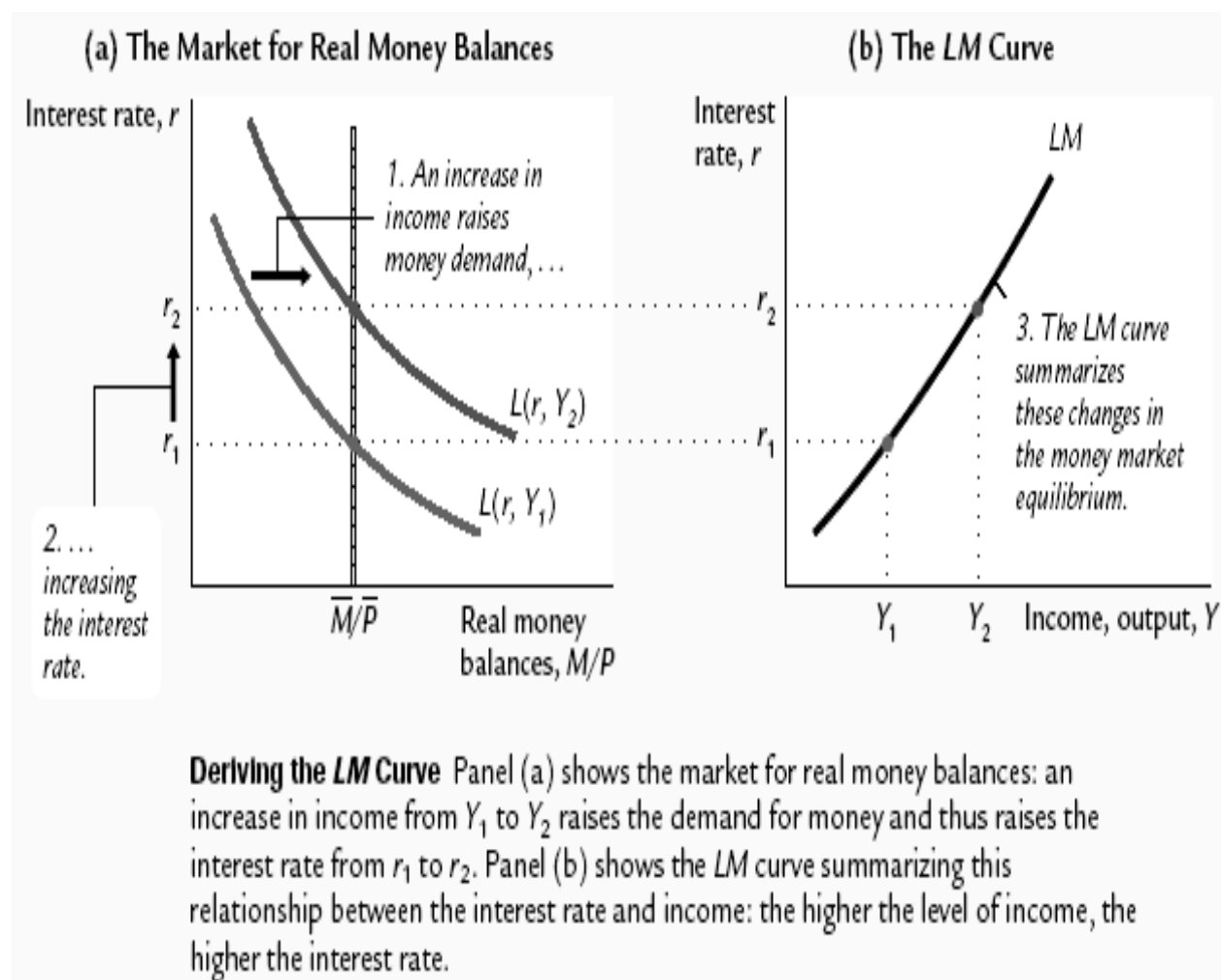


Fig.3.9 The derivation of the LM curve

As panel (a) in the above graph illustrates, this increase in income shifts the money demand curve to the right. With the supply of real money balances unchanged, the interest rate must rise from r_1 to r_2 to equilibrate the money market. Therefore, according to the theory of liquidity preference, higher income leads to a higher interest rate.

The LM curve plots this relationship between the level of income and the interest rate.

The higher the level of income, the higher the demand for real money balances, and the higher the equilibrium interest rate. For this reason, the LM curve slopes upward, as in panel (b) of Figure above.

(C) How Monetary Policy Shifts the LM Curve

The LM curve tells us the interest rate that equilibrates the money market at any level of income. Yet, as we saw earlier, the equilibrium interest rate also depends on the supply of real money balances, M/P . This means that the LM curve is drawn for a given supply of real money balances. If real money balances change—for example, if the Fed alters the money supply—the LM curve shifts.

Suppose that the National Bank decreases the money supply from M_1 to M_2 , which causes the supply of real money balances to fall from M_1/P to M_2/P . Figure below shows what happens. Holding constant the amount of income and thus the demand curve for real money balances, we see that a reduction in the supply of real money balances raises the interest rate that equilibrates the money market. Hence, a decrease in the money supply shifts the LM curve upward.

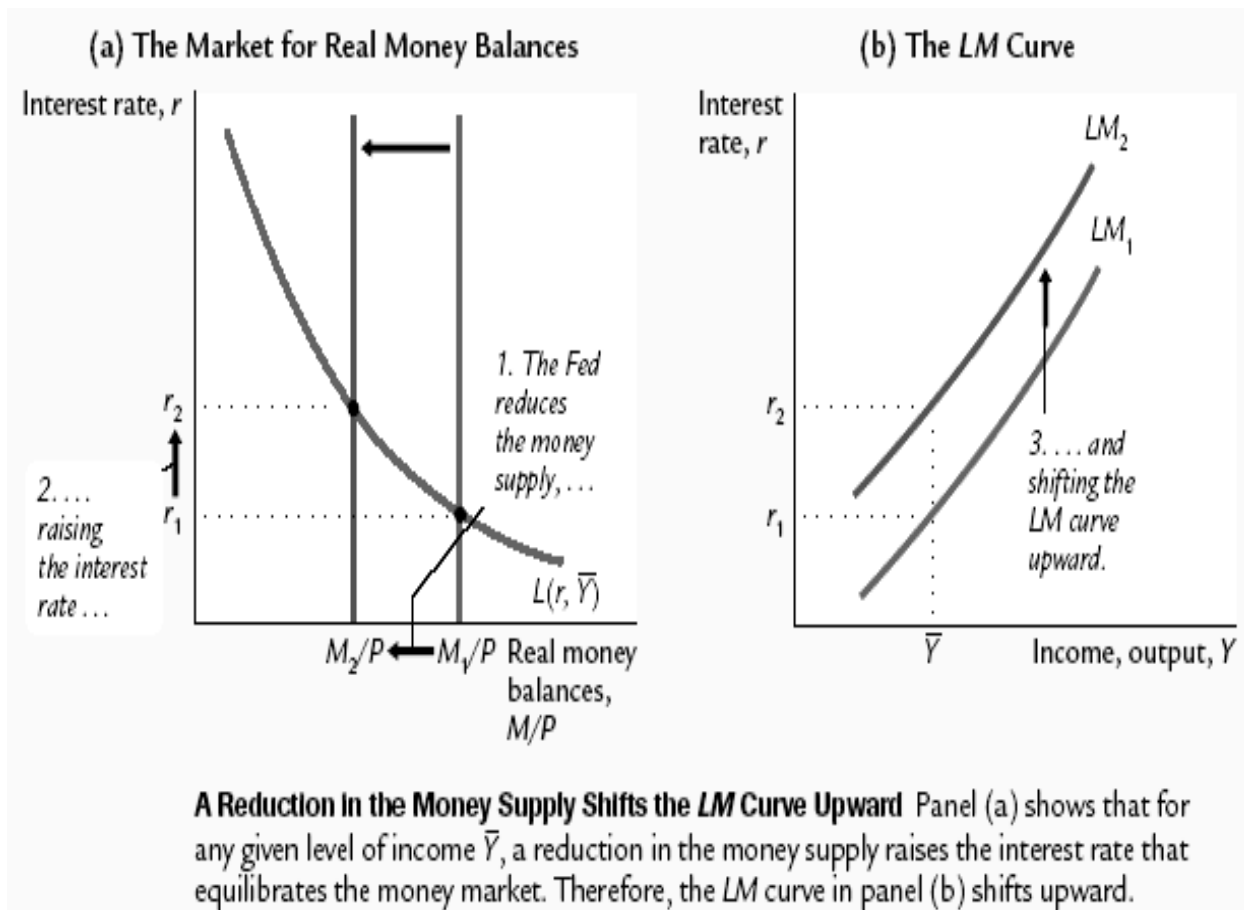


Fig. 3.10 A shift in the LM curve

In summary, the LM curve shows the combinations of the interest rate and the level of income that are consistent with equilibrium in the market for real money balances. The LM curve is drawn for a given supply of real money balances. Decreases in the supply of real money balances shift the LM curve upward. Increases in the supply of real money balances shift the LM curve downward.

3.5. The Short-Run Equilibrium

We now have all the pieces of the IS–LM model. The two equations of this model are

$$\begin{aligned} Y &= C(Y - T) + I(r) + G && \text{IS,} \\ M/P &= L(r, Y) && \text{LM.} \end{aligned}$$

The model takes fiscal policy, G and T , monetary policy M , and the price level P as exogenous. Given these exogenous variables, the IS curve provides the combinations of r and Y that satisfy the equation representing the goods market, and the LM curve provides the combinations of r and

Y that satisfy the equation representing the money market. These two curves are shown together in Figure below.

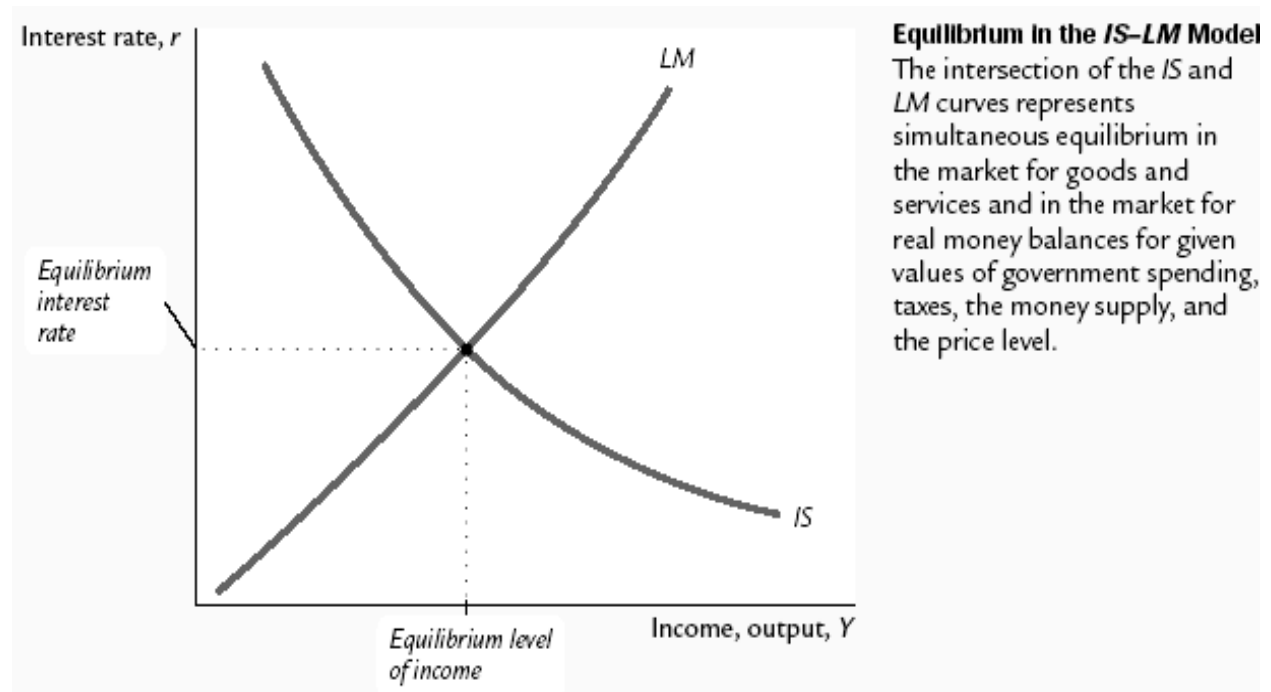


Fig 3.11. Equilibrium in IS-LM

The equilibrium of the economy is the point at which the IS curve and the LM curve cross. This point gives the interest rate r and the level of income Y that satisfy conditions for equilibrium in both the goods market and the money market. In other words, at this intersection, actual expenditure equals planned expenditure, and the demand for real money balances equals the supply.

3.5.1. Explaining Fluctuations with the IS – LM Model

✚ **How do changes in exogenous variables (government purchase, taxes, and money supply) affect endogenous variables (interest and income)?**

- ✓ The intersection of the IS curve and the LM curve determines the level of national income **for given values of government spending, taxes, and the money supply.**
- ✓ When one of these curves shifts, the short-run equilibrium of the economy changes and national income fluctuates. In this section, we examine how changes in policy and shocks to the economy can cause these curves to shift.

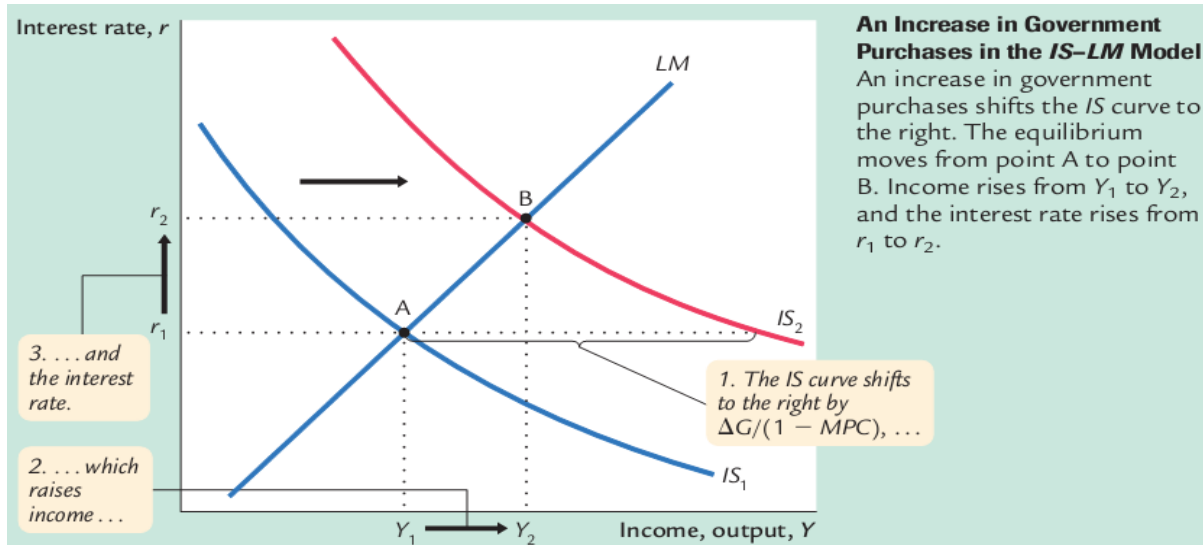
A) Fiscal Policy and the Short-Run Equilibrium

- How do Fiscal Policy Shifts the IS Curve and Changes the Short-Run Equilibrium?

i. Changes in Government Purchase

- $\uparrow G$ by $\Delta G \rightarrow \uparrow Y$ by $\frac{\Delta G}{1-MPC}$ but also $\uparrow r$ in the money market countering the full multiplier effect. How? $\uparrow r \rightarrow \downarrow I$ (investment). **That is, expansionary government spending policy has a crowding out effect on investment. The extent of crowding out is greater the more the interest rate increases when government spending rises.**

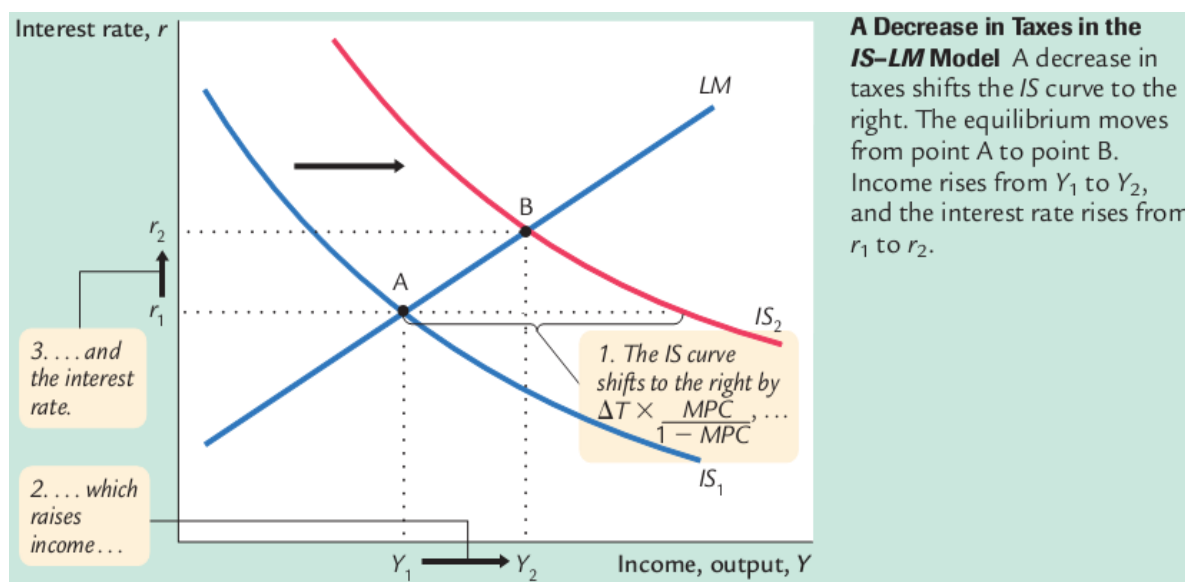
Figure 3.14: Government purchase and IS-LM model



ii. Changes in Taxes: In the IS-LM model, changes in taxes affect the economy much the same as changes in government purchases do.

- $\downarrow T$ by $\Delta T \rightarrow \uparrow Y$ by $\Delta T * MPC/(1-MPC)$, which will shift IS curve to the right $\rightarrow \uparrow r$ curbing the full effect of the policy on income i.e., **it is not as big as in the Keynesian cross.**

Figure 3.15: Taxes and IS-LM model



B) Monetary Policy and the Short-Run Equilibrium

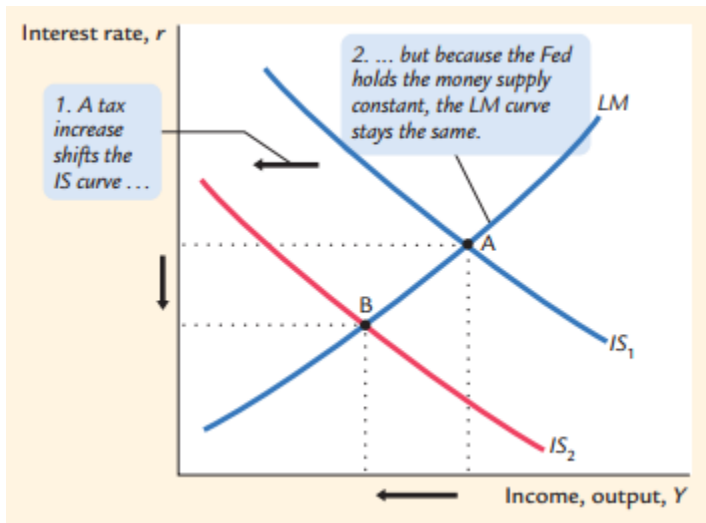
➤ How Monetary Policy Shifts the LM Curve and Changes the Short-Run Equilibrium?

- $\uparrow M \rightarrow \uparrow \left(\frac{M}{P}\right)^S$ (b/c prices are fixed) $\rightarrow \uparrow Y$, by shifting LM curve to the right $\rightarrow \downarrow r$ with fixed price in short run $\uparrow$ planned investment, I , $\rightarrow \uparrow Y$.
- **Monetary transmission mechanism:** monetary policy influences income by changing the interest rate.

3.4.1. Interaction Between Fiscal and Monetary Policy

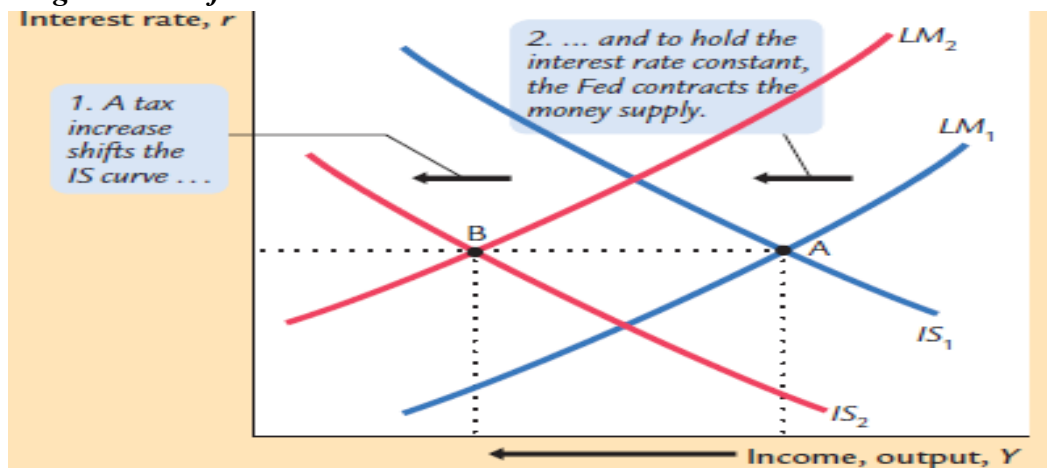
- The **interdependence** between the **two policies** making organs National Bank (NB) and Ministry of Finance and Development (MoFED) alters the impact of the policies.
- For example: **If MoFED raise the tax rate to curb the overheating of the economy**, its results depends on the reaction of National Bank of Ethiopia (NB). NB has three options:
 - a) Holds the money supply constant
 - b) Holds the level of interest rate constant
 - c) Holds the level of income constant

a) *Holding Money Supply Constant*



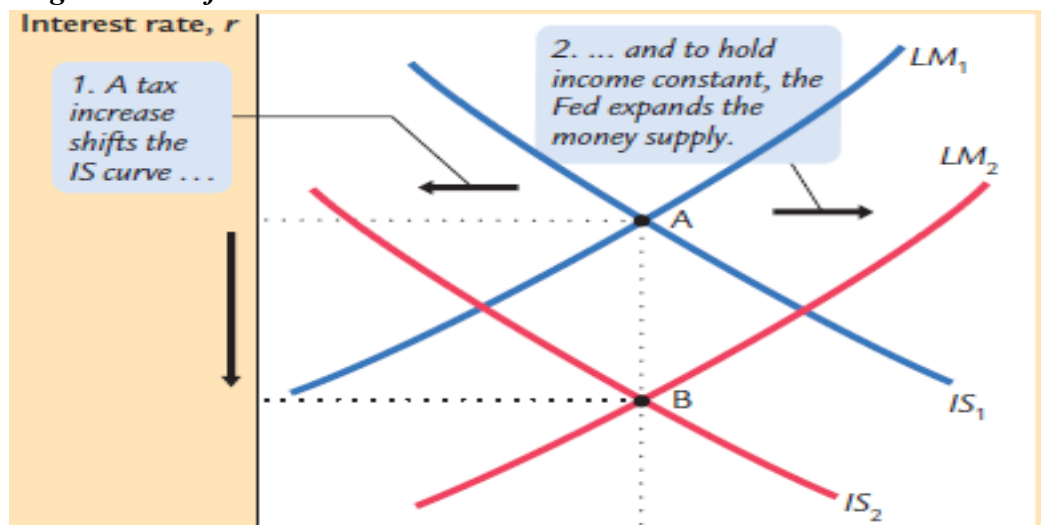
- ✓ The tax increase shifts the IS curve to the left (but LM curve remained the same b/c money supply held constant). **Income falls** (because higher taxes reduce consumer spending), and the **interest rate falls** (because lower income reduces the demand for money). The fall in income indicates that the tax hike causes a recession (I.e. the equilibrium of the economy moves from A to B).

b) *Holding the level of interest rate constant:*



- ✓ In this case, when the tax increase shifts the IS curve to the left, the National bank must decrease the money supply to keep the interest rate at its original level. This fall in the money supply shifts the LM curve upward. The interest rate does not fall, but income falls by a larger amount than if the NB had held the money supply constant.

c) *Holding the level of income constant*



- ✓ In case C, the NB wants to prevent the tax increase from lowering income. It must, therefore, raise the money supply and shift the LM curve downward enough to offset the shift in the IS curve. In this case, the tax increase does not cause a recession, but it does cause a large fall in the interest rate.
- ✓ Although the level of income is not changed, the combination of a tax increase and a monetary expansion does change the allocation of the economy's resources. The higher taxes depress consumption, while the lower interest rate stimulates investment. Income is not affected because these two effects exactly balance.

3.6. From IS-LM to Aggregate Demand

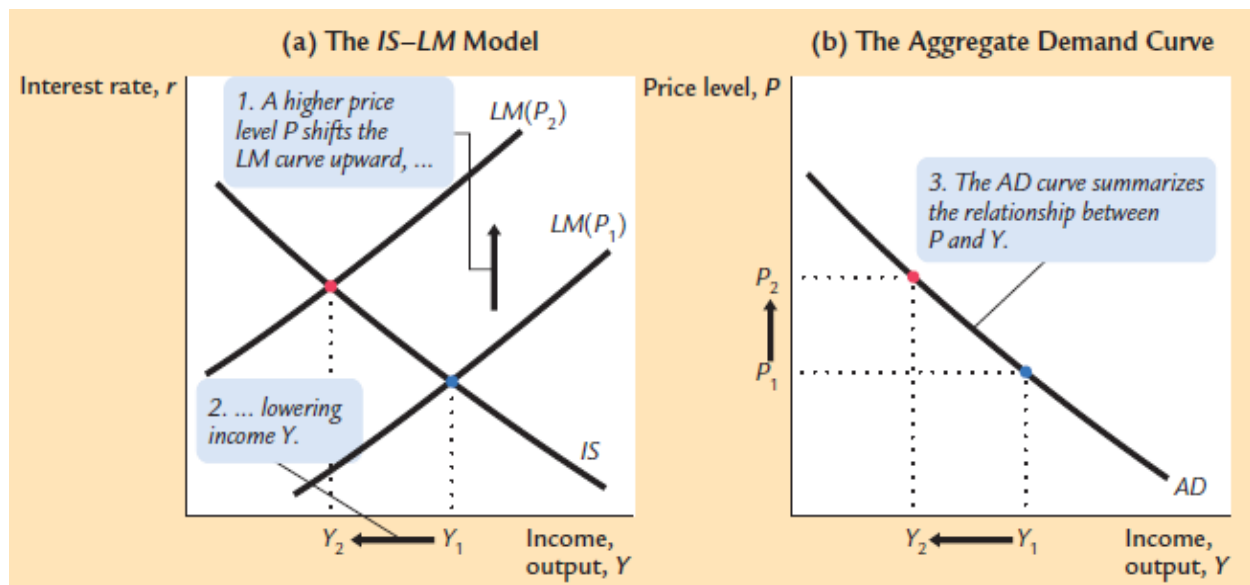
We now consider how the IS-LM model can also be viewed as a theory of aggregate demand.

- ✓ Recall AD describes the relationship between the price level and the level of income.
- ✓ Recall that the IS-LM model is constructed on the basis of a fixed price level.
- ✓ For a given value of the price level & the nominal money supply, the position of the LM curve is fixed.
- ✓ The real money supply (M/P) changes if either the nominal money supply (M) or the price level (P) changes.
- ✓ Thus, we can see that changes in the price level are associated with changes in the equilibrium level of output and interest rates.
- ✓ This is the relationship that is summarized by the aggregate demand curve.

✚ Why is the AD downward sloping?

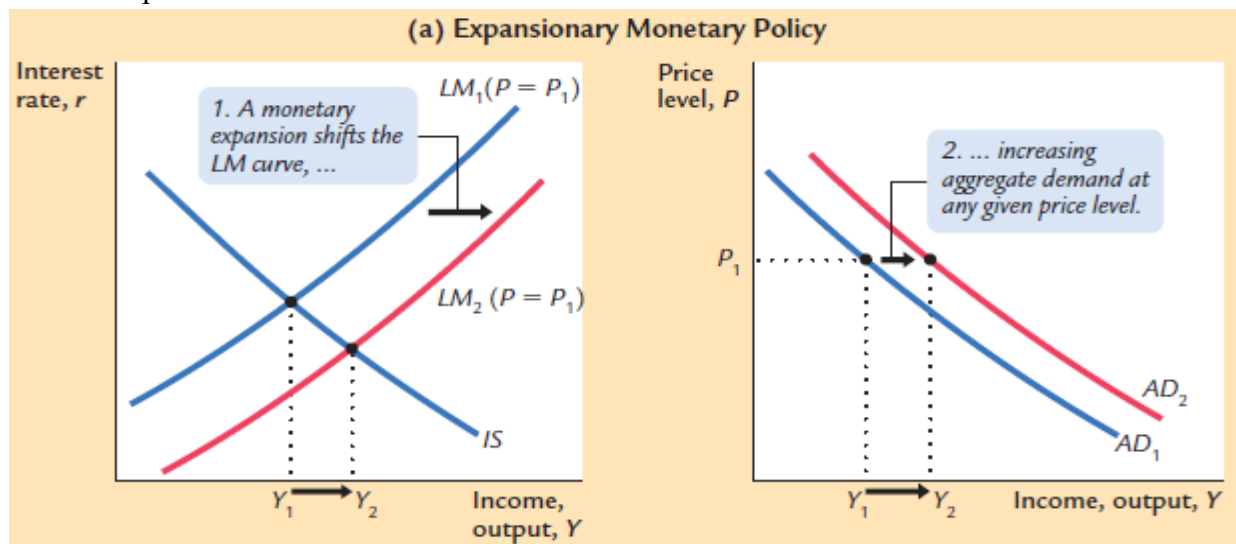
- $\uparrow P$ (for given money supply) $\rightarrow \downarrow M/P \rightarrow$ shifts LM upward $\rightarrow \uparrow r \rightarrow \downarrow Y$.
- As shown in figure below, this negative relation is captured by AD curve in panel B.

Figure: Derivation of the AD model from the IS-LM model

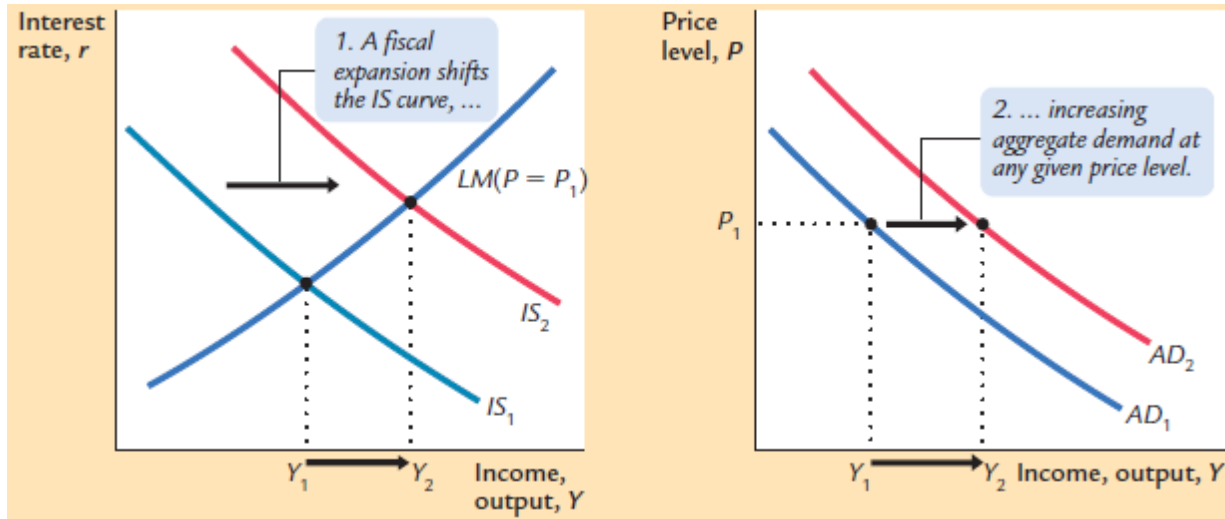


3.5.1. What causes shifts in AD?

A. Expansionary monetary policy: $\uparrow M \rightarrow \uparrow M/P \rightarrow$ shifts LM curve outward to the right $\rightarrow \downarrow r \rightarrow \uparrow$ planned I $\rightarrow \uparrow Y$ and vice versa.



B. Expansionary Fiscal policy → shifts IS curve outward → $\uparrow Y$ and vice versa.



3.5.2. IS-LM Model in the Short-Run and Long-Run

- ✓ We can compare the short-run and long-run equilibrium using either the IS –LM model in panel (a) or the aggregate supply–aggregate demand model in panel (b).
- ✓ In the short run, the price level is stuck at P_1 . The short-run equilibrium (Keynesian) of the economy is therefore point K.
- ✓ In the long run, the price level adjusts so that the economy is at the natural rate (full employment). The long-run equilibrium (Classical) is therefore point C.

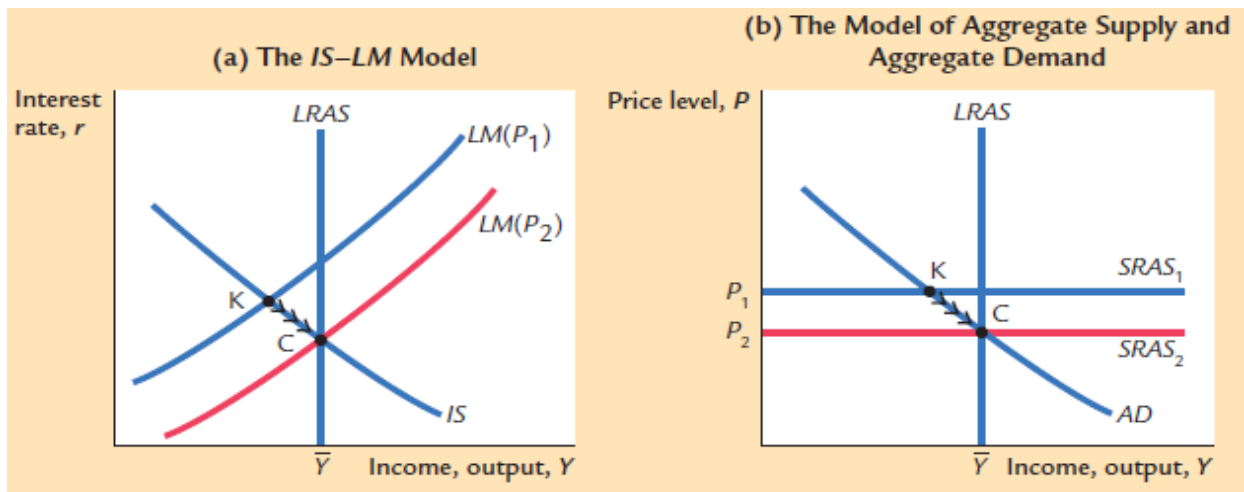
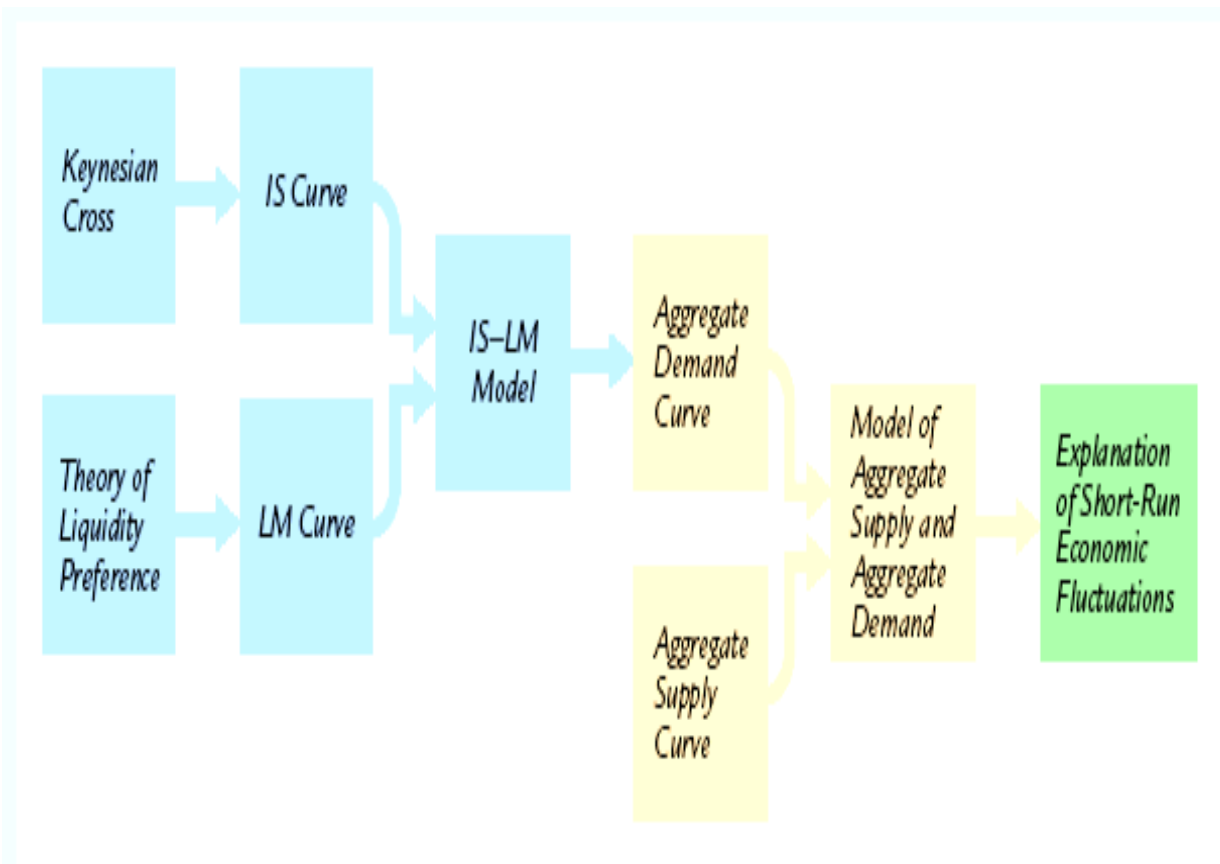


Figure: The Short-Run and Long-Run Equilibrium

We can diagrammatically present the whole of the model in the following diagram.



The Theory of Short-Run Fluctuations This schematic diagram shows how the different pieces of the theory of short-run fluctuations fit together. The Keynesian cross explains the *IS* curve, and the theory of liquidity preference explains the *LM* curve. The *IS* and *LM* curves together yield the *IS-LM* model, which explains the aggregate demand curve. The aggregate demand curve is part of the model of aggregate supply and aggregate demand, which economists use to explain short-run fluctuations in economic activity.

Review Exercises

1. Use the Keynesian cross to examine the effect of:
 - A. An increase in government purchases.
 - B. An increase in taxes.
 - C. An equal increase in government purchases and taxes.
2. In the Keynesian cross, assume that the consumption function is given by

$$C = 200 + 0.75 (Y - T)$$

Planned investment is 100; government purchases and taxes are both 100.

- A. Graph planned expenditure as a function of income.
- B. What is the equilibrium level of income?
- C. If government purchases increase to 125, what is the new equilibrium income?
- D. What level of government purchases is needed to achieve an income of 1,600?

3. Suppose that the money demand function is $(M/P)^d = 1,000 - 100r$, where r is the interest rate in percent. The money supply M is 1,000 and the price level P is 2.

- A. Graph the supply and demand for real money balances.
- B. What is the equilibrium interest rate?
- C. Assume that the price level is fixed. What happens to the equilibrium interest rate if the supply of money is raised from 1,000 to 1,200?
- D. If the Central bank wishes to raise the interest rate to 7 percent, what money supply should it set?

3. According to the IS –LM model, what happens to the interest rate, income, consumption, and investment under the following circumstances?

- a. The central bank increases the money supply.
- b. The government increases government purchases.
- c. The government increases taxes.
- d. The government increases government purchases and taxes by equal amounts.

4. Consider a hypothetical economy.

- a. The consumption function is given by

$$C = 200 + 0.75(Y - T).$$

The investment function is

$$I = 200 - 25r$$

Government purchases and taxes are both 100.

For this economy, graph the IS curve for r ranging from 0 to 8.

- b. The money demand function of the economy is

$$(M/P)^d = Y - 100r.$$

The money supply M is 1,000 and the price level P is 2.

For this economy, graph the LM curve for r ranging from 0 to 8.

- c. Find the equilibrium interest rate r and the equilibrium level of income Y .

- d. Suppose that government purchases are raised from 100 to 150. How much does the IS curve.
5. [**IS-LM Model**] Assume the following model of a closed economy in the short run, with the price level, **P**, fixed at **3.0**:

$$C = 50 + 0.75(Y - T) \quad - \text{Consumption function}$$

$$I = 200 - 25r \quad - \text{Investment function}$$

$$G = 3500, \quad T = 3000 \quad - \text{Exogenous Fiscal policy variables}$$

$$(M/P)^d = 0.5Y - 150r \quad - \text{Money demand Function}$$

$$M^S = 6000 \quad - \text{Nominal money supply}$$

- A. Write a numerical formula for the **IS curve**?
- B. Write a numerical formula for the **LM curve**?
- C. Find the equilibrium interest rate r and the equilibrium level of income Y in this **IS – LM** model?
- D. Now, suppose that the monetary authority increases the money supply from 6000 to 6600. Determine the new short-run equilibrium national income, Y (*Hint: Use the IS – LM model and support your computation with graphical presentation*)?

CHAPTER FOUR

AGGREGATE DEMAND IN THE OPEN ECONOMY

4.1. International Flows of Capital and Goods

In the previous chapter, we simplified our analysis by assuming a closed economy. In actuality, however, most economies are open: they export goods and services abroad, they import goods and services from abroad, and they borrow and lend in world financial markets.

Open Versus Closed Economies

- A closed economy is one that does not interact with other economies in the world.
 - There are no exports, no imports, and no capital flows.
- An open economy is one that interacts freely with other economies around the world.
 - ✓ An open economy interacts with other countries in two ways:
 - It buys and sells goods and services in world product markets.
 - It buys and sells capital assets in world financial markets.

4.1.1. International Flows of Goods: Exports, Imports, and Net Exports

Consider the expenditure on an open economy's output of goods and services, which can be divided into four components:

- C^d , consumption of domestic goods and services,
 - I^d , investment in domestic goods and services,
 - G^d , government purchases of domestic goods and services,
 - **EX**, exports of domestic goods and services.
- The division of expenditure into these components is expressed in the identity:

$$Y = C^d + I^d + G^d + EX$$

- Note that domestic spending on all goods and services is the sum of domestic spending on domestic goods and services and on foreign goods and services. Therefore,

$$C = C^d + C^f \quad I = I^d + I^f$$

$$G = G^d + G^f$$

- We substitute these three equations into the identity above:

$$Y = C^d + I^d + G^d + EX \text{ and rearranging}$$

$$Y = (C - C^f) + (I - I^f) + G - G^f + EX$$

$$Y = C + I + G + EX - (C^f + I^f + G^f)$$

$$Y = C + I + G + EX - IM$$

The national income identity in an open economy is therefore:

$$Y = C + I + G + Nx$$

$$\text{Or } Nx = Y - (C + I + G),$$

Net Exports = Output – Domestic Spending.

- ✓ The national income identity shows how domestic output, domestic spending, and net exports are related.
- It shows that in an open economy, domestic spending need not equal the output of goods and services. If output exceeds domestic spending, we export the difference: net exports are positive. If output falls short of domestic spending, we import the difference: net exports are negative.

Note: **Exports**: are domestically produced goods and services that are sold abroad.

Imports: are foreign produced goods and services that are sold domestically.

Net exports (NX): -also called the **trade balance**: refers to the values of a nation's exports minus the value of its imports.

Factors That Affect Net Exports

- The tastes of consumers for domestic and foreign goods.
- The prices of goods at home and abroad.
- The exchange rates at which currencies are exchanged.
- The incomes of consumers at home and abroad.
- The costs of transporting goods from country to country.
- International trade policies, etc...

4.1.2. International Capital Flows

Net foreign investment /Net capital outflow/- refers to the purchase of foreign assets by domestic residents minus the purchase of domestic assets by foreigners.

- For instance, the Ethiopian resident buys stock in the Chinese corporation and a Chinese buys stock in the Ethiopian corporation.
 - When Ethiopian resident buys stock in Telmex, the Mexican phone company, the purchase **raises** the Ethiopian net foreign investment.
 - When a Japanese resident buys a bond issued by the Ethiopian government, the purchase **reduces** the Ethiopian net foreign investment.
- **Financial/Capital markets and goods markets are closely related.**

Begin with the identity: $Y = C + I + G + NX$.

Subtract C and G from both sides to obtain:

$$Y - C - G = I + NX$$

Note: $Y - C - G$ is national saving S , the sum of private saving, $Y - T - C$, and public saving, $T - G$. Therefore,

$$S = I + NX.$$

Subtracting I from both sides of the equation, we can write the national income accounts identity as

$$S - I = NX \Rightarrow (\text{trade balance} = \text{net capital outflow})$$

- This identity shows that an economy's net exports must always equal the difference between its saving and its investment.
- ✓ $S - I$ is termed as **net capital outflow/net foreign investment** (domestic saving less domestic investment).
- If **net capital outflow is positive**, our saving exceeds our investment, and we are lending the excess to foreigners /I.e., **When $S > I$, the country is a net lender**
- If the **net capital outflow is negative**, our investment exceeds our saving, and we are financing this extra investment by borrowing from abroad /I.e., **When $S < I$, the country is a net borrower/**.
- Thus, net capital out flow equals the amount that domestic residents are lending abroad minus the amount that foreigners are lending to us.
- ✓ If $S - I$ and NX are positive, we have a **trade surplus**. In this case, we are net lenders in world financial markets, and we are exporting more goods than we are importing.

- ✓ If $S - I$ and NX are negative, we have a **trade deficit**. In this case, we are net borrowers in world financial markets, and we are importing more goods than we are exporting.
- ✓ If $S - I$ and NX are exactly zero, we are said to have **balanced trade** because the value of imports equals the value of exports.

The national income accounts identity shows that the international flow of funds to finance capital accumulation and the international flow of goods and services are two sides of the same coin.

- **On the one hand**, if our saving exceeds our investment, the saving that is not invested domestically is used to make loans to foreigners. Foreigners require these loans because we are providing them with more goods and services than they are providing us. That is, we are running a **trade surplus**.
- **On the other hand**, if our investment exceeds our saving, the extra investment must be financed by borrowing from abroad. These foreign loans enable us to import more goods and services than we export. That is, we are running a trade deficit.

Factors that Influence Net Foreign Investment

- The real interest rates being paid on foreign assets.
- The real interest rates being paid on domestic assets.
- The perceived economic and political risks of holding assets abroad.
- The government policies that affect foreign ownership of domestic assets.

4.2. Exchange Rates

Having examined the international flows of capital and of goods and services, we now extend the analysis by considering the prices that apply to these transactions.

- The exchange rate (E) is the rate at which one currency exchanges for another.
- One may view the exchange rate as indicative of the relative price of goods and services denominated in the currencies of the two countries concerned. Alternatively, the exchange rate can be regarded as the relative price of assets denominated in the currencies of a pair of countries. In any case, conversion from one currency unit to another is the job of the exchange rate.
- There are two conventions for measuring the exchange rate, the distinction between which can be the source of serious confusion:

A. Domestic currency units per unit of foreign currency

- For example, if the birr is the home currency and the dollar (\$) is the foreign Currency, and 20-birr exchanges for \$1, then the exchange rate is 20. The domestic currency is on the numerator of the ratio.
- Here, whenever E rises the home currency gets weaker; **it depreciates**. For example, arise from 20 to 21 means that 21birr exchanged for \$1, less than before.
- Conversely, when E falls, the domestic currency gets stronger, it appreciates.

B. Foreign currency units per unit of domestic currency

This is exactly the converse of the first convention.

- The domestic currency is on the denominator. When 20birr is exchanged for \$1, the exchange rate is 0.05.
- **If E rises, the home currency gets stronger, and vice versa.**

Real and Nominal Exchange Rates

- The **Nominal Exchange Rate (e)** is the relative price of the currency of two countries. For example, if the exchange rate between the Ethiopian birr and the U.S dollar is 20 birr per dollar, then you can exchange one dollar for 20 birrs in the world markets for foreign currency.
- An Ethiopian who wants to obtain dollars would pay 20 birrs for each dollar he bought.
- An American who wants to obtain birr would get 20 birrs for each dollar he paid.
- The **Real Exchange Rate** is the relative price of the goods of two countries.
- That is, the real exchange rate tells us the rate at which we can trade the goods of one country for the goods of another. It measures a country's competitiveness in the international trade.

The real exchange rate R, is usually defined as

$$\begin{array}{ccccc} \text{Real} & & \text{Nominal} & & \text{Ratio of} \\ \text{Exchange} & = & \text{Exchange} & \times & \text{Price} \\ \text{Rate} & & \text{Rate} & & \text{Levels} \\ \epsilon & = & e & \times & (P/P^*). \end{array}$$

We can write the nominal exchange rate as

$$e = \epsilon \times (P^*/P).$$

This equation shows that the nominal exchange rate depends on the real exchange rate and the price levels in the two countries. Given the value of the real exchange rate, if the domestic price

level P rises, then the nominal exchange rate e will fall: because a dollar is worth less, a dollar will buy fewer yen. However, if the Japanese price level P^* rises, then the nominal exchange rate will increase: because the yen is worth less, a dollar will buy more yen. It is instructive to consider changes in exchange rates over time.

The exchange rate equation can be written

$$\% \text{ Change in } e = \% \text{ Change in } \epsilon + \% \text{ Change in } P^* - \% \text{ Change in } P.$$

The percentage change in ϵ is the change in the real exchange rate. The percentage change in P is the domestic inflation rate π , and the percentage change in P^* is the foreign country's inflation rate π^* . Thus, the percentage change in the nominal exchange rate is

$$\% \text{ Change in } e = \% \text{ Change in } \epsilon + (\pi^* - \pi)$$

$$\begin{array}{l} \text{Percentage Change in} \\ \text{Nominal Exchange Rate} \end{array} = \begin{array}{l} \text{Percentage Change in} \\ \text{Real Exchange Rate} \end{array} + \begin{array}{l} \text{Difference in} \\ \text{Inflation Rates.} \end{array}$$

This equation states that the percentage change in the nominal exchange rate between the currencies of two countries equals the percentage change in the real exchange rate plus the difference in their inflation rates. If a country has a high rate of inflation relative to the United States, a dollar will buy an increasing amount of the foreign currency over time. If a country has a low rate of inflation relative to the United States, a dollar will buy a decreasing amount of the foreign currency over time.

This analysis shows how monetary policy affects the nominal exchange rate.

We know that high growth in the money supply leads to high inflation. Here, we have just seen that one consequence of high inflation is a depreciating currency: high p implies falling e .

In other words, just as growth in the amount of money raises the price of goods measured in terms of money, it also tends to raise the price of foreign currencies measured in terms of the domestic currency.

- The rate at which we exchange foreign and domestic goods depends on the prices of the goods in the local currencies and on the rate at which the currencies are exchanged.
- If the exchange rate equals 1, currencies are at purchasing power parity (PPP).
- A high real exchange rate shows that domestic goods are dearer. Thus, exports are discouraged while imports are encouraged so, the trade balance deteriorates.
- A low real exchange rate shows that domestic goods are cheaper. Thus, exports are encouraged while imports are discouraged so, the trade balance improves.

Exchange rate Determination/Fixed versus Flexible Exchange rate Regimes/

A. Flexible Exchange Rates:

- Under this system the exchange rate is completely market- determined, without any interference from government authorities (the central bank).
- The balance of payments balances: surpluses are eliminated via exchange rates appreciation; deficits are cured by exchange rate depreciation.
- In other words, balance of payments disequilibria is self-correcting.

B. Fixed Exchange Rates:

- Here the rate of exchange is a policy parameter fixed by the authorities. They have to meet excess demand for a foreign currency (not financed by sales in that foreign country) from their reserves of foreign currency.
- Net inflows of foreign currency swell the domestic money supply, as private economic agents who don't use it to purchase goods abroad convert foreign currency to domestic money at the fixed rate of exchange.
- Fixed exchange rates make the domestic money Supply dependent on changes in foreign exchange reserves. Also, balance of payments deficits cause reserve losses and surpluses cause reserve to be built up: **there is no self-correcting mechanism to balance of payments disequilibria.**
- These two exchange rate regimes are the two polar extremes in the economics of exchange rate. In reality we have a pastiche of arrangement, somewhere in between the two. Even with flexible rates, central banks do interfere with the market-determined exchange rate, often in an internationally concerted fashion. This is known as a managed float (consider the Ethiopian case). Fixed Exchange rates also allow for realignments and changes in the exchange rate, usually devaluation.

Devaluation Versus Revaluation.

- ✓ Devaluation takes place when the price of foreign currencies under fixed exchange rate regime is increased by official action (you can consider the action of the Ethiopian government to devalue the currency in 1992. The Birr was devalued in 1992 and this is expected to promote exporters and discourage importers. This was expected to narrow the gap between exports and imports and consequently improve the current account.
- ✓ Devaluation thus means that foreigners pay less for devalued currency and that residents of the devaluing country pay more for foreign currencies. The opposite of devaluation is revaluation.

Depreciation versus Appreciations

A currency is said to be depreciate when under floating rates it becomes less expensive in terms of foreign currency. By contrast, currency appreciates when it becomes more expensive in terms of foreign money.

4.3. Monetary and Fiscal Policy Analysis in an Open Economy with Perfect Capital Mobility

4.3.1 Saving and investment in a small open economy

So far in our discussion, we have considered the international flows of goods and capital, and their relationships. Our next step is to develop a model that explains the behaviour of these variables. We can then use the model to answer questions such as how the trade balance responds to changes in policy.

Basic Assumptions:

- ✓ **Small open economy**- it is to mean that the economy is a small part of the world market and thus, by itself, can have only a negligible effect on the world interest rate.
- ✓ **Perfect capital mobility**- residents of the country have full access to world financial markets. Meaning the government does not impede international borrowing/lending.
- Owing to the above assumptions, the interest rate r , must equal the world interest rate r^* (the real interest rate prevailing in world financial markets): $r = r^*$

Residents of the small open economy need never borrow at any interest rate above r^* , because they can always get a loan at r^* from abroad. Similarly, residents of this economy need never lend at any interest rate below r^* because they can always earn r^* by lending abroad. Thus, the world interest rate determines the interest rate in our small open economy.

✚ What determines the world real interest rate?

- The equilibrium of world saving and world investment determines the world interest rate.
- Our small open economy has a negligible effect on the world real interest rate because, being a small part of the world, it has a negligible effect on world saving and world investment. Hence, our small open economy takes the world interest rate as exogenously given.

The Model: To build the model of the small open economy, we make three assumptions

- i) Production function: $Y = \bar{Y} = F(\bar{K}, \bar{L})$:The economy's output Y is fixed by the factors of production
- ii) Consumption function: $C = C(Y - T)$
- iii) Investment function: $I = I(r)$

We can now return to the accounting identity and write it as

$$N_x = (Y - C - G) - I$$

$$N_x = S - I$$

Substituting our three assumptions and the condition that the interest rate equals the world interest rate, we obtain

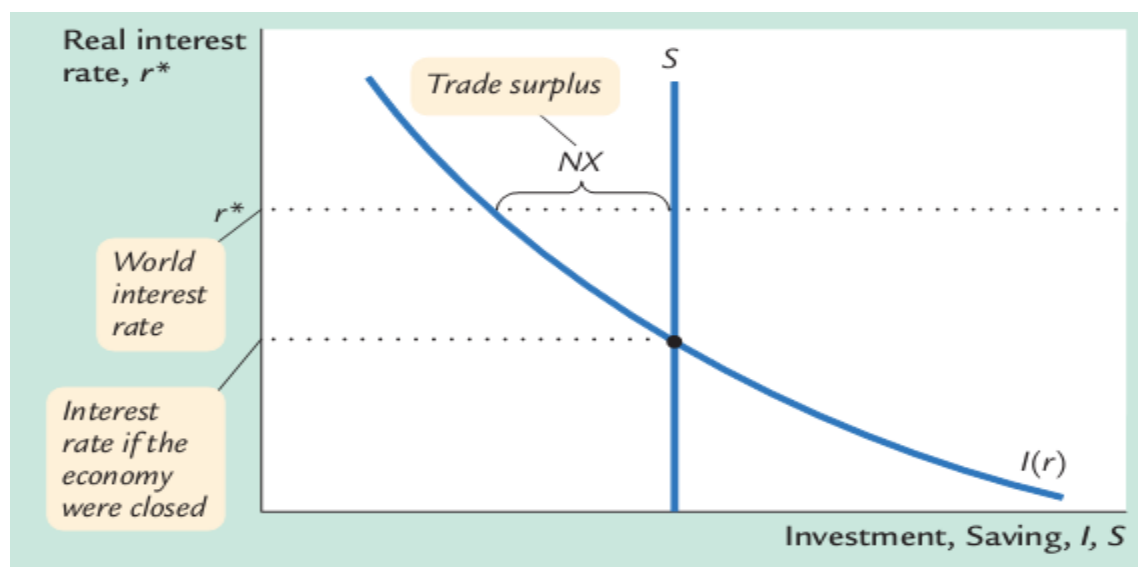
$$N_x = [\bar{Y} - C(\bar{Y} - T) - G] - I(r^*)$$

$$N_x = \bar{S} - I(r^*)$$

This equation shows what determines saving S and investment I, and thus the trade balance— N_x .

- **Saving depends on fiscal policy:** lower government purchases G or higher taxes T raise national saving.
- **Investment depends on the world real interest rate r^* :** high interest rates make some investment projects unprofitable. Therefore, the trade balance depends on these variables as well.
- The **trade balance** is determined by the difference between saving and investment at the world interest rate.

Figure: 4.1: Saving and Investment in a Small Open Economy



- In a closed economy, the real interest rate adjusts to equilibrate saving and investment.
- But, **in a small open economy**, the interest rate is determined in world financial markets. The difference between saving and investment determines the trade balance. Here there is a trade surplus, because at the world interest rate, saving exceeds investment.

4.3.2. How Policies Influence the Trade Balance

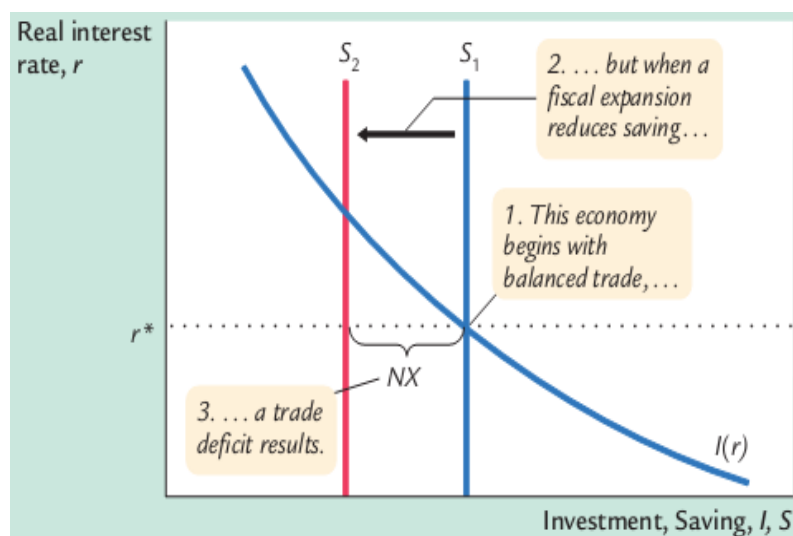
Suppose that the economy begins in a position of balanced trade. That is, at the world interest rate, investment I equal saving, S , and net exports Nx equal zero. Let's use our model to predict the effects of government policies at home and abroad.

A. Fiscal Policy at Home

What happens to the small open economy if the government expands domestic spending by increasing government purchases?

- The increase in G reduces national saving, because $S = Y - C - G$. **With an unchanged world real interest rate, investment remains the same.** Therefore, saving falls below investment, and some investment must now be financed by borrowing from abroad. Because $NX = S - I$, the fall in S implies a fall in Nx . The economy now runs a trade deficit.

Figure 4.2: A Fiscal Expansion at Home in a Small Open Economy



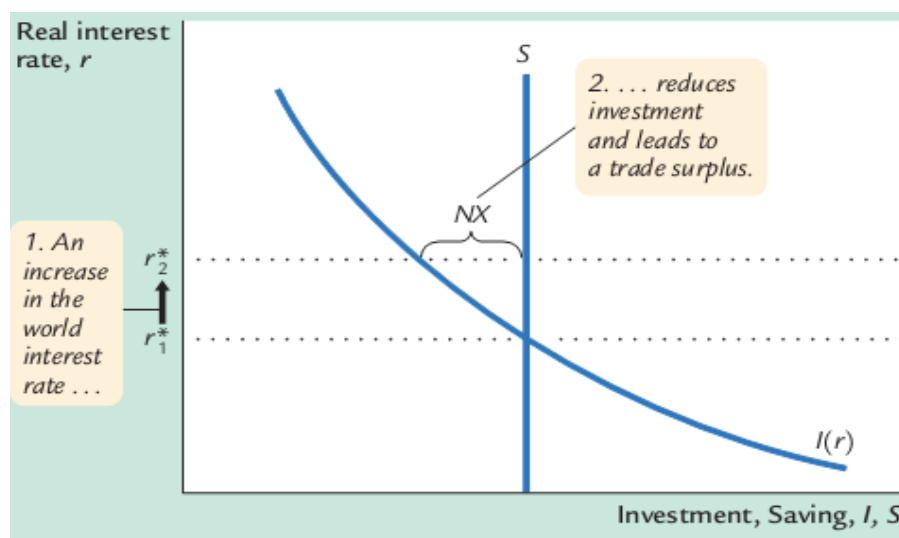
- ✓ An increase in government purchases or a reduction in taxes reduces national saving and thus shifts the saving schedule to the left, from S_1 to S_2 . The result is a **trade deficit**.

Because N_x is the distance between the saving schedule and the investment schedule at the world interest rate, this shift reduces N_x . **Hence, starting from balanced trade a change in fiscal policy that reduces national saving leads to a trade deficit.**

B. Fiscal Policy Abroad

- ✓ What happens to a small open economy when foreign governments increase their government purchases?
- Assuming these foreign countries are a large part of the world economy, and thus their increase in government purchases reduces world saving and causes the world interest rate to rise. The increase in the world interest rate raises the cost of borrowing and, thus, reduces investment in our small open economy.

Figure 4.3: A Fiscal Expansion at Abroad in a Small Open Economy



A fiscal expansion in a foreign economy large enough to influence world saving and investment raises the world interest rate from r_1^* to r_2^* .

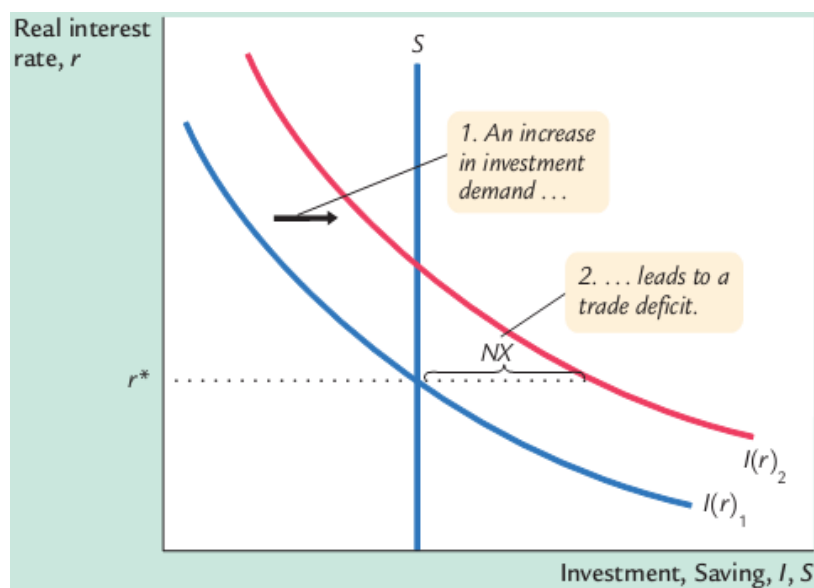
The higher world interest rate reduces investment in this small open economy, causing a trade surplus.

C) Shifts in Investment Demand

What happens to our small open economy if its investment schedule shifts outward—that is, if the demand for investment goods at every interest rate increases?

- This shift would occur if, for example, the government changed the tax laws to encourage investment.

Figure 4.4: A Shift in the Investment Schedule in a Small Open Economy



An outward shift in the investment schedule from $I(r)_1$ to $I(r)_2$ increases the amount of investment at the world interest rate r^* . As a result, investment now exceeds saving, which means the economy is borrowing from abroad and running a trade deficit.

Because $Nx = S - I$, the increase in I implies a decrease in Nx . Hence, an outward shift in the investment schedule causes a **trade deficit**.

4.3.3. The Small Open Economy under Floating Exchange Rates

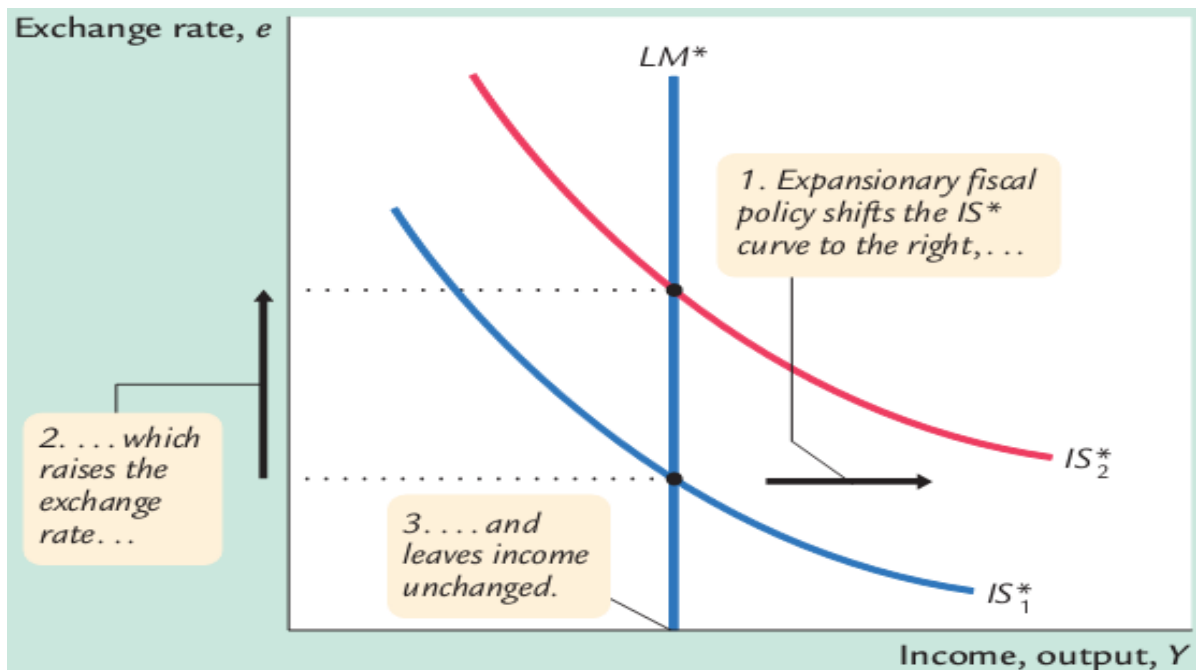
A) Fiscal Policy under Floating Exchange Rates

- Suppose that the government stimulates domestic spending by increasing government purchases or by cutting taxes. Because such expansionary fiscal policy increases planned expenditure, it shifts the IS^* curve to the right, as in Figure 4.10. As a result, the exchange rate appreciates, whereas the level of income remains the same.

Note the mechanism:

- When income rises in a small open economy, due to the fiscal expansion, the interest rate tries to rise but capital inflows from abroad put downward pressure on the interest rate. This inflow causes an increase in the demand for the domestic currency pushing up its value (the currency appreciates).
- The appreciation of the exchange rate makes domestic goods expensive for foreigners, and this reduces net exports. The fall in net exports offsets the effects of the expansionary fiscal policy on income.

Figure: 4.10: A Fiscal Expansion Under Floating Exchange Rates



- ✓ An increase in government purchases or a decrease in taxes shifts the IS^* curve to the right. This raises the exchange rate but has no effect on income.

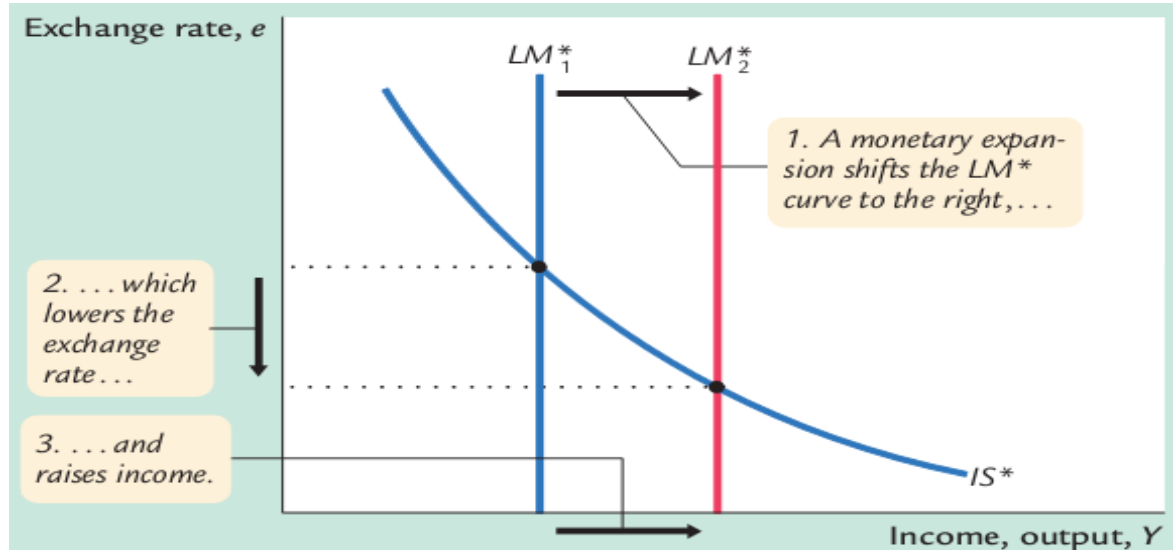
B) Monetary Policy under Floating Exchange Rates

- Suppose now that the central bank increases the money supply. Because the price level is assumed to be fixed, the increase in the money supply means an increase in real balances.
- The increase in real balances shifts the LM^* curve to the right, as in Figure 4.11. Hence, an increase in the money supply raises income and lowers the exchange rate.

Note the mechanism:

- ✓ When **increase in the money supply** puts **downward** pressure on the **domestic interest rate**, capital flows out of the economy as investors seek a higher return elsewhere. This capital outflow prevents the domestic interest rate from falling.
- ✓ In addition, because the **capital outflow** increases the supply of the domestic currency in the market for foreign-currency exchange, the **exchange rate depreciates**. The fall in the exchange rate makes **domestic goods cheap** relative to foreign goods and, thereby, **stimulates net exports**. Hence, in a small open economy, monetary policy influences income by altering the exchange rate.

Figure: 4.11: A Monetary Expansion Under Floating Exchange Rates



- ✓ An increase in the money supply shifts the LM^* curve to the right, lowering the exchange rate and raising income.

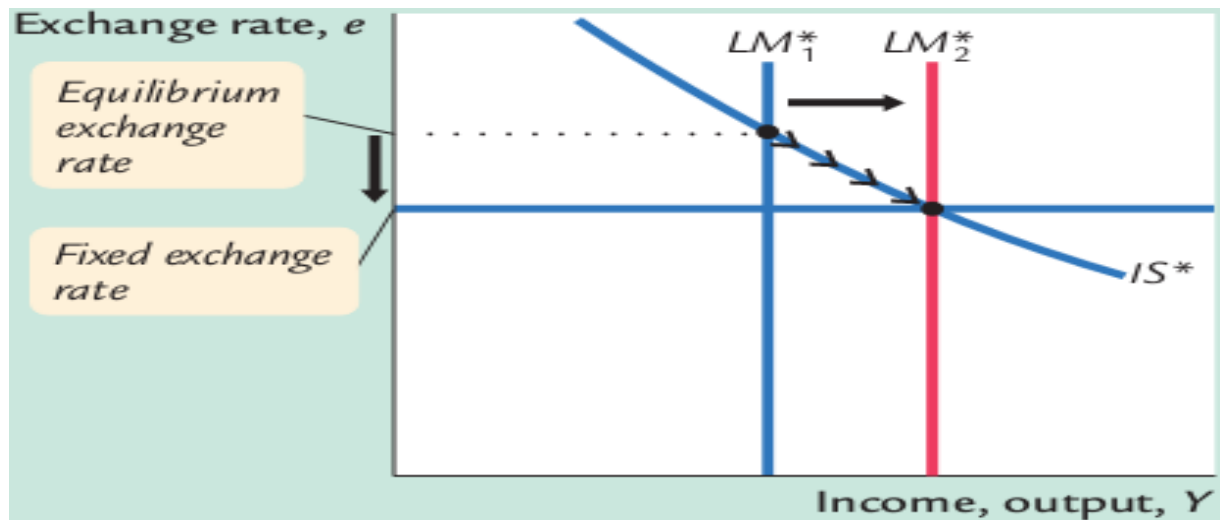
The Small Open Economy under Fixed Exchange Rates

- Under a fixed exchange rate, the central bank announces a value for the exchange rate and stands ready to buy and sell the domestic currency at a predetermined price to keep the exchange rate at its announced level.
- Fixed exchange rates require a commitment of a central bank to allow the money supply to adjust to whatever level will ensure that the equilibrium exchange rate in the market for foreign-currency exchange equals the announced exchange rate.
- i.e., the sole objective of monetary policy is to keep the exchange rate at the announced level.

How a Fixed-Exchange-Rate System Works

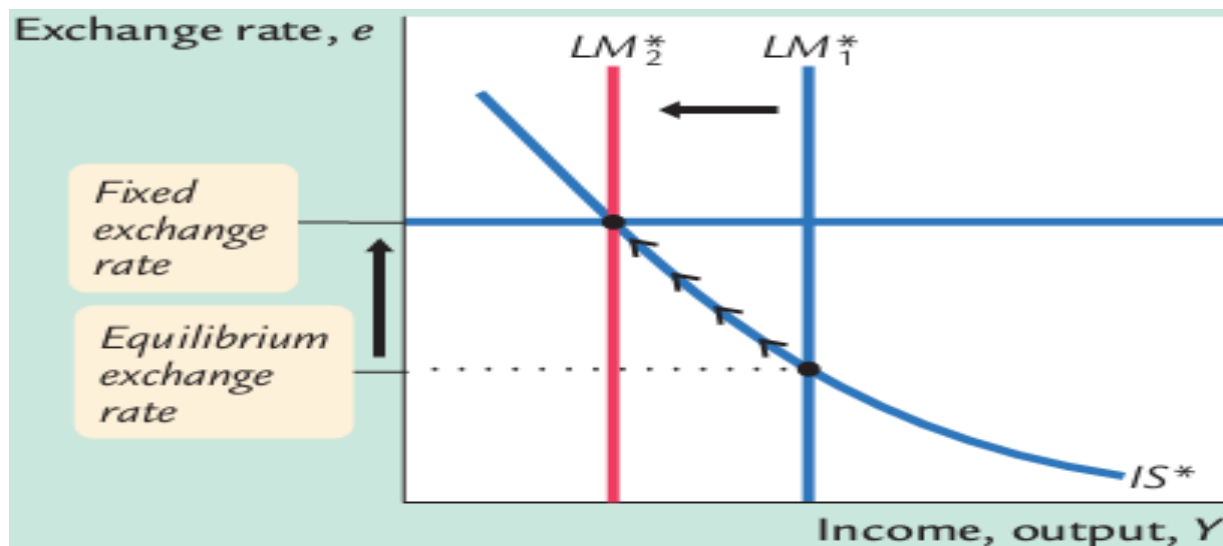
- Suppose that the Ethiopian national bank announces that it is going to fix the exchange rate at 32 birr per dollar (I.e., \$0.03125 per birr). It would then stand ready to give 32 birr in exchange for \$1 or to give \$1 in exchange for 32 birr.
- Hence, the NB would need a reserve of birr (which it can print) and a reserve of dollars (which must have been purchased previously)
- Suppose in the current equilibrium with the current money supply, the exchange rate is 20 birr per dollar (\$0.05 per birr) that is higher than 32 birr per dollar determined by the central bank. Arbitrageurs use their birr to buy foreign currency in foreign-exchange markets and sell it to the domestic national bank for a profit. **This process automatically increases the money supply (the base money)-shifting the LM* curve to the right and lowers the exchange rate as the table below presents.**

Figure: 4.12: When the equilibrium exchange rate is greater than the fixed exchange rate



Suppose the equilibrium exchange rate is lower than the fixed exchange rate. Arbitrageurs will buy birr in foreign-exchange markets and use them to buy foreign currency from national bank of Ethiopia. **This process automatically reduces the money supply, shifting the LM^* curve to the left and raises the exchange rate as figure 4.13, presents.**

Figure: 4.13: When the equilibrium exchange rate is less than the fixed exchange rate



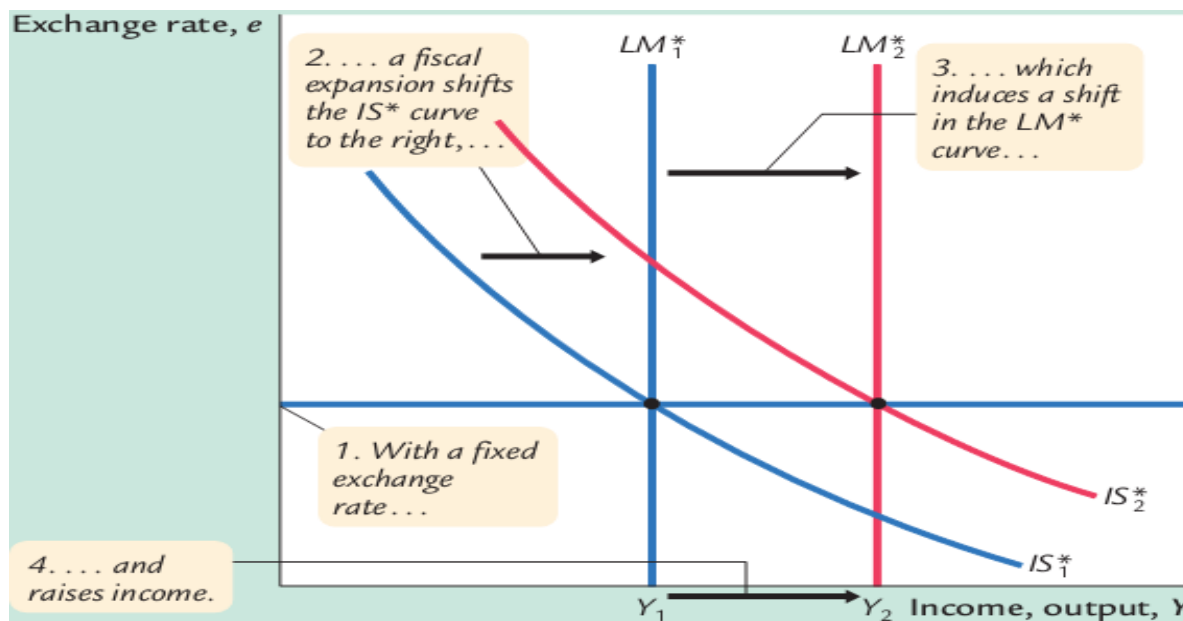
Figures 4.12 and 4.13 shows how a fixed exchange rate governs the money supply.

- ✓ Let's now examine how economic policies affect a small open economy with a fixed exchange rate.

A) Fiscal Policy under Fixed Exchange Rates

- Suppose that the government stimulates domestic spending by **increasing government purchases or by cutting taxes**. This policy shifts the IS^* curve to the right, as in Figure 4.12, **putting upward pressure on the exchange rate**.
- But because the central bank stands ready to trade foreign and domestic currency at the fixed exchange rate, arbitrageurs quickly respond to the rising exchange rate by selling foreign currency to the central bank, leading to an **automatic monetary expansion**.
- The rise in the money supply (in the **base money**) shifts the LM^* curve to the right.
- Thus, under a fixed exchange rate, a fiscal expansion raises aggregate income.

Figure: 4.14: A Fiscal Expansion Under Fixed Exchange Rates



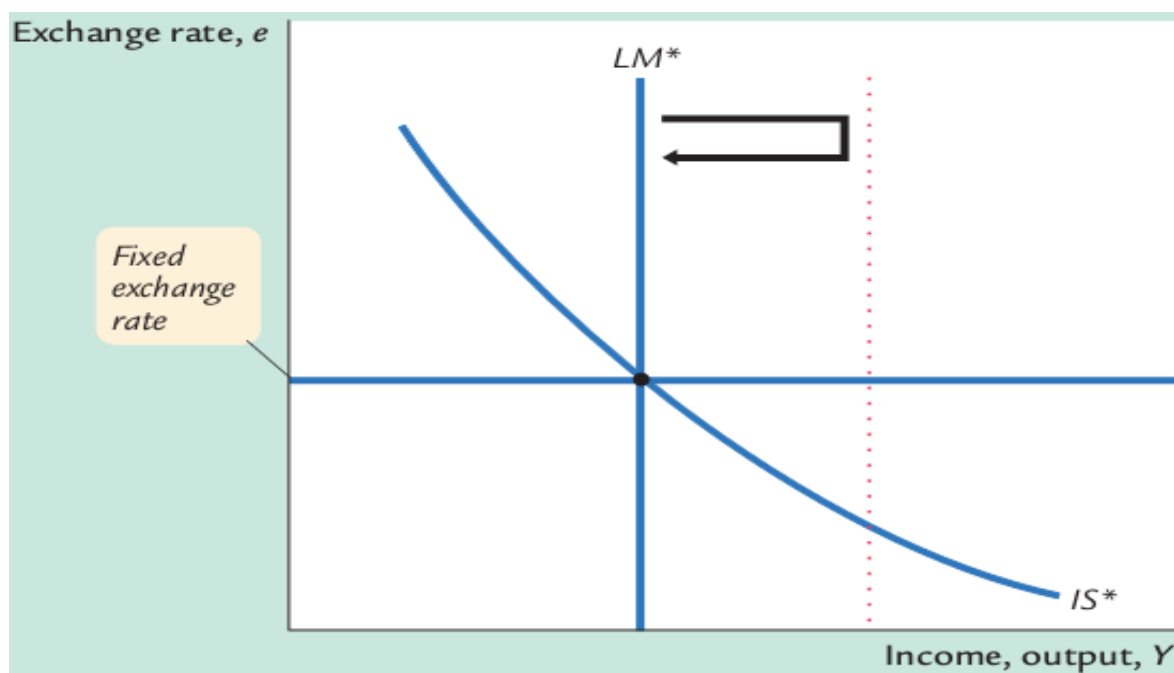
- ✓ A fiscal expansion shifts the IS^* curve to the right. To maintain the fixed exchange rate, the central bank must increase the money supply, thereby shifting the LM^* curve to the right. Hence, in contrast to the case of floating exchange rates, under fixed exchange rates a fiscal expansion raises income.

B) Monetary Policy under Fixed Exchange Rates

- Imagine that a central bank operating with a fixed exchange rate were to try to increase the money supply—for example, by buying bonds from the public.
- What would happen? The initial impact of this policy is to shift the LM^* curve to the right, lowering the exchange rate, as in Figure 4.13.

- But, because the central bank is committed to trading foreign and domestic currency at a fixed exchange rate, arbitrageurs quickly respond to the falling exchange rate by selling the domestic currency to the central bank, causing the money supply and the LM^* curve to return to their initial positions.
- Hence, monetary policy is ineffectual under a fixed exchange rate. By agreeing to fix the exchange rate, the central bank gives up its control over the money supply.

Figure: 4.15: A Monetary Expansion Under Fixed Exchange Rates



- ✓ If the National bank tries to increase the money supply -for example, by buying bonds from the public—it will put downward pressure on the exchange rate. To maintain the fixed exchange rate, the money supply and the LM^* curve must return to their initial positions. Hence, under fixed exchange rates, monetary policy is ineffectual.

Fixed vs. Floating Exchange Rate Conclusions

Fixed Exchange Rate	Floating Exchange Rate
➤ Fiscal Policy is Powerful.	➤ Fiscal Policy is Powerless.
➤ Monetary Policy is Powerless.	➤ Monetary Policy is Powerful

4.4. The Mundell-Fleming Model

- The model developed in this chapter, called the Mundell–Fleming model, is an open-economy version of the IS – LM model.
- This model owes its origins to papers published by James Flemming (1962) and Robert Mundell (1962, 1963).
- Their major contribution was to incorporate international capital movements into formal macroeconomic models based on the Keynesian IS-LM framework.
- Their papers led to some dramatic implications concerning the effectiveness of fiscal and monetary policy on national income.

Notice the Relationship b/n IS-LM and Mundell–Fleming model:

- Both models stress the interaction between the goods market and the money market.
- Both models assume that the price level is fixed and then show what causes short-run fluctuations in aggregate income (or, equivalently, shifts in the aggregate demand curve).
- The key difference is that the IS –LM model assumes a closed economy, whereas the Mundell–Fleming model assumes an open economy. The Mundell–Fleming model extends the short-run model of national income by including the effects of international trade and finance.

The Mundell–Fleming model makes one important and extreme assumption: It assumes that the economy being studied is a small open economy with perfect capital mobility. That is, the economy can borrow or lend as much as it wants in world financial markets and, as a result, the economy's interest rate is determined by the world interest rate.

One virtue of this assumption is that it simplifies the analysis: once the interest rate is determined, we can concentrate our attention on the role of the exchange rate.

In this section, we will build the Mundell –Fleming model, and use the model to examine the impact of various policies to discern how the economy operates under floating and fixed exchange rate regimes and notice whether a floating or fixed exchange rate is better - /an important question in recent years, as many nations around the world have debated what exchange-rate system to adopt/.

Small Open Economy with Perfect Capital Mobility

- r is determined by the world interest rate r^* - the small open economy being under consideration do not affect world interest rate. Mathematically: $r = r^*$
- The $r = r^*$ equation represents the assumption that the international flow of capital is rapid enough to keep the domestic interest rate equal to the world interest rate.
- Since r is fixed, fiscal and monetary policies will affect the IS and LM curves through the exchange rates.

4.4.1. The Goods Market and the IS* Curve

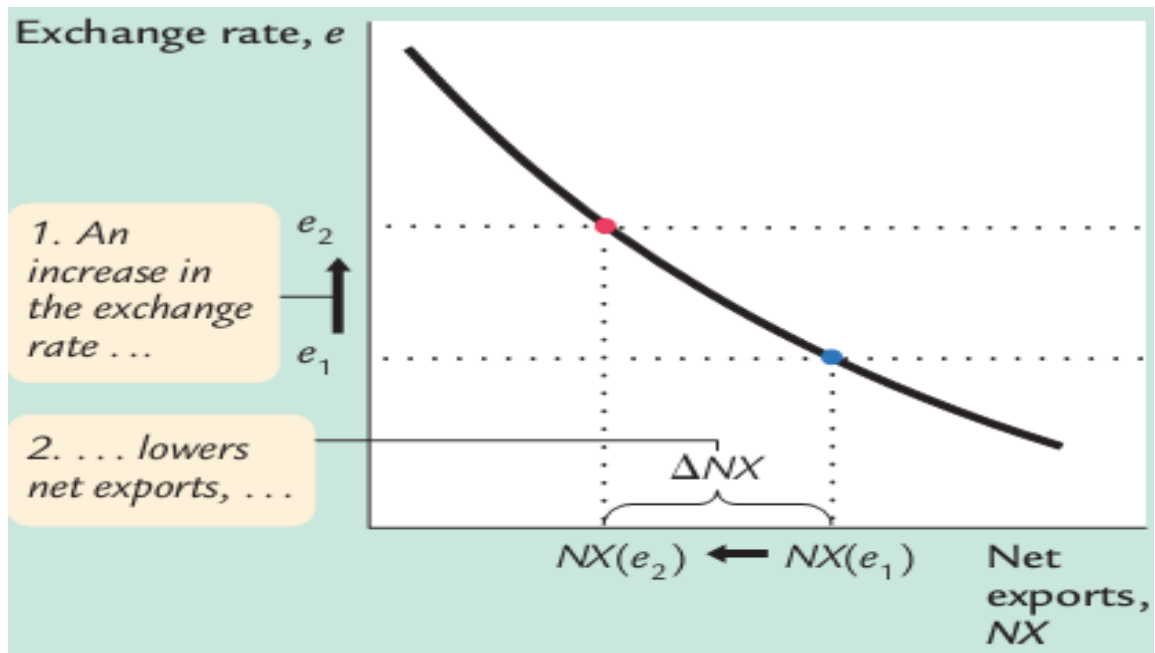
The Mundell–Fleming model describes the market for goods and services much as the IS–LM model does, but it adds a new term for net exports. In particular, the goods market is represented with the following equation:

$$Y = C(Y - T) + I(r^*) + G + Nx(e)$$

- The net exports depend on the real exchange rate. The Mundell-Fleming model, however, assumes that the price levels at home and abroad are fixed, so the real and nominal exchange rates are proportional.
- When the nominal exchange rate appreciates foreign goods become cheaper compared to domestic goods, and this causes exports to fall and imports to rise, and lowers the aggregate income.

The relation of the exchange rate and aggregate income is called the IS* curve.

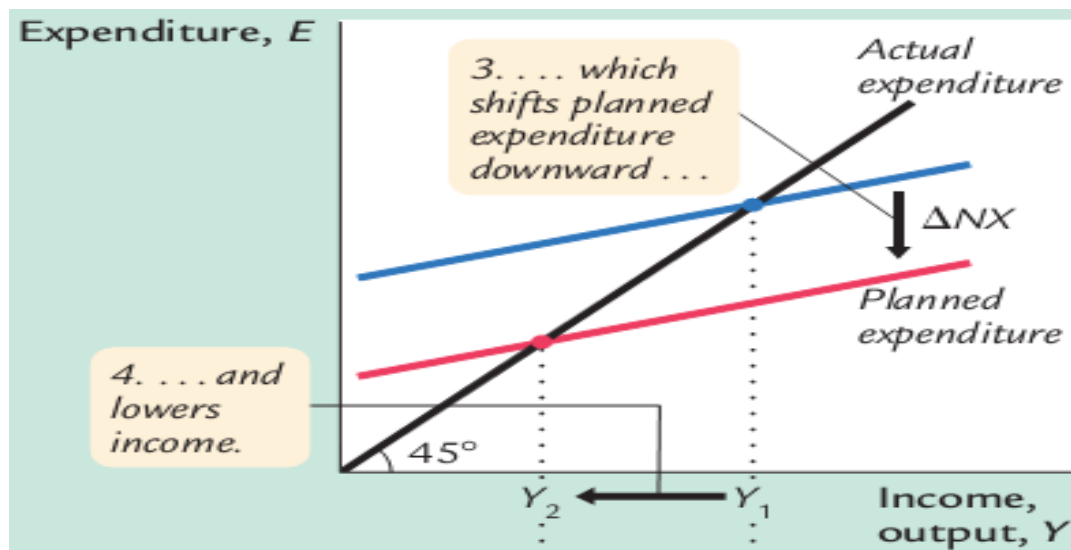
Figure:4.5: The Net-Exports Function



- ✓ An increase in the exchange rate from e_1 to e_2 lowers net exports from $Nx(e_1)$ to $Nx(e_2)$.

How changes in the exchange rate do affect income?

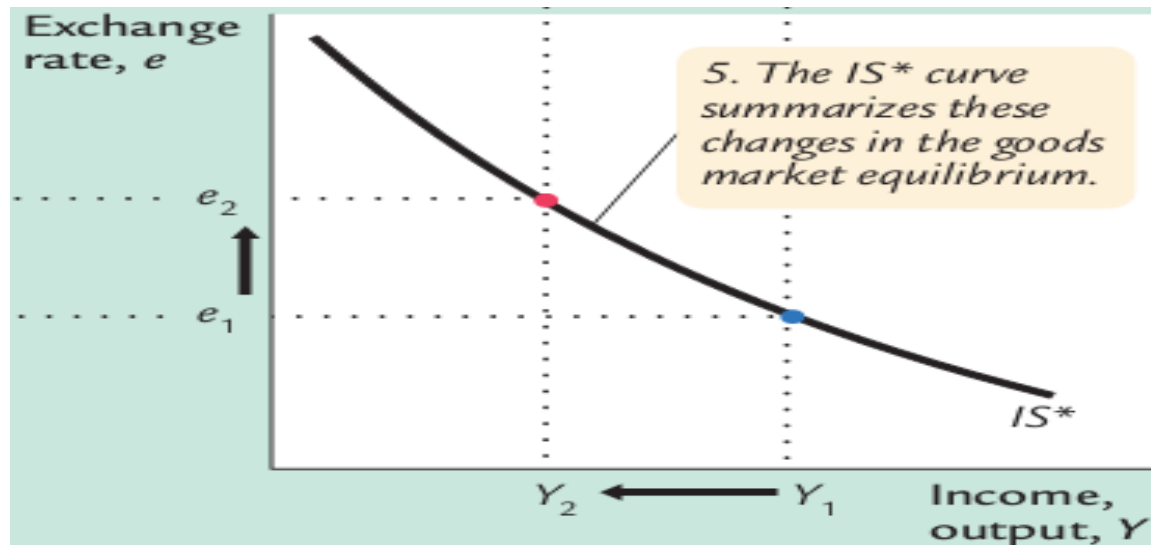
Figure: 4.6: The Keynesian Cross



- ✓ A decrease in net exports from $Nx(e_1)$ to $Nx(e_2)$ shifts the planned-expenditure schedule downward and reduces income from Y_1 to Y_2 as indicated above.

The IS* curve summarizing this relationship between the exchange rate and income: the higher the exchange rate, the lower the level of income.

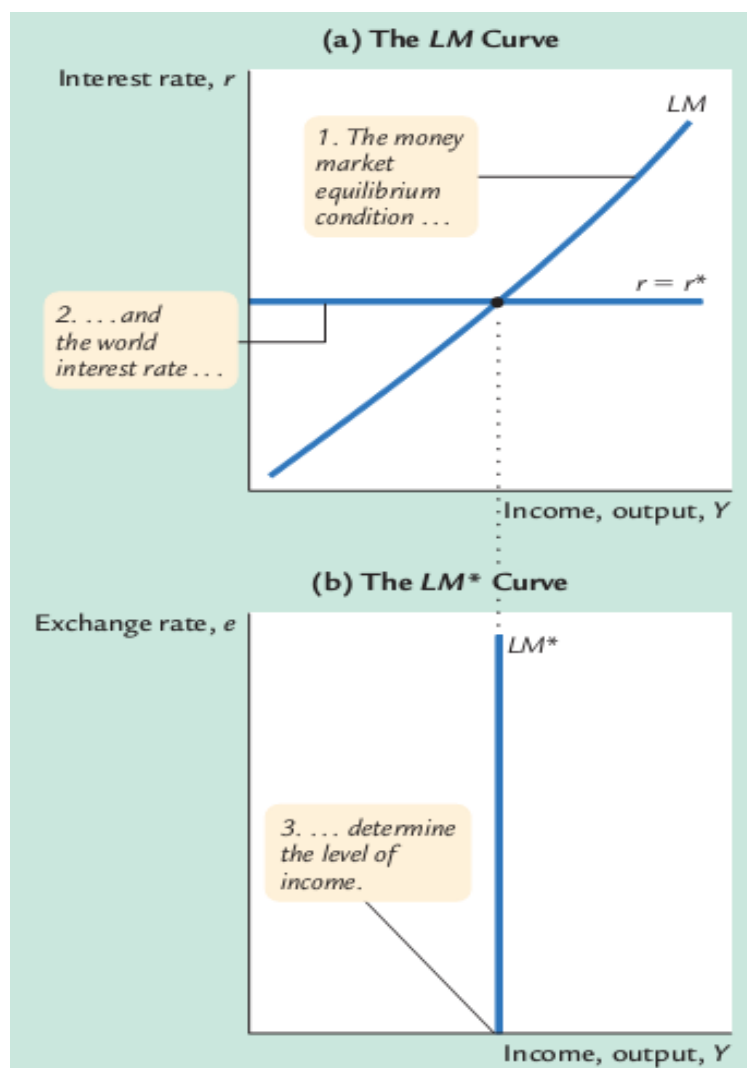
Figure: 4.7: The IS* Curve



4.4.2. The Money Market and the LM* Curve

- LM curve is derived from $M/P = L(r, Y)$
- The model assumes M as fixed policy variable chosen by central banks and P is also fixed in the short-run, and thus LM curve draws combinations of r and Y satisfying the equation written above.
- Since r is fixed by the world interest rate in an open economy, the equation finds a fixed amount of income Y .
- This is to say Y is not affected by the exchange rate. Hence, the LM curve, which draws the relation of the exchange rate and aggregate income, is vertical.

Figure: 4.8: The LM* Curve:



✓ Panel (a) shows the standard LM curve [which graphs the equation $M/P = L(r, Y)$] together with a horizontal line representing the world interest rate r^* .

✓ The intersection of these two curves determines the level of income, regardless of the exchange rate. Therefore, as panel (b) shows, the LM^* curve is vertical.

4.4.3. The Equilibrium/Combining the IS^* and LM^* Curves

According to the Mundell–Fleming model, a small open economy with perfect capital mobility can be described by two equations:

$$Y = C(Y - T) + I(r^*) + G + Nx(e) \quad IS^*$$

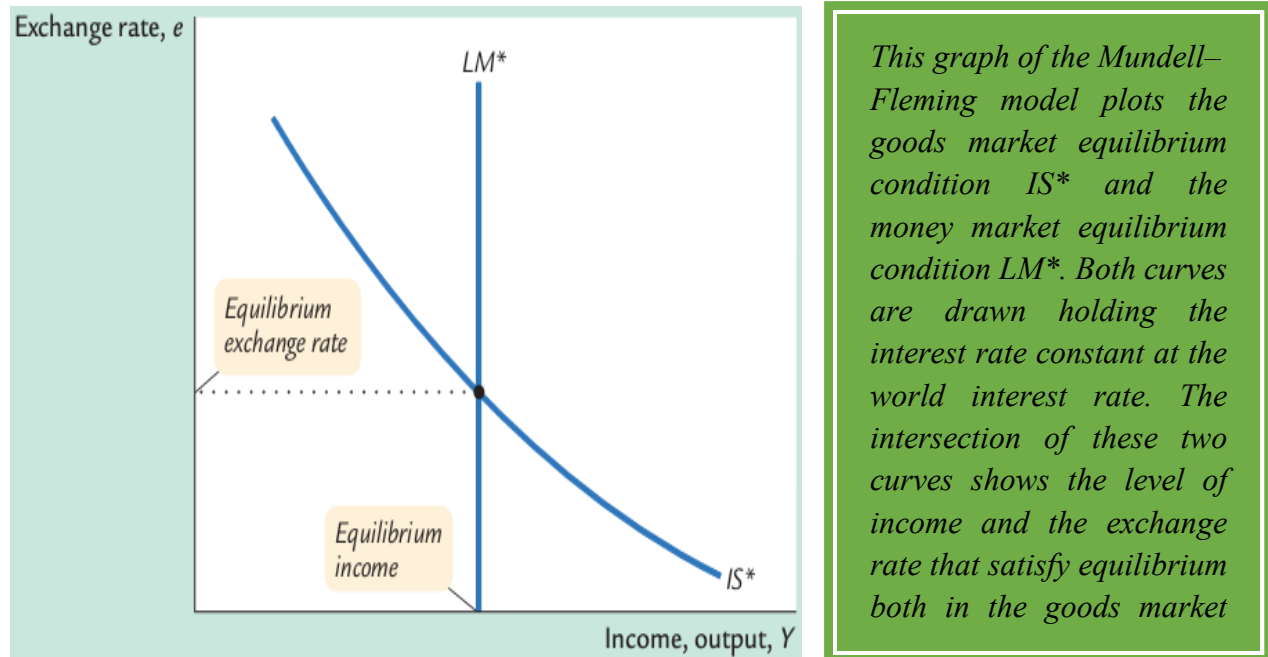
$$M/P = L(r^*, Y) \quad LM^*$$

- The first equation describes equilibrium in the goods market, and the second equation describes equilibrium in the money market. The exogenous variables are fiscal policy G and T , monetary policy M , the price level, P and the world interest rate, r^* .
- The endogenous variables are income Y and the exchange rate e .

Figure 4.9 illustrates these two relationships. The equilibrium for the economy is found where the IS^* curve and the LM^* curve intersect. This intersection shows the exchange rate and the level of income at which both the goods market and the money market are in equilibrium.

With this diagram, we can use the Mundell–Fleming model to show how aggregate income Y and the exchange rate e respond to changes in policy.

Figure 4.9: Mundell–Fleming model



4.4.4 The Mundell-Fleming Model with a Changing Price Level

Recall the two equations of the Mundell-Fleming model:

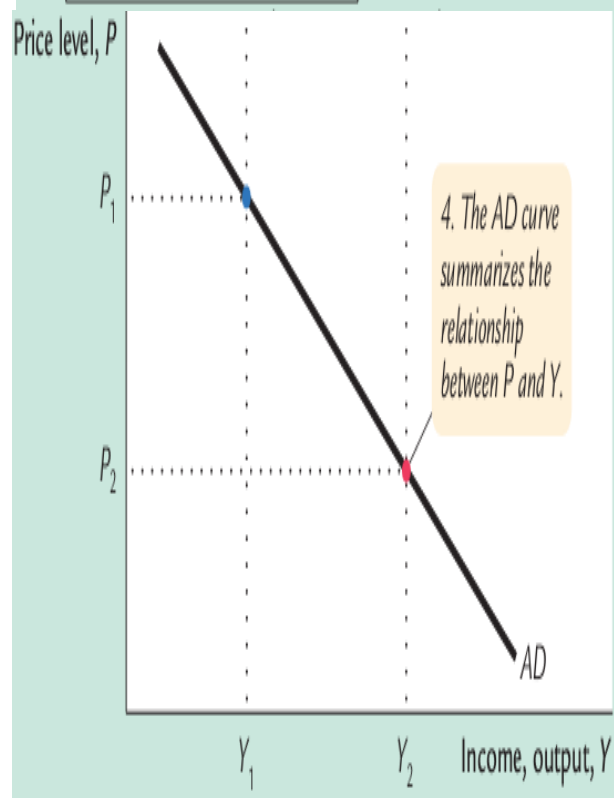
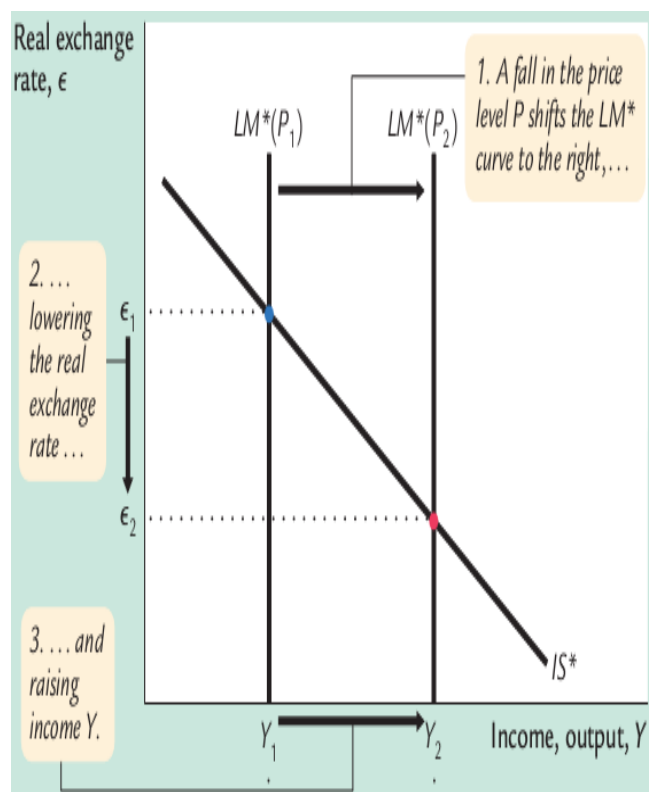
$$Y = C(Y - T) + I(r^*) + G + Nx(e) \quad IS^*$$

$$M/P = L(r^*, Y) \quad LM^*$$

- ✓ The $IS^* - LM^*$ model is constructed for fixed price level. However, a change in the price level shifts the LM^* curve and changes the equilibrium income.
- ✓ Consider reduction in the price level. When the price level falls, the LM^* curve shifts to the right. The equilibrium level of income rises.
- ✓ The second graph displays the negative relationship between P and Y , which is summarized by the aggregate demand curve.

i) The Mundell–Fleming Model

b) The Aggregate Demand Curve



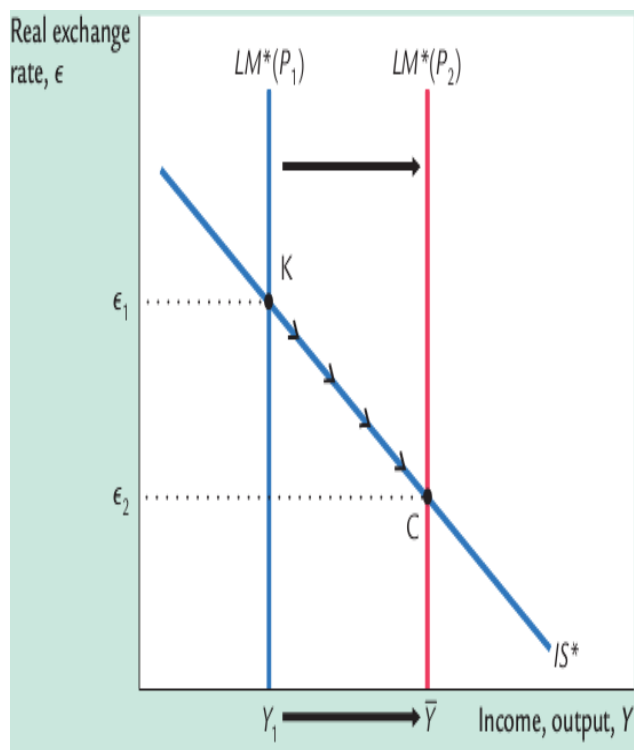
The Short-Run and Long-Run Equilibria in a Small Open Economy

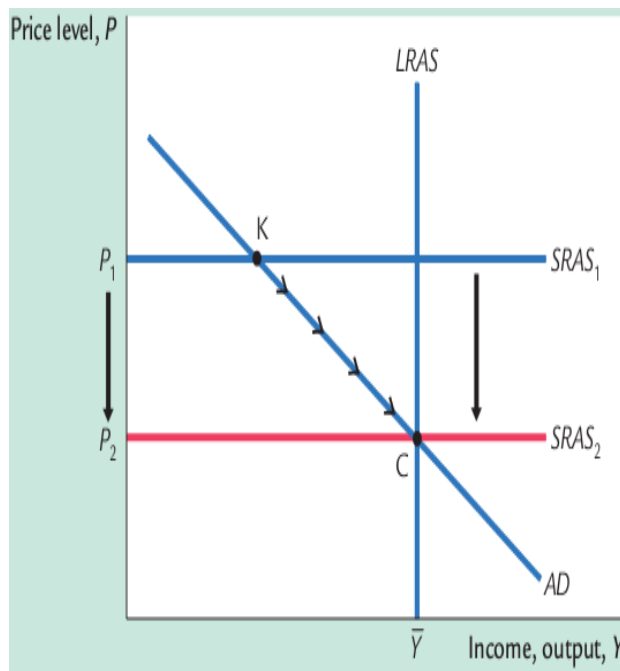
Now let's compare the short-run and the long-run equilibrium of the economy using the IS^* – LM^* and aggregate demand-aggregate supply models.

- ✓ Point **K** in both panels shows the equilibrium under the Keynesian assumption that the price level is fixed at P_1 .
- ✓ Point **C** in both panels shows the equilibrium under the classical assumption that the price level adjusts to maintain income at its natural rate $\bar{Y}$.

a) The Mundell –Fleming Model

b) The AD-AS model





❖ Policy in the Mundell-Fleming Model: A Summary

- ✓ The Mundell-Fleming model shows that the effect of almost any economic policy on a small open economy depends on whether the exchange rate is floating or fixed.
- ✓ To be more specific, the Mundell –Fleming model shows that the power of monetary and fiscal policy to influence aggregate income depends on the exchange-rate regime.
 - Under floating exchange rates, only monetary policy can affect income. The usual expansionary impact of fiscal policy is offset by a rise in the value of the currency.
 - Under fixed exchange rates, only fiscal policy can affect income. The normal potency of monetary policy is lost because the money supply is dedicated to maintaining the exchange rate at the announced level.

4.4.4. Limitations of the Mundell-Fleming Model

1. **Neglect of the long run budget constraints:** the model fails to take account of long run constraints that govern both the private and public sector. In the long run private sector spending has to equal its disposable income, while in the absence of money creation government expenditure has to equal its revenue from taxation. This means that in the long run the current account has to be in balance. One implication of these budget constraints is that a forward-looking private sector would realize that increased government expenditure will imply higher taxation for them in the future, and this will induce increased private sector savings today that will undermine the effectiveness of fiscal policy.

2. **Wealth Effect:** the model does not allow for wealth effects that may help in the process of restoring long run equilibrium. A decrease in wealth resulting from a fall in foreign assets associated with a current account deficit will ordinarily lead to a reduction in import expenditure which should help to reduce the current account deficit. While such an omission of wealth effects on the import expenditure function may be justified as being small significance in the short run, the omission nevertheless again emphasizes the essentially short-term nature of the model.
3. **Neglect of supply side factors:** one of the obvious limitations of the model is that it concentrates on the demand side of the economy and neglects the supply side. There is an implicit assumption that supply adjust in accordance with changes in demand. In addition, because the aggregate supply curve is horizontal up to full employment, increases in aggregate demand do not lead to changes in the domestic price level, rather they are reflected solely by increases in real output.
4. **Treatment of capital flows:** one of the biggest problems of the model concerns the modelling of capital flows. It is assumed that a rise in the domestic interest rate leads to a continuous capital inflow from abroad. However, to expect such flows to continue indefinitely is unrealistic because after a point international investor will have rearranged the stocks of their international portfolios to their desired content and once this happens the net capital inflows into the country will cease. The only way that the country could then continue to attract capital inflows would be a further rise in its interest rate until once again international portfolios are restored to their desired content. Hence a country that needs a continuous capital inflow to finance its current account deficit has to continuously raise its interest rate. In other words, capital inflows are a function of the change in the interest differential rather than the differential itself.
5. **Exchange rate expectations:** A major problem with the model is the treatment of exchange rate expectations. The model does not explicitly model these and implicitly presumes that the expected change is zero, which is known as static exchange rate expectation. While this might not seem to be an unreasonable assumption under fixed exchange rates, it is less tenable under floating exchange rates. According to the model a monetary expansion leads to a depreciation of the currency under floating exchange rates- in such circumstances it seems unreasonable to assume that economic agents do not expect depreciation as well. If agents expect depreciation this may require a rise in the domestic interest rate to encourage them to continue

to hold the currency which will have an adverse effect on domestic investment- implying a weaker expansionary effect of monetary policy than is suggested by the model. Indeed, the need to maintain market confidence in exchange rates can severely restrict the ability of government to pursue expansionary fiscal and monetary policies.

Review exercise

1. In the Mundell–Fleming model with floating exchange rates, explain what happens to aggregate income, the exchange rate, and the trade balance when
 - A. Taxes are raised.
 - B. Government expenditure increases
 - C. Central bank increased money supply
2. In the Mundell–Fleming model with fixed exchange rates, explain what happens to aggregate income, the exchange rate, and the trade balance when
 - A. taxes are reduced.
 - B. Government expenditure decreases
 - C. Central bank decreased money supply
3. Consider a **closed economy** where the following quantitative relationships are given. Using the IS-LM model, work out the problems that follow.
 $C = 200 + 0.5Y$
 $M_s = 500$
 $P = 1$
 $G = 540$
 $M_d = 0.5Y + 150 - 80r$
 $I = 170 - 65r$

- a) Determine the **equilibrium interest rate**
- b) Calculate the equilibrium level of income

b) A small open economy is described by the following equations

$$C = 50 + 0.65(YD)$$

$$I = 100 - 10r$$

$$NX = 160 - 42e$$

$$M/P = Y - 50r$$

$$G = 400$$

$$T = 100$$

$$M = 2500$$

$$P = 10$$

$$r^* = 5$$

- A) Derive and graph the IS* and LM* curves.
- B) Calculate the equilibrium exchange rate, level of income, and net exports.
- C) assume a fixed exchange rate. Calculate what happens to the exchange rate, the level of income, net exports, and the money supply if the government increased Tax by 100. Use a graph to explain what you find.
- D) Assume a floating exchange rate. Calculate what happens to the exchange rate, the level of income, net exports if Central bank increased nominal money supply to 3100. Use a graph to explain what you find.

.....

CHAPTER FIVE

AGGREGATE SUPPLY ANALYSIS

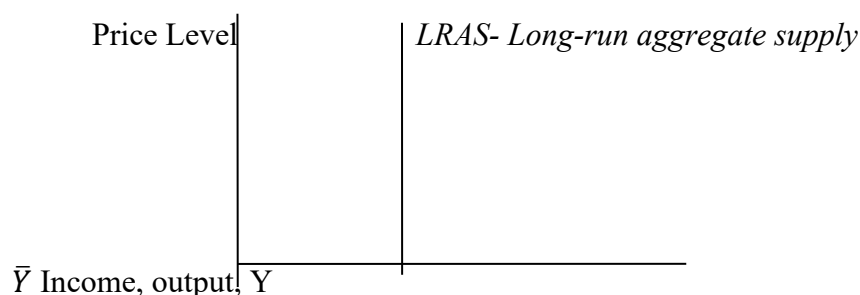
5.1 INTRODUCTION

- ❖ Economists usually analyze short-run fluctuations in aggregate income, employment and the price level using the model of aggregate demand and aggregate supply.
- ✓ Up to now an attempt was made to analyze the demand side of the economy taking the price level, P as exogenously determined.
- ✓ This section develops the supply side of macroeconomics. This will give us a supply curve to add to our demand curve, so that by equating supply and demand in the economy we obtain endogenously determined equilibrium values of output and the price level.
- ✓ More focus will also be given to overview different aggregate supply models and how the shape of the aggregate supply curve will take in different models.
 - **Aggregate supply** is the relationship b/n total quantity of output (i.e., GDP) that firms will produce and the price level. *Because the firms that supply goods and services have flexible prices in the long run but sticky prices in the short run, the aggregate supply relationship depends on the time horizon.*
- ✓ Here, we need to discuss **two different extreme aggregate supply curves**:
 - ✚ The long run aggregate supply curve, LRAS (the classical supply curve) and
 - ✚ The short-run aggregate supply curve, SRAS (the Keynesian supply curve).
- ✓ Finally, an attempt will be made to present three prominent models of short-run aggregate supply curve.

5.1. The Classical Approach to Aggregate Supply /The LR Vertical AS Curve/

- ✓ Because the classical model describes how the economy behaves in the long run, we derive the long-run aggregate supply curve from the classical model.
- ✓ **The classical aggregate supply curve is vertical**, indicating that the same amount of goods will be supplied whatever the price level i.e. **output does not depend on the price level** (see the figure below).

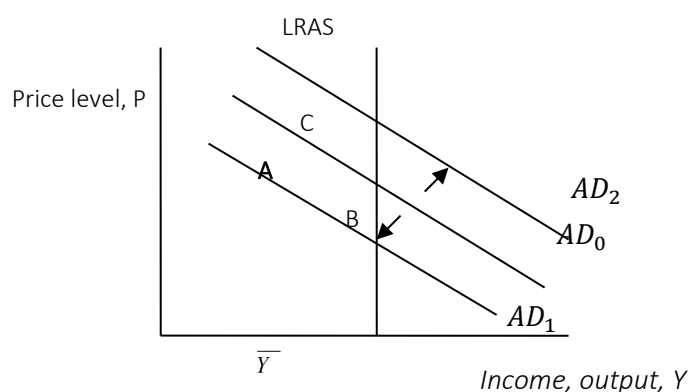
- ✓ *The classical supply curve is based on the assumption that the labor market is in equilibrium with full employment of the labor force.*



The position of the vertical long-run aggregate supply curve (*long run level of output*) is often called **potential output**, **full employment output**, or the **natural level of output**. That means the position of the LRAS curve shows the output that results when the unemployment rate is at its natural rate, or normal level.

- ✓ If the aggregate supply curve is vertical, then changes in aggregate demand affect prices but not output.
- ✓ For example, if the money supply falls, the aggregate demand curve shifts downwards, as in figure below. The economy moves from the old intersection of aggregate supply and aggregate demand, point A to the new intersection, point B. If it is an increase in money supply, the economy moves from A to C. *The shift in aggregate demand affects only prices.*

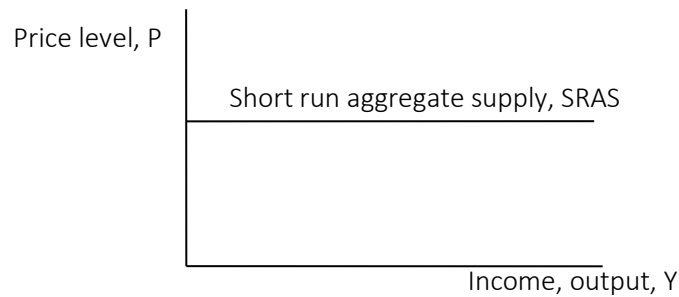
Thus, the vertical aggregate supply curve satisfies the **classical dichotomy** (separation of economic variables into **real variables** and **nominal variables**); because it implies that the level of output is independent of the money supply.



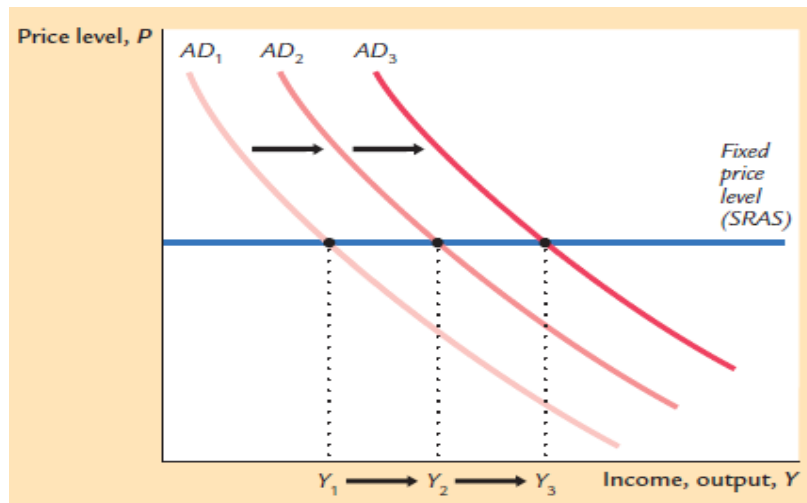
5.2. The Keynesian Approach to Aggregate Supply/The SR- Horizontal AS Curve/

- ✓ It is clear from the above discussion that the classical model and the vertical aggregate supply curve apply only in the long run.

- ✓ In the short run, some prices are sticky and, therefore, do not adjust to changes in demand. Because of price stickiness, short run aggregate supply curve is not vertical.
- ✓ This is what we call the **Keynesian aggregate supply curve**. *In the extreme Keynesian case aggregate supply curve is horizontal*; indicating that firms will supply whatever amount of goods is demanded at the existing price level (see the figure below).



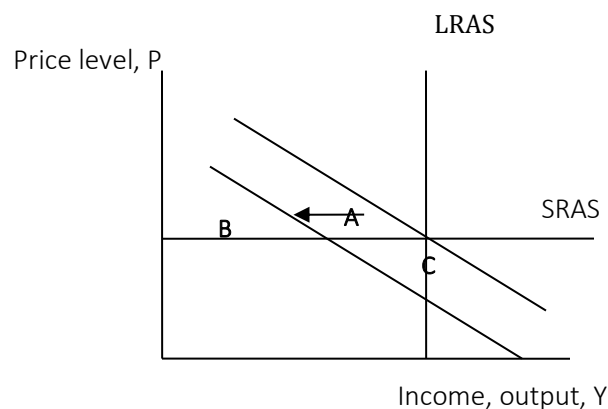
- ✓ **The idea underlying the Keynesian aggregate supply curve is that:**
 - *Because there is unemployment, firms can obtain as much labor as they want at the current wage. Their average costs of production therefore are assumed not to change as their output levels change. They are accordingly willing to supply as much as is demanded at the existing price level.*
- ✓ The short run equilibrium of the economy is obtained at the intersection of the aggregate demand curve and the horizontal short run aggregate supply curve (as in the following figure).
- ✓ *In this case changes in aggregate demand either through fiscal policy or monetary policy do affect the level of output in the economy (that is why Keynesian economists are in favor of government policy intervention towards fighting recession).*
- ✓ *For instance, as the following figure shows, right-ward shift in AD curve from AD_1 to AD_2 and to AD_3 (be it due to expansionary fiscal policy or monetary policy) causes output or real GDP to expand from Y_1 to Y_2 and to Y_3 , respectively, without influencing the price level.*



Movement

from the Short-Run to the Long-Run

- ✓ Suppose for instance the central bank reduces the money supply and the aggregate demand curve shifts downward as in the following figure.
- ✓ In the short run, prices are sticky, so the economy moves from point A to point B.
- ✓ Output and employment fall below their natural levels, which means the economy, is in recession.
- ✓ Over time, in response to the low demand, wages and prices fall. The gradual reduction in the price level moves the economy downward along the aggregate demand curve to point C, which is the new long run equilibrium.



5.3. Alternative Models of Aggregate Supply

- ✓ When we introduce the aggregate supply curve above, we established that aggregate supply behaves differently in the short run and in the long run.
- ***In the long run, prices are flexible, and the aggregate supply curve is vertical.*** In this case, shifts in the aggregate demand curve affects the price level, but output remains unchanged.
- By contrast, ***in the short run, prices are sticky, and the aggregate supply curve is not vertical. In this case, shifts in the aggregate demand do cause fluctuations in output.***
- In the last section, we look at the extreme Keynesian case where aggregate supply curve is horizontal in which all prices are fixed.
- ✓ ***Our task in this section is to refine this understanding of short-run aggregate supply.***
- Here, we will present three prominent models of short-run aggregate supply curve.
- Although these models differ in some significant details, they are also related in an important way: they share a common theme about what makes the short-run and the long-run aggregate supply curves differ and a *common conclusion that short-run aggregate supply curve is upward sloping i.e. the final destination is a short run aggregate supply equation of the form*

$$Y = \bar{Y} + \alpha(P - P^e), \quad \alpha > 0$$

Where Y is output, $\bar{Y}$ is the natural rate of output, P is the price level, and P^e is the expected price level.

- ***This equation states that output deviates from its natural rate when the price level deviates from the expected price level.***
- The parameter α indicates how much output responds to unexpected changes in the price level.

Although each of the three theories adheres to the given functional form, each highlights a different reason why unexpected movements in the price level are associated with fluctuations in aggregate output.

5.3.1. The Sticky- Wage Model

- To explain why the short-run aggregate supply curve is upward sloping, many economists stress the sluggish adjustment of nominal wages.
- **The Reasons:**
 - i. In many industries, nominal wages are set by long-term contracts, so wages cannot adjust quickly when economic conditions change.

ii. Even in industries where there is no such formal contract, implicit agreements between workers and firms may limit wage changes.

✓ **For these reasons, many economists believe that nominal wages are sticky in the short run.**

○ The sticky-wage model shows what a sticky nominal wage implies for aggregate supply.

○ ***To understand the model let's consider what happens to the amount of output produced when the price level rises:***

a. *When nominal wage is stuck, a rise in the price level lowers the real wage (W/P), making labor cheaper.*

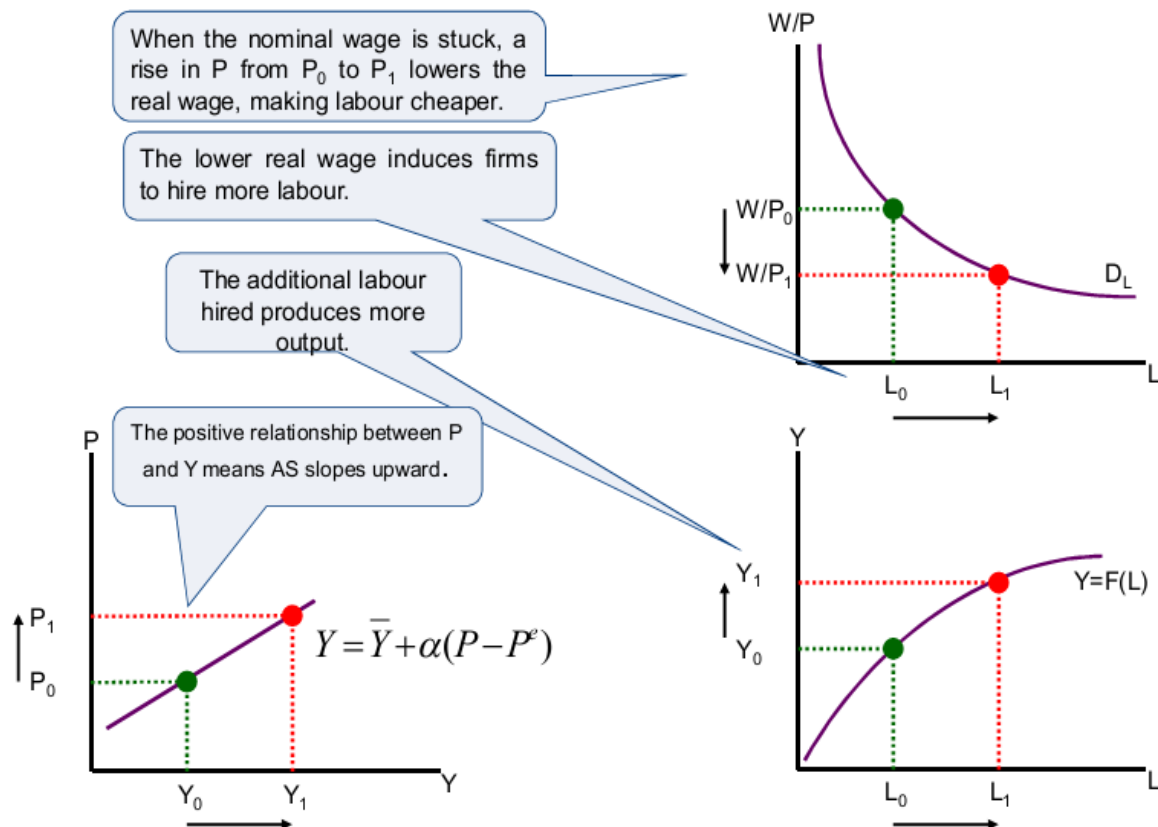
b. *The lower real wage induces firms to hire more labor because labor demand is a function of (W/P). I.e., $L^d = L(W/P)$,*

c. *The additional labor hired produces more output since output (Y) is a function of employment (L).*

○ ***This positive relationship between the price level and the amount of output means that the aggregate supply curve slopes upward during the time when nominal wage cannot adjust.***

○ *The sticky-wage model can now be analyzed by using the following graphs.*

The Sticky Wage Model



5.3.2. The Imperfect-Information Model

Unlike the sticky-wage model, this model assumes that market clear i.e. all wages and prices are free to adjust to balance supply and demand. In this model, the short run and long-run aggregate supply curves differ because of temporary misperceptions about prices.

The imperfect-information model assumes that each supplier in the economy produces a single good and consumes many goods. Because the number of goods is so large, suppliers cannot observe all prices at all times. They monitor closely the price of what they produce but less closely the prices of all the goods they consume. Because of imperfect information, they sometimes confuse changes in the overall level of prices with changes in the relative prices. ***This confusion influences decisions about how much to supply, and it leads to a positive relationship between the price level and output in the short-run.***

Consider for instance, if all prices in the economy (unobserved) increase including the supplier's own price (observed) and the supplier expected it then $P = P^e$ and output remains unchanged. The perception is that the relative price for the supplier has not changed. However, if the supplier did

not expect the overall price changes, then the supplier perceives mistakenly that the *relative price of their own product has increased* ($P > P^e$). The supplier then produces more output.

To sum up, the imperfect-information model says that when actual prices exceed expected prices, suppliers raise their output. The model implies an aggregate supply curve of the form:

$$Y = \bar{Y} + \alpha(P - P^e)$$

- ✓ *The equation states that output deviates from its natural rate when the price level deviates from its expected level.*

5.4.3. The Sticky-Price Model

- ❖ This model explains an upward sloping AS curve by assuming that some prices are sticky because:

1. *Firms have long term contracts with customers,*
2. *Firms hold prices steady in order not to annoy regular customers with frequent price changes, and*
3. **Menu Costs:** *once a firm has printed and distributed its catalog or price list, it is costly to alter prices.*

To see how sticky prices can help explain an upward sloping aggregate supply curve, we first consider the pricing decisions of individual firms and then add together the decisions of many firms to explain the behavior of the economy as a whole.

Consider the pricing decision facing a typical firm. The firm's desired price P depends on two macroeconomic variables:

- The overall level of prices P . A higher price level implies that the firm's costs are higher. Hence, the higher the overall price level, the more the firm would like to charge for its product.
- The level of aggregate income Y . A higher level of income raises the demand for the firm's product. Because marginal cost increases at higher levels of production, the greater the demand, the higher the firm's desired price.

We write the firm's desired price as

$$p = P + \beta(Y - \bar{Y})$$

This equation says that the desired price p depends on the overall level of prices P and on the level of aggregate output relative to the natural rate $Y - \bar{Y}$. The parameter β (which is greater than zero) measures how much the firm's desired price responds to the level of aggregate output.

✚ Now assume that there are two types of firms.

- ✓ Some have flexible prices: they always set their prices according to this equation.

- ✓ Others have sticky prices: they announce their prices in advance based on what they expect economic conditions to be. Firms with sticky prices set prices according to:

$$p = P^e + \beta(Y^e - \bar{Y}^e)$$

Where, as before, a subscript “e” represents the expected value of a variable.

For simplicity, assume that these firms expect output to be at its natural rate, so that the last term $\beta(Y^e - \bar{Y}^e)$, is zero. Then these firms set the price, $p = P^e$.

That is, firms with sticky prices set their prices based on what they expect other firms to charge.

We can use the pricing rules of the two groups of firms to derive the aggregate supply equation. To do this, we find the overall average price level in the economy, which is the weighted average of the prices set by the two groups. If s is the fraction of firms with sticky prices and $1 - s$ the fraction with flexible-prices, then the overall average price level is:

$$P = sP^e + (1 - s)[P + \beta(Y - \bar{Y})]$$

The first term is the price of the sticky-price firms weighted by their fraction in the economy, and the second term is the price of the flexible price firms weighted by their fraction. Now subtract $(1 - s)P$ from both sides of this equation to obtain

$$sP = sP^e + (1 - s)[\beta(Y - \bar{Y})]$$

Divide both sides by s to solve for the overall price level:

$$P = P^e + [(1 - s)\beta/s](Y - \bar{Y})$$

The two terms in this equation are explained as follows:

- When firms expect a high price level, they expect high costs. Those firms that fix prices in advance set their prices high. These high prices cause the other firms to set high prices also. Hence, a high expected price level P^e leads to a high actual price, P .
- When output is high, the demand for goods is high. Those firms with flexible prices set their prices high, which leads to a high price level.
- The effect of output on the price level depends on the proportion of firms with flexible prices.

Hence, the overall price level depends on the expected price level and on the level of output. Algebraic rearrangement puts this aggregate pricing equation into a more familiar form:

$$Y = \bar{Y} + \alpha(P - P^e)$$

where $\alpha = s/[1 - s)\beta]$.

Like the other models, the sticky-price model says that the deviation of output from the natural rate is positively associated with the deviation of the price level from the expected price level.

Exercises

1. Why Nominal wages are is sticky in the short run?
2. Explain why the short-run aggregate supply curve is upward sloping?
3. Differentiate between the Keynesian and the Classical approach to aggregate supply?